

UNIVERSITY OF PADUA

MASTERS THESIS

Modelling Infectious Diseases

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Declaration of Authorship

I, Daniel GUTIERREZ VELEZ, declare that this thesis titled, “Modelling Infectious Diseases” and the work presented in it are my own. I confirm that:

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- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work.
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Abstract

Department of Mathematics

Laurea Magistrale in Data Science

Modelling Infectious Diseases

by Daniel GUTIERREZ VELEZ

This thesis develops and evaluates a comprehensive set of statistical and machine learning models to explain and forecast the spread of infectious diseases, motivated by applications in public health and the insurance industry. Using diverse datasets for COVID-19 across ten countries and endemic diseases like Dengue in Colombia, this work integrates epidemiological records with meteorological, environmental, behavioural (Google Trends), and policy data. The analysis spans three distinct modelling paradigms: long-term forecasting of cumulative cases with time series models (SIR, SEIR, ARIMA, GGM, GRU); short-term prediction of new cases using multivariate regression (Lasso, Random Forest, DNNs); and high-resolution spatial analysis of Dengue drivers via Generalized Additive Models (GAMs).

Key findings reveal that for long-term forecasting, simpler data-driven models like ARIMA and diffusion models often provide more robust trend predictions than traditional mechanistic models, which struggle with the multi-wave nature of modern pandemics. In the multivariate context, penalized linear models such as Lasso offer a compelling balance of predictive accuracy and interpretability, consistently identifying lagged cases and Google Trends as significant predictors. The spatial GAM analysis successfully identified significant, non-linear relationships between Dengue risk and environmental factors, such as precipitation and high temperature. However, it also critically exposed poor predictive when compared to ensemble models.

Ultimately, this thesis demonstrates that the optimal modelling strategy is contingent on the specific analytical goal. It provides a practical framework for selecting appropriate models—from interpretable regression for policy insights to spatial models for descriptive risk mapping—thereby contributing a versatile toolkit for managing the multifaceted challenges posed by infectious diseases.

Contents

Declaration of Authorship	ii
Abstract	iii
1 Introduction	1
1.1 Motivation	1
1.2 History	2
2 Literature Review	4
2.1 Review	4
3 Data	12
3.1 Dates	12
3.2 Infectious Diseases Data	12
3.2.1 COVID-19	12
3.2.2 Epidemics	13
3.2.3 Stringency Index	13
3.2.4 Vaccines	14
3.3 Air Quality	14
3.4 Mobility Index	14
3.5 Google Trends	15
3.6 Population	15
3.7 Meteorological Data	15
3.8 Merging	16
3.9 Summary	16
4 Exploratory Data Analysis	17
4.1 Time Series	17
4.1.1 COVID-19	17
4.1.2 Epidemics	17
4.2 Targets	18
4.2.1 Time series Models	18
4.2.2 Multivariate Models	18
4.2.3 Remark	19
4.2.4 Correlation	19
4.2.5 Analysis of Variance - ANOVA	20
4.2.6 Variance Inflation Factor - VIF	21
4.2.7 Wrapper methods	21
4.2.8 Embedded Methods	22
4.2.9 Sparse PCA + PFA	22
4.2.10 Vector Autoregression (VAR)	23
4.2.11 Selected Features - COVID	23
4.3 Covariates - Epidemics	23

4.3.1	Correlation	24
4.3.2	ANOVA + VIF	25
4.3.3	Wrapper Methods	25
4.3.4	Embedded Methods	25
4.3.5	Sparse PCA + PFA	25
4.3.6	VAR	25
4.3.7	Selected Features - Epidemics	25
5	Time Series Models	29
5.1	Mechanistic Models	29
5.1.1	SIR	29
	Model Equations	30
	Cumulative Cases	30
	Numerical Simulation	30
	Parameter Fitting	31
	Results	32
5.1.2	SEIR	33
	Model Definition	33
	Differential Equations	34
	Cumulative Cases	34
	Numerical Simulation	34
	Curve Fitting	35
	Results	35
5.2	Diffusion Models	36
5.2.1	Bass diffusion Model	37
5.2.2	Remarks	38
5.2.3	Implementation	38
5.2.4	Results	38
5.2.5	Guseo Guidolin Model - GMM	39
5.2.6	Adaptation to Model Infectious Diseases	40
5.2.7	Implementation	40
5.2.8	Results	40
5.3	Linear Models	41
5.3.1	ARIMA	41
5.3.2	Prophet	42
5.3.3	Implementation	42
	Results	42
5.4	Neural Networks	44
5.4.1	GRU-based Recurrent Neural Network Architecture	44
	Advantages of the GRU Architecture	46
	Results	47
5.5	Results	48
5.5.1	COVID	48
	Evaluation Metrics	49
5.5.2	Epidemics	50
	Evaluation Metrics	51
5.6	Conclusions	52

6	Multivariate Models	54
6.1	Data	54
6.2	Linear Models	54
	Coefficients	55
6.3	Non-Linear Models	58
	Feature Importance	58
6.3.1	Neural Networks	59
6.3.2	Results	61
6.4	Results	62
6.4.1	COVID	62
6.4.2	Epidemics	63
6.5	emarks	63
6.6	Conclusions	64
7	Spatial Statistics	67
7.1	Data	67
7.1.1	Target	67
7.1.2	Geographic location	67
7.1.3	Weather	67
7.1.4	Summary	68
7.2	Motivation	68
7.3	Mathematical Framework	68
7.3.1	From Linear Model to GAM	68
	Basis Functions	69
7.4	Data Pre-Processing	69
7.5	Model Fitting	70
7.5.1	Model Structure	70
	Remark	70
7.5.2	Link Function	71
7.5.3	Final Predictor Specification	72
7.6	Results	73
7.6.1	Smooth Terms	73
7.6.2	Parametric Effects	74
7.6.3	Smooth Interactions	74
7.6.4	Predictive power	76
7.7	Conclusions	77
8	Conclusions	80
8.0.1	Applications of this work	81
A	EDA Details	84
A.1	Analysis of Variance - ANOVA	84
A.2	Variance Inflation Factor - VIF	84
A.3	Wrapper methods	85
A.4	Embedded Methods	85
A.4.1	Lasso	85
A.4.2	Adaptive Lasso	86
A.4.3	Decision Trees	86
A.5	Principal Component Analysis - PCA	86
A.5.1	Sparse PCA	87
A.5.2	Principal Feature Analysis (PFA)	88

A.6 Vector Autoregression (VAR)	88
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List of Figures

4.1	Weekly COVID-19 cases by country.(cumulative)	18
4.2	Weekly COVID-19 cases by country.	19
4.3	Weekly Epidemics Cases (cumulative) - Colombia.	20
4.4	Weekly Epidemics Cases - Colombia.	21
4.5	Correlation of features with target	22
4.6	Correlation of features with target	24
5.1	Prophet Components COVID cumulative cases (Colombia)	43
5.2	Prophet Components Dengue cumulative cases	44
5.3	GRU Training and Validation Loss - COVID Colombia	47
5.4	GRU Training and Validation Loss - Dengue	48
5.5	Model fitted values vs Actual COVID cumulative cases (Colombia)	49
5.6	Ranking of RMSE by Country-Model - COVID	51
5.7	Model fitted values vs Actual Dengue cases	52
5.8	Ranking of RMSE by Epidemic-Model	53
6.1	Feature Importance COVID-Colombia	59
6.2	Feature Importance Dengue	60
6.3	Training and Validation Loss FFN Covid- Colombia	61
6.4	Training and Validation Loss FFN Dengue	62
6.5	Model fitted values vs Actual COVID new cases (Colombia)	63
6.6	Ranking of RMSE by Country-Model (COVID)	64
6.7	Model fitted values vs Actual Dengue new cases	65
6.8	Ranking of RMSE by Epidemic-Model	66
7.1	GAM Link Gaussian with count covariate	71
7.2	GAM Link Gaussian without count covariate	72
7.3	GAM Effects Base	75
7.4	GAM Effects autoregressive	76
7.5	Coordinates Effect Base	77
7.6	Coordinates Effect autoregressive	78

List of Tables

3.1	Weekly Country-level Variables by Model Usage	16
4.1	ANOVA F-test results.	20
4.2	Variables selected by wrapper methods using AIC as criterion	22
4.3	Significant lagged variables selected by the VAR model for COVID - Colombia (target: new_cases)	23
4.4	Variable selection across different methods: COVID-19	26
4.5	Significant lagged variables selected by the VAR model for Dengue (target: new_cases)	27
4.6	Variable selection across different methods: Dengue	28
5.1	Parameters SIR-COVID-19	32
5.2	Parameters for each Epidemic	33
5.3	Parameters SEIR-COVID-19	36
5.4	Parameters SEIR-Epidemics	36
5.5	Parameters Bass-COVID-19	39
5.6	BM coefficients by epidemic	39
5.7	GGM coefficients by country	41
5.8	GGM coefficients by epidemic	41
6.1	OLS parameter estimates COVID Colombia	56
6.2	Model coefficients (original scale) COVID-Colombia	57
6.3	OLS parameter estimates Dengue	57
6.4	Model coefficients (original scale) Dengue	58
7.1	Descriptive statistics of the dataset.	68
7.2	Comparison of RMSE for GAM Models Link Functions	72
7.3	Significance of Smooth Terms and Model Fit - Base Model	73
7.4	Significance of Smooth Terms and Model Fit - Model with count variable	73
7.5	Parametric Coefficients of the GAM	74
7.6	Comparison of Model Performance on the Test Set	77
7.7	Comparison of models with and without count	78

Chapter 1

Introduction

1.1 Motivation

The main goal of this thesis work is to find a way to model and explain the spread of infectious diseases using statistics and machine learning. Infectious diseases can get out of hand and evolve to become a pandemic. That is why it is very important to monitor them and have the appropriate tools to explain and forecast their evolution through space and time. In this way we can maybe hopefully identify factors that enable outbreaks so we address them directly and prevent a new pandemic or even epidemic outbreak.

As part of my data science master's degree I intend to develop and explore a set of tools that better explain and forecast the spread of infectious diseases through space and time, making use of novel data and techniques. Infectious diseases have been studied since more than a hundredth years, but thanks to the development of statistical tools, machine learning, and deep learning, and the wide availability of data nowadays, we find ourselves in a position where we can leverage data and knowledge from different sciences and build a novel approach on modelling infectious diseases.

A secondary goal of this document lies in the context this topic plays in the insurance industry. Health and life insurance can benefit greatly from anticipation and precision in the modelling of infectious diseases. Losses can be estimated with more confidence via both frequency and severity of the diseases at hand. Then more appropriate pricing can take place and the insurance carriers will manage better the underlying risk of infectious diseases in their businesses. This business context comes as part of my job as a strategic risk modeller and data scientist for an insurance carrier in Latin America called Sura. The company does business in seven different countries spread in Latin America. In all countries, both health and life policies are a key product for the company. The company plays a special role in Colombia, as part of the private-public system of both health and workers compensation insurance. The company is the largest insurance agency in the country which makes it responsible for hundredths of thousands of lives directly. The multi-country presence and its role as a pillar of the health system in Colombia makes for this company a very high priority being able to forecast the spread of infectious diseases.

The spread of Infectious diseases has been studied since there was little knowledge of the biological nature of them. Still, mathematical models have been useful to describe and forecast the spread of infectious diseases, pandemics and epidemics.

Some of the key questions we aim to find in this document are:

- Which class of time series models provides the most robust long-term forecasts for multi-wave pandemics?
- What are the most significant short-term predictors across infectious diseases?

- Which multivariate models offer the best balance of accuracy and interpretability?
- How do environmental and geographic factors influence disease spread risk at a local level in Colombia?

1.2 History

Throughout history, mathematical and statistical models have been used to understand and manage the spread of infectious diseases. Dating back to the 1700's to our last pandemic spread in 2020, epidemiological models have evolved significantly, adapting sophisticated methodologies to incorporate more information and different natures of disease outbreaks. The foundation of what we know today as mathematical modelling for infectious diseases originated in 1760. Back then, Daniel Bernoulli introduced one of the earliest epidemiological models to study smallpox, this is one of the earliest and most significant steps for the family of quantitative disease analysis. Almost a hundredth years later, in 1854, John Snow made the first contribution to spatial epidemiology through his investigation of a cholera outbreak in London. He provided a deeper understanding of what it meant for an epidemic to spread in space by mapping disease incidence and identifying contaminated water sources. This was incidently a very early contribution to what we now know as Network Science. The epidemiological studies and network science have grown hand to hand ever since.

Then we jump to the next century, the 20th. At this time we see one of the main breakthroughs of epidemiological modelling. Particularly we see the arrival of mechanistic or compartmental modelling: the still widely used SIR model. In 1927, Kermack and McKendrick developed the Susceptible-Infectious-Recovered (SIR) model. This model was used to analyse the spread of the Spanish flu pandemic from 1918. This model has been used ever since for various disease outbreaks. Later influenza pandemics, such as the Asian flu in 1957 and Hong Kong flu 1968, inspired epidemiologists to improve the stochastic models. The Reed-Frost and chain binomial models gained strength. They were able to effectively capture randomness inherent in diseases spread. Later on the 20th century a currently active epidemic was widely spread: the HIV/AIDS crisis of the 1980s. This period helped for the rise of network-based models, the same topic originated from the London cholera outbreak. Here, the importance of social connectivity patterns in disease transmission dynamics was acknowledged and more deeply modelled. The social connectivity patterns and spatial spread of the diseases are key to the mitigation and treatment of the diseases, that is why more spatial statistics should focus on this angle. In this document I intend to contribute in this field to some extend.

The 21st century saw a very fast shift in the world in several aspects. Globalization scientific and technological advancements enabled us to move around the world more, have faster and easier access to data and information. With this also came some new and more complex infectious diseases. Epidemiological modelling continued to evolve in response to these new diseases. The SARS outbreak (2002-2004) introduced opened the door for more sophisticated techniques, such as Agent-Based Models (ABMs), and saw the application of the SEIR (Susceptible-Exposed-Infectious-Recovered) model, an extension of the classic SIR framework that accounts explicitly for incubation periods. Then came the H1N1 pandemic in 2009, which accelerated the adoption of Bayesian inference methods and phylogenetic

models. This approaches enhanced the precision of disease forecast predictions and the tracking of virus evolution.

As we move closer to today it seems that there are more observed outbreaks. In 2014- 2016 we saw the Ebola crisis. During this period spatial models incorporating mobility data were extensively implemented, showcasing that human movement significantly impacts disease transmission patterns. Coincidentally with the technological advancements, during this period machine learning was becoming more popular in science, so we see an introduction of machine learning approaches into epidemiology, paving the way for their broader application in subsequent pandemics.

Most recently, during the COVID-19 pandemic we saw a lot epidemiological modelling methods and approaches, combining diverse data sources, models coming from different scientific backgrounds and a community effort to understand contain the spread of the pandemic. Among the most notable innovations we see the diverse application of neural networks with Bayesian inference, metapopulation modelling , and the application of advanced deep learning techniques like Graph Neural Networks (GNNs). This reflects the shift towards a more computational intensive approach on infectious disease modelling, diverging from the traditional mechanistic models saw from the 18th century. The COVID-19 pandemic came at a time where machine learning had become more widespread and available, thanks to advances in hardware and more easily programable software. That is why a lot of literature uses advanced neural network architectures for the forecast of cases of this disease. In this document we will see that these types of architectures behave very well for some problems but might not be the most suitable tool for longer term forecasting and understanding of the diseases spread.

This historical trajectory briefly outlines the role of some mathematical models and applications in shaping responses to infectious disease threats. Pandemics will continue to emerge and evolve, and the nature of the next one is not known yet. This is why it is important to for us to find a way to model and understand the spread of infectious diseases in a general sense, so we can at least be half-prepared for the next pandemic.

Chapter 2

Literature Review

2.1 Review

This section reviews academic contributions focused on epidemiological modelling and forecasting techniques used to understand and manage pandemics and epidemics, highlighting diverse methodologies and their unique characteristics.

Sarker, Jadbabaie, and Shah, 2022 tackle the problem of COVID-19 modelling using a mixed approach. They introduce a mixture-based model which takes part of the mechanistic approaches and part of the non-mechanistic approaches, retaining the benefits of both. Sarker, Jadbabaie, and Shah, 2022 say that “The model represents time series of cases and fatalities as a mixture of Gaussian curves, providing a flexible function class to learn from data compared to traditional mechanistic models. Although the model is non-mechanistic, we show that it arises as the natural outcome of a stochastic process based on a networked SIR framework.” This mixture allows to learn parameters that have meaningful interpretation, similar to the non-mechanistic models. Their model measures the number of cases $N(t)$ at time t as a sum of gaussian-like terms, with parameters of the gaussian curves, and a small noise term. Then the algorithm to learn the parameters of the gaussian curves is based on detecting the peak in each time series, then fit individual gaussian components around that peak. This should arise naturally from a network-based SIR model where people reduce contacts over time and the diseases spreads through a stochastic block model, i.e different communities.

Bozzuto and Ives, 2023 analyze the forecasting capabilities of models applied to COVID data in the USA. He shows that prediction of new cases can be done at most 8-9 weeks ahead via statistical models. They also find that high predictability is associated with high cyclicity of the nature of the spread of disease. Finally, they conclude that predictability is hindered by measures to avoid high contagion: in places where strong precaution measures to prevent contagions were taken, models fail to predict the natural course of the spread. Their analysis is mostly done using auto-regressive models and simulations of the SIR model.

Other than the pure dynamics of the disease spread as a time series, spatial dynamics have been analyzed and found relevant in the spread of infectious diseases. Sun, 2012 addresses the spatial epidemiology via risk and incidence. The author states that pathogens might be highly localized, spatially restricted or susceptible hosts can be clustered. The author investigates spatial patterns of epidemics with non-linear incidence rates. Evidence of statistical significance is found in the coefficients associated with spatial patterns. The author uses a Susceptible Infected Reaction-Diffusion Model with non linear incidence. This is similar to the SIR models, where there are a set of partial differential equations and the solution to these

are the paths that the Susceptible and Infected will take through time. Among the parameters there is a β , which is the force of infection, or nonlinear incidence. Then Bifurcation analysis and Linear Stability Analysis are made. Here spatial pattern formation takes relevance, as some of the parameters of the system are spatial (D_1 and D_2 which are diffusion coefficients of how the systems spreads in space, not only in time. These findings can be extended to more general disease spreads or also applied to some other contexts. In a later section of this document, this logic will be applied to the specific case of epidemics in Colombia to investigate the spatial effects on disease spread, where diseases often cluster in specific cities or regions thanks to some meteorological, salubrity and other conditions. One could use spatial SI reaction-diffusion models to simulate disease spread between regions and estimate the critical β values to determine thresholds where spatial patterns (localized outbreaks) might happen.

Ranjan, 2020 finds that traditional epidemiological models like SIR, SEIR, Logistic models: underestimate the duration and total size of COVID-19 pandemic because there is an assumption of symmetric rise and fall of new cases, around a single-main peak. He shows with data from US, Europe and India that there is a slower decay in new-cases after the peak, which traditional epidemiological models fail to capture. To address this issue he proposes two data driven models that better fit the actual observed data, especially the slower decay phases. These are a modified gaussian decay model and a piecewise-linear decay model. The key ideas presented are that to keep the rapid growth phase but modify the decline phase to match real-world data where decay is in fact slower). And to use a better fitting model that matches the “stair-step” or slow plateau effect seen in real world cases.

Chowell et al., 2022 illustrate the importance of models that can capture dynamics beyond a single-peak trajectory for worldwide forecasting. They demonstrate a five-parameter sub-epidemic wave modeling framework that addresses the different trajectories of COVID-19 epidemics across various spatial scales. Their approach captures the rise to an initial peak followed by a wide range of post-peak behaviors, from typical decline to steady incidence levels and repeated small waves. They show that their model outperforms the Richards model in terms of forecasting performance, proving useful for short- and intermediate-term forecasting.

Simple and parsimonious approaches have been used for the modeling of pandemics and epidemics. Particularly the Bass model, a widely used model in many diffusion models of products, has been also adapted to understand the dynamic of disease spread. Gurumurthy and Mukherjee, 2020 present early works on modeling and forecasting the mortality (cases) of the COVID-19. They tackle the modeling via a Bass diffusion model, and they compare it with several running models at the time, used by universities to forecast the future cases of the COVID pandemic. Most of them were based on the SEIR, SEIRS methodology. Other relevant models for the comparison were the IHME model which is a statistical model fitting a Gaussian Error Function to the cumulative number of deaths. The fitting is done via non-linear regression to estimate the trajectory of the cumulative and daily death rate as a function of the implementation of social distancing measures. The authors find that the Bass model compares favorably to other notable forecasting models and, in some cases, even outperforms them in predictive accuracy. A takeaway from this study is that simple diffusion models can be a very versatile tool, which are easily implemented and produce good results when compared to other more complicated

models. This makes it a good candidate for using it in several disease spreading scenarios and deepen a bit further into it, as I do later on the document.

Gorbachuk, Dunaievskiy, Syrku, et al., 2022 examine some of the prerequisites and assumptions of the classical Bass diffusion model when applied to the COVID-19 pandemic. They get estimations of the Bass model parameters when used for the adoption in vaccination in Ukraine and Belarus. Their work was based on weekly data from the first semester of 2021.

Sidorov, Faizliev, Balash, et al., 2021 propose a new innovation diffusion model that is an extension of the existing ones: logistic model, the Bass model and the Gompertz model. Their proposal is based on the case when the observed process is the result of interactions of other unobserved processes. As an example of innovation diffusion models they mention the spread of disease, so their new proposal could fit this type of problem.

Di Lucchio and Modanese, 2024 implement agent-based network simulations to the classical Bass diffusion model and its extensions. They do it by combining software packages network and NetLogo. Their implementation would allow us to use more complex models than the Barabasi-Albert and small world networks, which give more flexibility to the model simulation. This implementation can be leveraged to understand the network dynamics of infectious diseases, via fitting the spread of the disease with a Bass model or any of its extensions and then simulate the network generating it with their approach.

Other approaches like incorporating external information, other than the pure time series of cases has been done with success. Information like the number of tests done can have a great effect on "simple" models like the non-linear diffusion models. Furlan and Mortarino, 2021 analyze the effect of the variation in the number of tests done on the effective cases of COVID-19. Their approach confirms that the number of confirmed cases is strongly dependent on the number of tests performed. Particularly they use a nonlinear asymmetric diffusion model. The model is found to be more efficient in comparison with other existing models like the SIRD and the logistic model, combining the properties of innovation diffusion models with the information on the testing. In particular their model extends classical innovation diffusion models (like Bass and its extensions) using a Dynamic Market Potential (DMP) framework. They have an intervention function $\omega(t)$, which is linked to the number of tests or swabs $B(t)$. Their work is relevant to this document because it leverages the external information (swabs performed) to improve a diffusion model approach. This can be extended to the particular use case in this document, leveraging external information like social media or google searches for improving diffusion-like models.

Further diffusion models have been modified and improved to model the behavior of infectious diseases. In their work, Eryarsoy et al., 2021 argue that SIR type models are not good enough for the COVID pandemic. For addressing the shortcomings of the SIR models, they propose an approach that is based on a modified Bass diffusion model. Their novel approach lies in the fact that they do not require initial conditions and assumptions, or data pre-processing. Their additions to the base Bass model are: First, Direct modelling of daily infections using real incidence

data. Second, scaling for under-reporting, Time delay to model fatalities after infections. And lastly, the crude fatality rate in cases.

There is a trend on building upon classical mechanistic models like the SIR and SEIR. The idea is to incorporate analogies from other fields like physics. This gives the models another layer of interpretability, since one can then simulate their trajectory based on physical properties. For example, in their paper, Bender et al., 2024 propose a simplified SIR model that derives quasi-analytical solutions and fitting functions based on the SIR. This is intended to capture the dynamics of the COVID-19 spread. In their work, they compare the spread of a disease to the motion of a particle sliding down a potential energy curve, where the particle's position over time can be seen or interpreted as the fraction of the population that is infected and grows over time. This methodology shows the adoption of incorporating other scientific knowledge background to the spread of infectious diseases modelling.

Simplification of existing models has been done in this application. In their work, Raheja et al., 2023 propose a model called the diffusion prediction model. Here, two types of spread are considered. The first one where transmission occurs after a delay following infection. The second one where transmission happens immediately. The authors give evidence of their model outperforming state-of-the-art methods of COVID-19 cases over a four-week horizon. They fit the model's parameters to observed cases to optimize predictive performance over time. When compared to other models like support vector machines (SVMs), logistic regression, and convolutional neural networks (CNNs), their approach showed higher accuracy and lower error. These types of work are evidence that it is possible to address pandemics and epidemics effectively through mathematically well-designed frameworks, when clearly understanding the goal and data. This can result in methodologies that are less computationally expensive but generalize well to real-world data.

Another approach involves boosting the known mechanistic models. This can be done by combining them with network-based frameworks. This is actually one of the most directly observable ways of how are supposed diseases to spread. These models are supposed to improve explainability by simulating how the diseases spreads from individual to individual in the form of a link between two nodes in a graph. Their ultimate goal is to capture in a more detailed way the dynamics of infectious disease spread. Rafiq et al., 2023 highlight that traditional epidemiological models, such as SIR, often assume random diffusion of infection, overlooking the structured interactions between individuals. To account for this, the authors numerically simulate the SIR model on both Barabási-Albert and Erdős-Rényi networks to study the spread of COVID-19 in Pakistan. The network-based modelling allows them to include the underlying interaction structure of individuals in the connected graph, and with this capture key spatio-temporal features of disease transmission in a localized setting.

Further examples of network-based models can be found in the existing literature. Authors often focus on the pure network properties like the growth dynamic and how are the nodes connected to each other. For instance in the particular case of pandemics and epidemics Xenikos and Constantoudis, 2024 work explores how sub-exponential power law can generate networks observed in real-world epidemics.

They use a simulation based approach to identify cases where networks grow following an extension of the Bass diffusion model. The authors propose a network-based SIR percolation model, that extends the classical Bass diffusion model to describe the spread of an epidemic. The model considers sub-exponential power law growth in the early stages, Weibull dynamics as a natural diffusion pattern, and the incorporation of spatial heterogeneity via modular networks. They simulate the network via 2D lattices and modular networks (representing cities and countries). They find in the empirical application that animal epidemics grow with a $\beta \approx 2$, while COVID-19 spread shows a value larger than 2, even up to 6.

One of the main challenges of modelling an epidemic or pandemic using networks is the computational cost and the existing methods can lead to inaccuracies in some cases. For simulating a mechanistic model like the SIR using an underlying graph, is costly, thanks to the size of the network where each node is an individual person. Some authors have tried to overcome this issue by optimizing the graph generation part of the modelling. For example Awolude, Don, and Cator, 2024 propose a method that incorporates correlations and results in more accurate approximations, using the Erdős-Rényi graph. They state that this method is computationally friendly and can be applied to large networks, which can lead to using graphs as a tool in modelling epidemics with more ease.

Diffusion models based on network growth have also been approached via optimization problems. Babay et al., 2022 introduce a methodological framework which is based on the MinINF problem. They tackle it via two bicriteria approximation algorithms. One is based on cut sparsification result of Krager and the other is a Sample Average Approximation algorithm. The takeaway of their work is that optimization is being introduced in the field of network-based epidemiology, because measures like the social distancing introduce the need to know what is the optimal social distancing measurement to be placed to reduce contagions. The optimization can be made more efficiently by incorporating spatial information into the graph structure. Using the geographic relationships between nodes, one could simplify the network before running computationally intensive algorithms. For example, spatial proximity (nodes that are nearby in actual distance) can result in the removal of less relevant long-distance nodes. This is based on the idea that far away nodes are less likely to contribute to local a local transmission. Additionally, spatial clustering methods can be used to group nearby nodes into larger “super-nodes,”. When there are less nodes, one is effectively reducing the size of the graph. A graph with less nodes and links results in a less complex optimization problem. Spatial information is a key ingredient in the optimization part of the network-based models. Later in the document, some spatial insights are introduced from another modelling focus.

Let us take a small shift in the methodology implemented for infectious diseases modelling into machine learning and deep learning approaches, but without losing the touch with network-based models. Liu et al., 2024 argue that mechanistic models have limitations when it comes to capturing the complexities of modern disease spread. They propose Neural Networks, specifically Graph Neural Networks (GNNs). Their work reviews what they think are the key research directions for applying GNNs to epidemiological modelling. The authors identify two main categories of GNN approaches. The first: common GNNs, which are purely data-driven

and learn from temporal and spatial patterns in their set of parameters without relying on traditional epidemiological constraints. This provides flexibility to the models, since they do not need to fit a rigid shape like the S-shaped curve seen in mechanistic models. The second approach focuses on hybrid GNN models, which combine GNNs with mechanistic models (e.g., SIR) to estimate parameters such as recovery rates. The authors list some advantages when using GNNs over mechanistic models. Some of the mentioned are:

- Their ability to capture relational information (how individuals interact with each other in a network).
- Their flexibility in handling diverse data types. Thanks to the fact that graph can hold information not only about connection, but also inside each node in a tensor form.
- The capacity to adapt to dynamic graph structures, where multiple dimensions and time is considered.
- The interpretability when integrated with hybrid designs (compared to pure GNNs).

GNNs are not just superior in every sense, they also point out a few challenges that come when modelling infectious diseases with GNN-based models.

- GNNs require high-quality data to achieve good performance, involve significant model complexity, and may lack interpretability when the NN implemented grow deep.
- Finally, GNNs might not be ready to go and are computationally expensive, this could limit their effectiveness in real-time applications.

It is important to keep in mind that modelling the behaviour and spread of an infectious disease is not the definitive tool to forecast and prevent the risks of a pandemic or epidemic. We can understand how the disease spreads, but maybe we need some time to identify that the disease is an epidemic. Having an early alarm system is crucial, because we are observing the disease before it gets out of control. Many studies try to leverage machine learning to produce an early warning system for pandemics and epidemics. El Morr et al., 2024 make an exhaustive review of research related to pandemics and epidemic early warning systems. They try to evaluate whether machine learning applications are effective when predicting epidemiological phenomena. Their paper states that machine learning has effectively been implemented in various contexts to produce early alarms for epidemics and pandemics. They cover a wide range of diseases, methodologies and input data. Both regression and classification problems were analysed and found that the most effective techniques in their review were non-linear models architectures: Deep Neural Networks (like LSTM) and Boosting models (like Random Forest). Support vector machines (SVMs), K-Nearest Neighbors (KNN), Logistic Regression and SARIMA models were other methodologies found effective for the tasks. Their review of the existing literature gives a good baseline to try some of those approaches that go beyond the mechanistic and network based models to forecast the behaviour of the infectious diseases.

One example of applying machine learning that goes beyond mechanistic traditional models for infectious diseases can be observed by the work of Gökler, 2024. He uses a hybrid methodology of SVM-Taguchi to predict the number of confirmed cases and deaths of the COVID-19 pandemic. The relevance of this work is the wide application to 51 countries, obtaining better results than some more complex models. This, with some of the previous works reviewed, seems to suggest that in some cases, simpler models can output better results than more complex ones. The results vary from study to study and datasets selected for each is a main driver of this variation. Still, there is a general trend: there is not a perfect model that generalizes well over all the datasets of infectious diseases, every research piece finds their best methodology for their specific set of data. This is why it is so important to understand different natures of modelling and know which strategy could be best applied under different data, circumstances and resources.

Traditionally, efforts to predict cases and infections have been fairly self-contained in terms of data. Meaning, that many studies focus on time series of reported cases to train models that generalize over time. However, additional external information can be included for more complex machine learning models. For example, deep neural networks and random forest architectures can incorporate multiple covariates to improve prediction. Commonly used variables, covered in the work by El Morr et al., 2024 include social or government-imposed restrictions (social distancing), socio-demographic characteristics, meteorological and environmental data. Meteorological data is widely used in the literature. Some variables considered include: elevation, temperature, humidity, rainfall, wind speed and direction, and air and water pollution. Other machine learning implementations cover extra-physical information. They go for the inclusion of digital behaviour, such as social media activity and internet usage. For instance, Lazebnik et al., 2022 use social network data for early prediction, and control of disease outbreaks. Their study focused on analyzing Twitter data to predict the next wave of COVID-19. They get high accuracy in their classification task and good balanced-performance metrics. This type of work opens the door for the integration non traditionally used variables like social media and digital behaviour into infectious diseases modelling.

In the family of social behaviour we find Google Trends. Studies have shown that Google Trends can act as an early warning system for infectious-disease outbreaks by detecting shifts in searches before official case reports emerge. Verma et al., 2018 used Google Trend indices for multiple diseases in India. The search behaviour showed to anticipate national health data, across diverse regions. Likewise, Wang, Lang, and Chen, 2024 reported that Google Trends about measles followed very closely the clinical case counts at regional levels, outperforming rank-based approaches and confirming Google Trends usefulness for infectious monitoring and modelling.

Another crucial part of predicting the behaviour of infectious diseases is to actually understand the pathogens and underlying risks that could arise in any time. Some studies focus on understanding which pathogens could cause a pandemic, in a more survey-type approach. Salmanton-García et al., 2024 conducted a research on which a survey was rolled out to several infectious diseases' experts. In the surveys, health experts had rank the pathogens that would most likely cause the next pandemic. The relevant finding in this study are that influenza viruses were ranked more frequently by experts to be a cause for the next pandemic (79% of surveyed). This finding aligns very strongly with the work detailed on the next chapters. If the

next pandemic is caused by an influenza virus, the models covered in this document can be applied to forecasting the behaviour and spread of the given disease.

Chapter 3

Data

In this chapter, we present a comprehensive overview of the data gathered for this thesis. Our goal is to examine each variable in detail: we explain the rationale for its inclusion and provide its source for those readers that want to find out a little more about the data used.

Countries The geographies covered in this document vary from models, data and chapters. But across the document the following geographies are covered in some form:

- **Latin America:** Brazil, Chile, Colombia, Dominican Republic, Mexico, Panama, Uruguay
- **North America:** United States
- **Europe:** Germany, Italy

Latin American countries were chosen based our familiarity with the data sources and conditions, being born in Latin America and having worked with Latin American data in the past. Continuing to understand data sources and characteristics in Latin America is key for the business motivation of this document: modelling infectious-disease dynamics for future applications in health and life insurance pricing and reserving. Other countries like Germany and Italy offer contrasting meteorological conditions, policy responses, and population behaviours, while the United States provides a global benchmark, thanks to the wide literature covering this region. Including it serves both as a sanity check and as a basis for comparative analysis.

3.1 Dates

All variables collected have an associated date. To simplify the analysis we homogenize everything to be on the weekly level, taking the first day of the weeks (Monday) as reference date. This reference date is for wrangling purposes and numerical usage. The dates are converted to a numeric format to be used as variable inside some of the models, specifically in the Chapter 6.

3.2 Infectious Diseases Data

3.2.1 COVID-19

The first package of data used is the well-known COVID-19 data. The data was collected and organized by The Johns Hopkins Coronavirus Resource Center. And

for the purpose of this document it was accessed via its official Center for Systems Science and Engineering (CSSE), Johns Hopkins University, 2025 GitHub repository. The original data has daily observations of COVID-19 at country level. The data was aggregated to the weekly level and filtered by the countries on our scope.

3.2.2 Epidemics

The other main package used is the Epidemics data. This is not as known as the COVID-19, but contains valuable information for Colombian epidemics. The National Health Institute of Colombia captures the reported cases of different epidemics throughout the country at a weekly level. These are reported at the testing-clinic-level. This attribute makes it very useful to leverage the spatial information in the data. The information was downloaded via the Epidemics watch web platform SIVIGILA at Sistema Nacional de Vigilancia en Salud Pública (SIVIGILA), Instituto Nacional de Salud de Colombia, 2025.

The epidemics collected are:

- Dengue
- Zika
- Chikungunya
- Varicella

3.2.3 Stringency Index

The Oxford Coronavirus Government Response Tracker (OxCGRT) produces a daily *Stringency Index*, which combines nine distinct policy indicators into a single score from 0 to 100. It can be interpreted as going from no measures of restriction at all to the maximum strictness regards COVID-19 control. The group of indicators that are part of the index are:

- School closures
- Workplace closures
- Cancellation of public events
- Restrictions on public gatherings
- Closure of public transport
- Stay-at-home requirements
- Public information campaigns
- Restrictions on internal movements
- International travel controls

Each of these metrics is scored on a 0-100 scale, and the Stringency Index for any given day is computed as the arithmetic mean of the nine values. When policies vary at the subnational level, the index reflects the strictest policy in force (by the authors construction). The data was accessed through the official GitHub repository Blavatnik School of Government, University of Oxford, 2025. For this document the index is averaged at country-week level.

3.2.4 Vaccines

The OxCGRT also consolidates in the repository the percentage of the population vaccinated against COVID-19. This information is also averaged at the country-week level for our work.

3.3 Air Quality

We obtain air pollution measurements from the World Air Quality Index (WAQI) project. The project offers a dedicated global dataset—updated three times daily—covering roughly 380 major cities since January 2015. For each city, pollutant concentrations are aggregated by taking the median across all available monitoring stations, and the dataset reports the daily minimum, maximum, median, and standard deviation for each pollutant. In this thesis we focus on the following six pollutants:

- Carbon monoxide (co)
- Nitrogen dioxide (no2)
- Ozone (o3)
- Particulate matter $\leq 10 \mu\text{m}$ (pm10)
- Particulate matter $\leq 2.5 \mu\text{m}$ (pm25)
- Sulfur dioxide (so2)

These have been averaged to the weekly level. and can be found at the project website World Air Quality Index Project, [2025](#).

3.4 Mobility Index

The mobility data used for this document is retrieved from MobMeter initiative, which is a data compilation as part of the paper by Finazzi, [2023](#). This work procures to offer an open alternative to the data that was shared by Google Mobility, Apple Mobility and Waze during the COVID-19 pandemic. . They use a country-level, daily mobility dataset—originally published as a supplement to their article derived from over 3.8 million anonymized smartphone trajectories. It covers 17 countries from March 11, 2020 to the present day, with the following characteristics:

- **Metrics:**
 - Estimated daily average travelled distance per person.
 - Estimated percentage of people who did not move during the 24 hours.
 - 95% bootstrap confidence interval for each estimate.
- **Countries (3-letter ISO codes):** ARG, CHL, COL, CRI, ECU, GRC, GTM, ITA, MEX, NIC, PAN, PER, PHL, SVN, TUR, USA, VEN
- **Temporal coverage:** March 11, 2020 – present (daily resolution).
- **Temporal smoothing options:**
 - None

- 7-day moving average
- 14-day moving average
- 21-day moving average
- 28-day moving average

For the purpose of this work we use the 14-day moving average. It is then averaged again at the country-week level. The repository of this work is found at Finazzi and Cremonesi, 2024.

3.5 Google Trends

Our data-collection strategy is motivated by the works already reviewed of Lazebnik et al., 2022, Verma et al., 2018 and Wang, Lang, and Chen, 2024, who showed that Google Trends search volumes and social network interactions can anticipate official case counts. We leverage Google Trends as an accessible index of public concern.

We therefore selected four influenza-related topics from the Google Trends platform Google LLC, 2025:

- Physician
- Hospital
- Pharmacy
- Symptom

For each topic, we extract weekly search-interest indices to be used in Chapter 6.

3.6 Population

This data piece is included to keep track of the population, which is needed specially in our implementation of the SIR and SEIR models. But it is also useful in the multivariate regression framework. We simply use the yearly population by country level. And repeat the entry for each weekly observation. The data is obtained in United Nations, Department of Economic and Social Affairs, Population Division, 2025.

3.7 Meteorological Data

Data pertaining meteorological conditions was reviewed in Chapter 2. For this work we use the Meteostat Development Team, 2025 to access the weather of the 9 biggest cities in Colombia, where the majority of the population lives. For the selected variables the weather is collected on a daily basis, and the weighted average by population is performed to obtain a global weekly observation for each of the following variables:

- `tavg`: Weighted average of daily mean temperature over the week.
- `tmin`: Weighted average of daily minimum temperature over the week.
- `tmax`: Weighted average of daily maximum temperature over the week.

- prcp: Total weekly precipitation (e.g., rainfall or snowfall).
- wdir: Weighted average wind direction over the week (in degrees from north).
- wspd: Weighted average wind speed over the week (e.g., meters per second).
- pres: Weighted average atmospheric pressure over the week (e.g., hPa).

3.8 Merging

For the Chapter 6 we need to merge the data. This is done by the country-week level. Obtaining a dataset which contains weekly observations for each country, serving as a multi-index column and the other columns are the features outlined so far.

3.9 Summary

Table 3.1 contains a summary of the variables used for both COVID and epidemics. Taking into account that for the target variable y_{t+1} , that is, cases for next week, the variable y_t (cases in this week) is used.

TABLE 3.1: Weekly Country-level Variables by Model Usage

Category	Variable	Model	Explanation	Source
Temporal & Cases	date	Both	Weekly dates	Each time series
	country	Covid	Country of cases reported	Johns Hopkins
	new_cases	Covid	Weekly new cases of covid	Johns Hopkins
	cases	Covid	Weekly cumulative cases of covid	Johns Hopkins
	cases-epidemics	Epidemics	Weekly cumulative cases of epidemics	Colombian Health Institute
Population	population	Covid	Population in thousands	United Nations
Air Pollution	co	Both	Avg. weekly CO concentration: country	Air Quality Open Data
	no2	Both	Avg. weekly NO2 concentration: country	Air Quality Open Data
	o3	Both	Avg. weekly O3 concentration: country	Air Quality Open Data
	pm10	Both	Avg. weekly PM10 concentration: country	Air Quality Open Data
	pm25	Both	Avg. weekly PM2.5 concentration: country	Air Quality Open Data
	so2	Both	Avg. weekly SO concentration: country	Air Quality Open Data
Weather	tavg	Epidemics	Weighted avg. weekly avg. temp	Meteostat.net
	tmin	Epidemics	Weighted avg. weekly min. temp	Meteostat.net
	tmax	Epidemics	Weighted avg. weekly max. temp	Meteostat.net
	prcp	Epidemics	Weighted avg. precipitation	Meteostat.net
	wdir	Epidemics	Weighted avg. weekly wind direction	Meteostat.net
	wspd	Epidemics	Weighted avg. weekly wind speed	Meteostat.net
	pres	Epidemics	Weighted avg. air pressure	Meteostat.net
	dew	Epidemics	Weighted avg. weekly dew point	Meteostat.net
	humidity	Epidemics	Weighted avg. weekly air humidity	Meteostat.net
Google Trends	google_physician	Both	Weekly trends for topic "physician"	Google Trends
	google_hospital	Both	Weekly trends for topic "hospital"	Google Trends
	google_pharmacy	Both	Weekly trends for topic "pharmacy"	Google Trends
	google_symptom	Both	Weekly trends for topic "symptom"	Google Trends
Mobility	mob_index	Both	Est. weekly avg. (14d rolling avg.) travelled distance by people,	MobMeter
Stringency & Vaccination	StringencyIndex_Average	Covid	Government response tracker for covid containment	Oxford University
	pop_vaccinated	Covid	Pct. of population vaccinated	Oxford University

Chapter 4

Exploratory Data Analysis

This chapter is dedicated to the exploratory data analysis. Before jumping into fitting all the different models at our disposal, we can see how the data is actually structured. In this way we can know beforehand which type of models would be more suitable to fit the data, what anomalies are there in the data and what type of pre-processing should be done, if any to properly have working models on our data.

4.1 Time Series

First we simply plot our target variables. Both the cumulative and new cases. We have two "packages" of data: COVID-19 and Epidemics.

4.1.1 COVID-19

Figure 4.1 displays the behaviour of cumulative cases of COVID-19 across the 10 countries.

In Figure 4.2 we can see more detail about the dynamics of the COVID-19. This figure shows new cases on a weekly basis. This plot would be equivalent as taking the first difference of the cumulative cases series. In some time series this would produce a stationary time series, but in plain sight, we see that the differenced version of COVID-19 cases is not a stationary series, where we observe some seasonality or structured spikes. This seasonality shows periods with peaks. Having more than one peak means a modelling challenge, because traditional mechanistic models are based on the a single-peak shaped curve, resembling a CDF of a Gaussian distribution. Additionally, simple diffusion models like the Bass can also capture a single peak, with a similar shaped curve of adoption. Under this context we think that some models that capture more than one single peak could be more powerful to explain the spread of the COVID-19. The GGM will be introduced in Chapter 5, which has some interesting properties to capture this multi-peak nature. Other models that are not based on differential equations can also capture multiple types of shapes.

4.1.2 Epidemics

The same overview is done as before for each epidemic. We see the plots of both cumulative cases and new cases in Figures 4.3 and 4.4. This time we see that each epidemic has a different date range, some have longer historical information while others have a shorter span. This is due to the different outbreaks happening in time. Zika and Chikungunya emerged between 2014 and 2016 and have been almost eradicated, while Varicella and Dengue have been around for a longer time and are still seeing new cases.

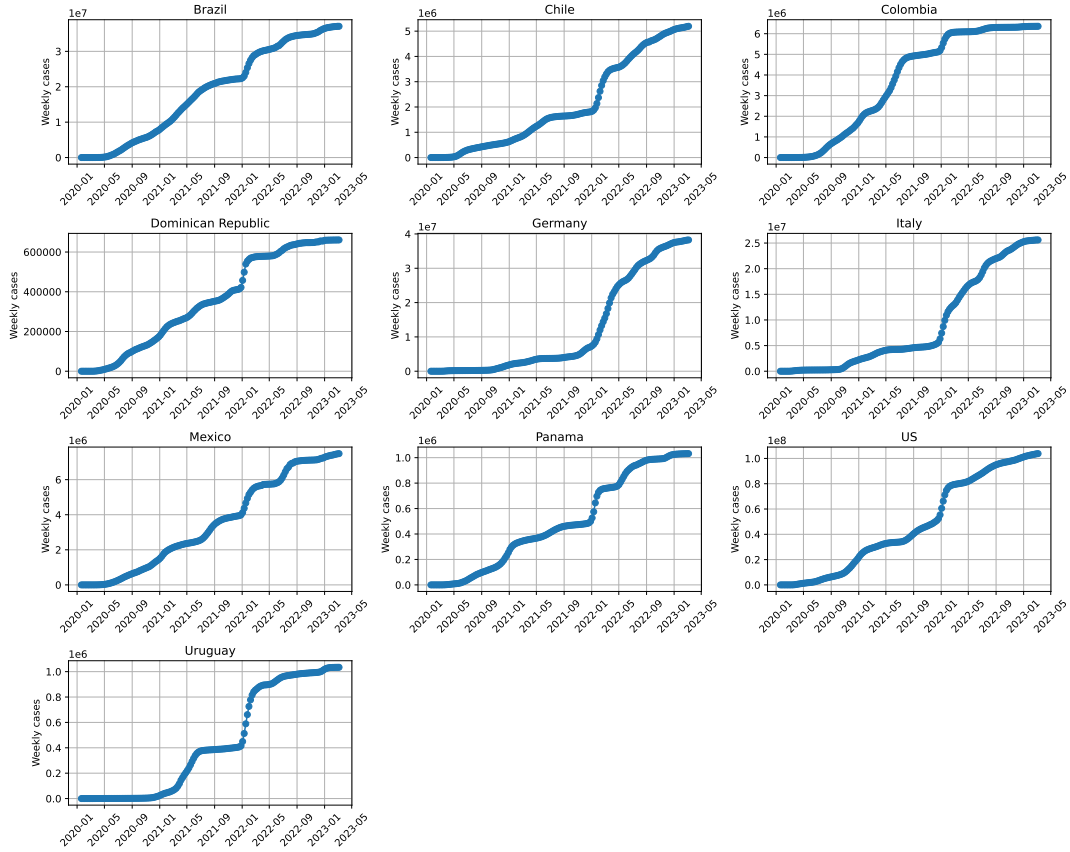


FIGURE 4.1: Weekly COVID-19 cases by country.(cumulative)

Refer to Figure 4.3 to see how both Chikungunya and Zika have a plateauing shape, while Dengue and Varicella/Chickenpox have a steady increase or trend, which suggests that they are not yet eradicated. In a more clear sight we see in Figure 4.4 that Zika and Chikungunya have a period that has a large peak followed by a smaller peak. Mechanistic models might be able to capture the behaviour of these two diseases, while might fail to capture the behaviour of the the other two trending diseases.

4.2 Targets

4.2.1 Time series Models

For this family of models, we select the cumulative contagions "cases" as a target, which will be forecasted in a test period. With this selection of target we can use different models that capture trends and seasonality in time series.

4.2.2 Multivariate Models

For the multivariate time series, we opt to focus on the "new_cases" at $t + 1$ as a target variable. We want to find variables that are explanatory of the disease spread for the short term, that is, next week. This variables are collected on a weekly basis and are not easily forecasted. Given that we cannot tell for certain their behaviour in the longer run, the intelligent path would be to not forecast them and stick to having a short term forecasting model for the new cases. That is why we keep the analysis

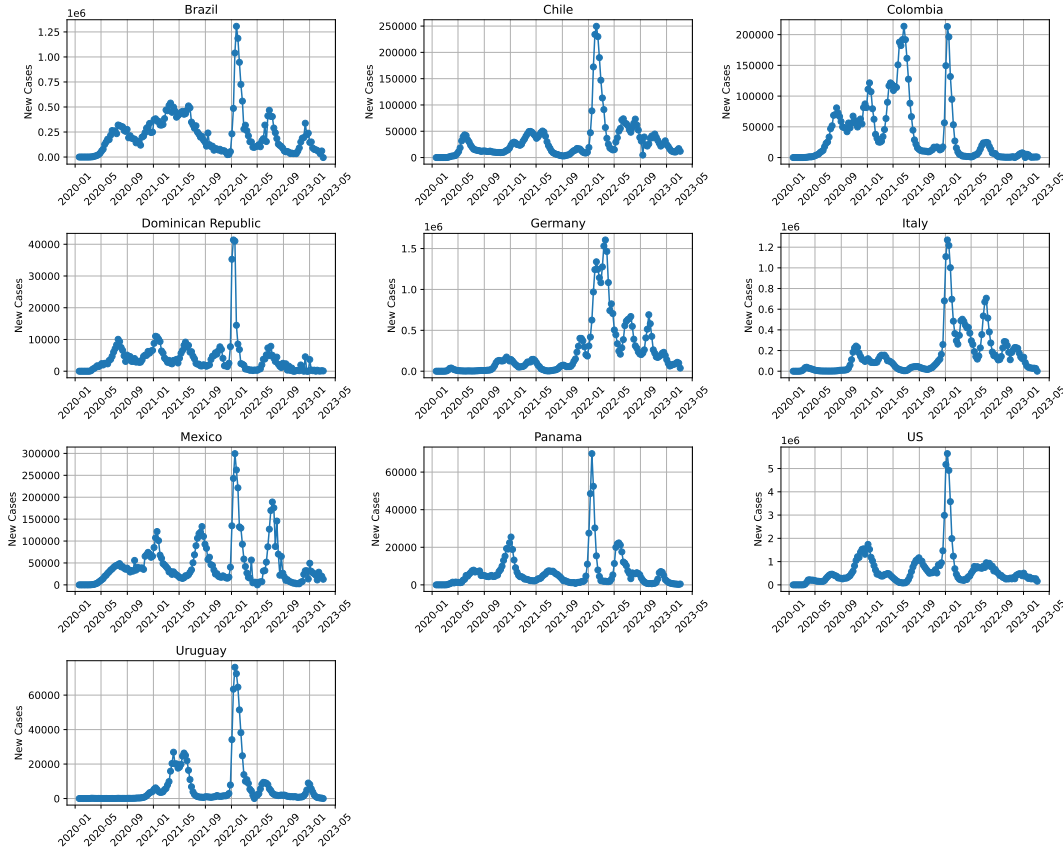


FIGURE 4.2: Weekly COVID-19 cases by country.

to just one step ahead (one week) and use new cases as target. Here we will capture the relation that the variations in the features have on the new cases.

4.2.3 Remark

Before deepening into the analysis, let us note that we are going to display in charts and figures the particular example of Colombia, so the interactions of variables for other countries might differ. We also did a train-test split before the EDA, to select the appropriate features without leaking information from the "future" to the modelling part.

Covariates - COVID Now let us observe the behaviour and relationships of the covariates selected for the COVID-19 modelling task. The relationship between COVID-19 new cases and their covariates interests us, especially the "forwarded" contagions variable, i.e., `new_cases` at time $t + 1$.

4.2.4 Correlation

In Figure 4.5, we show the correlation of the available features using two methods: Pearson and Spearman-Rank correlation. Spearman-Rank is used because of the possibility that among the variables there might be a monotonic relation which is not captured by Pearson's method. Note that the target variable is strongly correlated with its lagged values, which rules out that the target series is stationary. Including lagged-target as a covariate in the modelling phase can help to capture the fact that the series are non-stationarity and lead to well behaved residuals. Then we see that

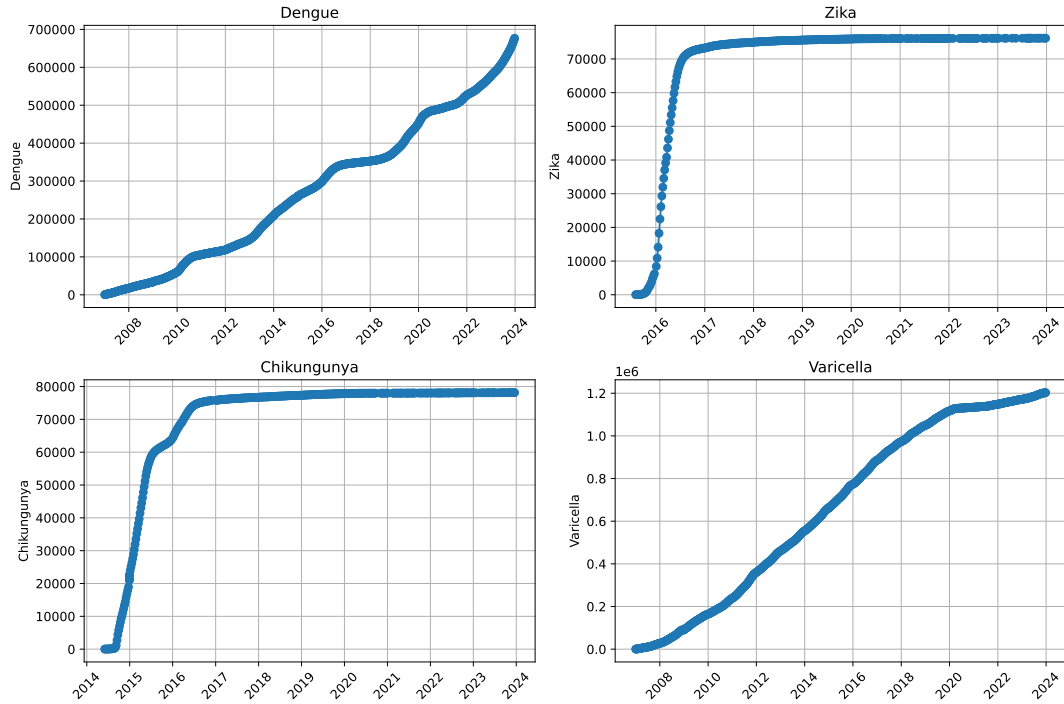


FIGURE 4.3: Weekly Epidemics Cases (cumulative) - Colombia.

Google trends of symptoms and physicians are among the most related features. Also, note that one of the air pollutants: NO₂ is found among the most correlated variables. Finally, the Stringency Index and the Population vaccinated are also found to be some of the ones that have higher correlation with our target variable.

4.2.5 Analysis of Variance - ANOVA

After looking at pure correlations, let us check the ANOVA (Analysis of Variance) to find the statistical relationship between the target variable and each individual feature.

Table 4.1 displays the results for the ANOVA.

TABLE 4.1: ANOVA F-test results.

Feature	F-score	p-value
new_cases	887.61	$2.42 \times 10^{-55*}$
google_symptom	36.30	$2.16 \times 10^{-8*}$
pop_vaccinated	8.40	$4.51 \times 10^{-3*}$
pm25	5.15	$2.52 \times 10^{-2*}$
no2	5.00	$2.72 \times 10^{-2*}$
o3	4.50	$3.60 \times 10^{-2*}$
google_physician	3.47	6.51×10^{-2}
StringencyIndex_Average	2.35	1.28×10^{-1}
co	0.89	3.47×10^{-1}
google_hospital	0.62	4.31×10^{-1}

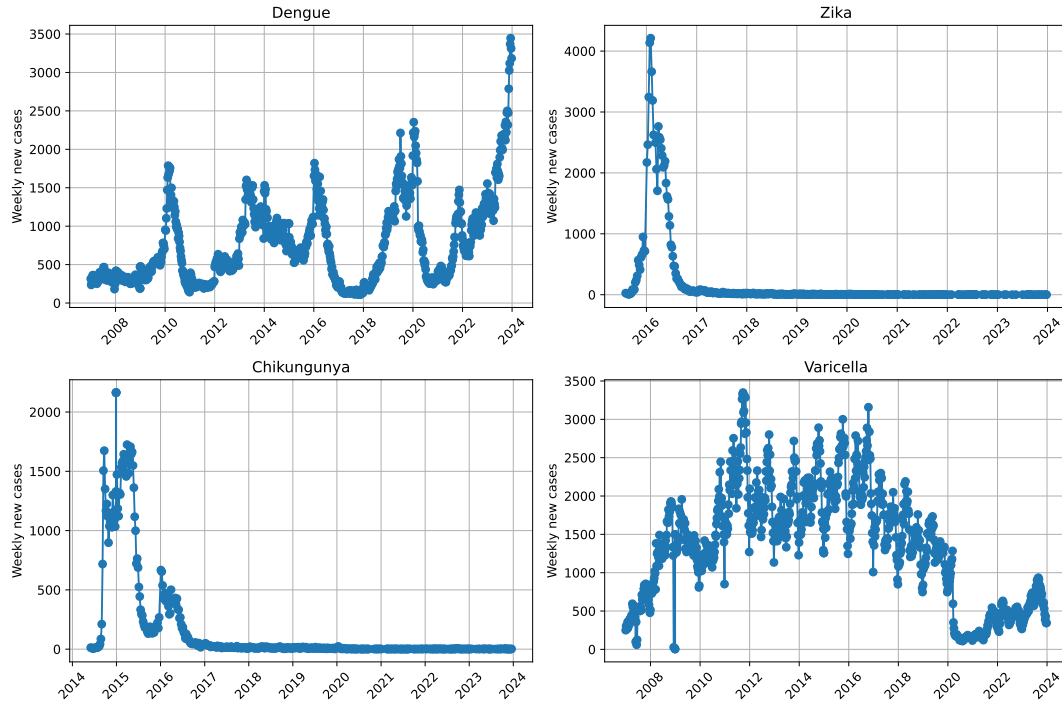


FIGURE 4.4: Weekly Epidemics Cases - Colombia.

4.2.6 Variance Inflation Factor - VIF

After selecting some candidates based on correlation and the F-test we continue to check the features. We note that some of the features are strongly correlated with each other. To control for potential multicollinearity among the selected features so far, we apply the **Variance Inflation Factor (VIF)** analysis. For our analysis all features from the ANOVA continued to be present after checking the VIF.

4.2.7 Wrapper methods

Wrapper methods work by directly training the model as they select the relevant subsets of features by adding or dropping variables in a loop and selecting the model that performs best based on the AIC. The main wrapper methods used for selecting relevant features are:

- Backward Elimination
- Forward Selection
- Add-Delete Stepwise

In this study, all three approaches were implemented to cross-validate the selection of a robust feature subset under AIC minimization. Table 4.2 shows the variables selected by each method, which are mainly consistent with each other.

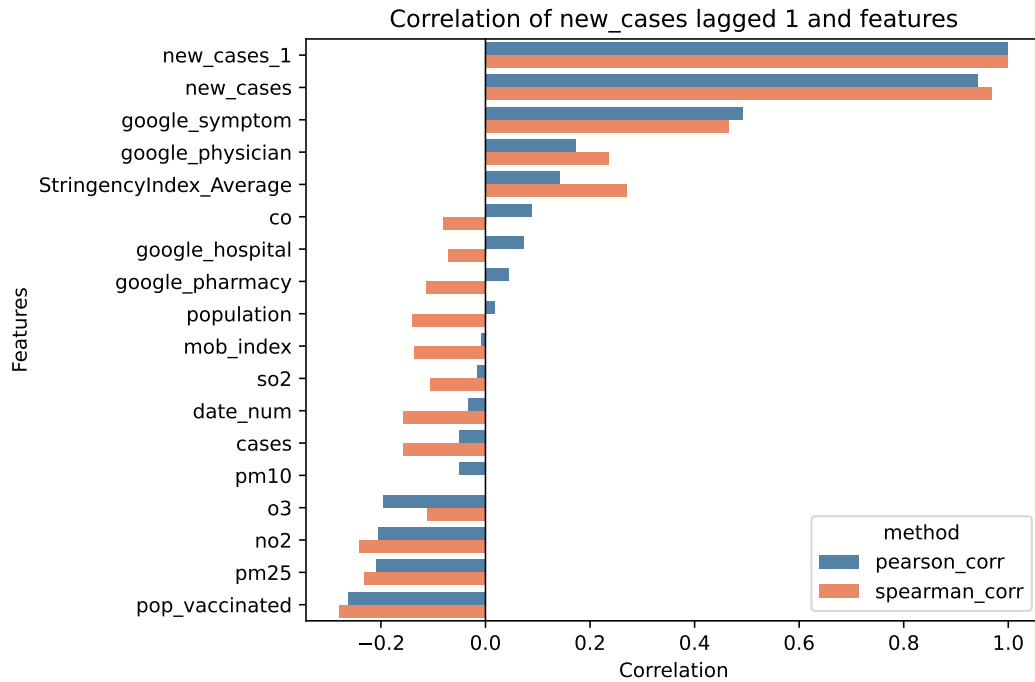


FIGURE 4.5: Correlation of features with target

TABLE 4.2: Variables selected by wrapper methods using AIC as criterion

Method	Selected Variables
Backward Elimination	new_cases, google_symptom, pop_vaccinated, no2, o3
Forward Selection	new_cases, google_symptom, o3, pop_vaccinated, no2
Add-Delete Stepwise	new_cases, google_symptom, o3, pop_vaccinated, no2

4.2.8 Embedded Methods

Embedded methods integrate variable selection into the model training process itself. This means that the model simultaneously learns the parameters and determines which features are most relevant to be selected. These methods are particularly appealing because they optimize both predictive performance and model simplicity in a single go.

In this work, we use embedded methods not only to fit predictive models, but also to perform variable selection. The subset of features selected by these models is later compared to the results of other methods. In Chapter 6, we evaluate the predictive performance of models including some of the ones described below.

- Lasso
- Adaptive Lasso
- Decision Trees

4.2.9 Sparse PCA + PFA

Principal Component Analysis (PCA) and Principal Feature Analysis (PFA) were used on the features matrix to obtain some probable strong predictors that explain

most of the variance of the features, and hopefully this will make them good predictors for the target.

4.2.10 Vector Autoregression (VAR)

Vector Autoregression (VAR) is a multivariate type of linear model based on the univariate autoregressive (AR) model. This variation is used multivariate time series where series have inter-temporal influence on each other. Stochastic processes can influence each other over time, so we use VAR models to capture both the temporal relationships and the contemporaneous interactions among multiple time series variables.

The model comes handy in our variable selection pipeline, because it allows us to evaluate how different predictors, across multiple time lags, contribute to forecasting the target variable. It also highlights indirect relationships: for example, an increase in new COVID-19 cases may lead to a rise in Google searches for symptoms, not necessarily because users feel sick, but because they are reacting to news coverage and seeking information about the disease.

Table 4.3 summarizes the lagged variables whose coefficients were found to be significant (p-value < 0.05), indicating their potential to be used in the modelling phase.

TABLE 4.3: Significant lagged variables selected by the VAR model for COVID - Colombia (target: new_cases)

Variable	Coefficient	P-value
L1.new_cases	0.3596	0.006375
L1.google_symptom	1.2810	1.4×10^{-5}
L2.StringencyIndex_Average	-1.1600	0.001172
L3.new_cases	0.3039	0.021269

4.2.11 Selected Features - COVID

After performing each of the variable selection methods, we end up with different results across each method. This is only natural, because each approach has a different way of determining which set of features are to be used on the regression task at hand. We collect all of the selected features by each method in Table 4.4. To be able to combine all methods we opt for a simple approach: out of all selected variables by each method, we get the more frequently selected across all methods. The top five variables of the table, which are bolded are the ones that will be used in the modelling chapter.

4.3 Covariates - Epidemics

For the variable selection of the Epidemics the same approach is performed. We single out one of the epidemics in this case Dengue. Then with the data for this epidemic the following feature selection methods are performed:

- **Correlation Analysis**
- **ANOVA with Variance Inflation Factor (ANOVA + VIF)**

- Backward Elimination
- Forward Selection
- Add-Delete Stepwise Selection
- Lasso Regression
- Adaptive Lasso Regression
- Decision Trees
- Sparse Principal Component Analysis (Sparse PCA)
- Principal Feature Analysis (PFA)
- Vector Autoregression (VAR)

The overall selected variables can be found in Table 4.6, where the same voting system as in the COVID case was carried out: the variables that are more frequently selected across each method are selected as predictors for the modelling phase.

4.3.1 Correlation

In the correlation analysis, the same behaviour is found: the series, even though is not cumulative, has strong autocorrelation with its firsts lags and the date variable: indicating a strong trend in the time series. Besides than auto correlation, the mobility index indicates strong correlation with the target. Google Trends, atmospheric pollution and weather conditions are also strongly correlated with the target. In Figure 4.6 the correlation of all variables is displayed, using both Pearson's and Spearman Rank correlation.

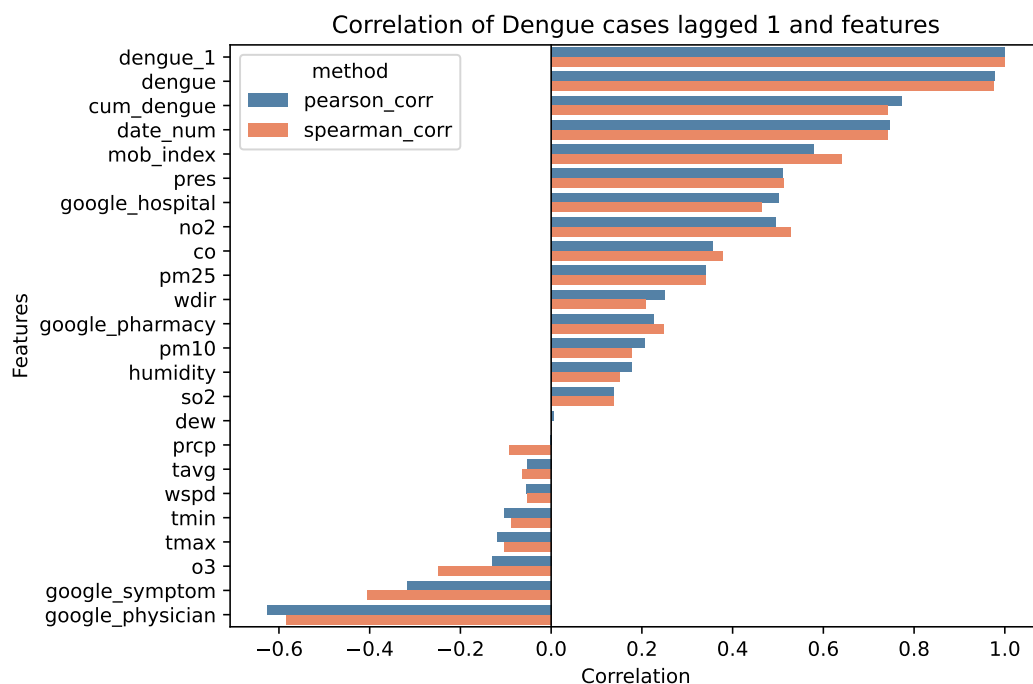


FIGURE 4.6: Correlation of features with target

4.3.2 ANOVA + VIF

In the case of feature selection for Dengue modelling, the VIF procedure rules out two variables that were initially pre-selected by ANOVA F-testing: `cum_dengue` and `date_num`. After applying both ANOVA and VIF filtering, 8 variables are retained.

4.3.3 Wrapper Methods

In the wrapper methods, further differences in variable selection arise. The Backward AIC method selects 9 features, while the Forward and Add-Delete (ADD-DEL) methods each retain only 3.

4.3.4 Embedded Methods

Lasso Lasso selects a set of 5 features, which include atmospheric variables, Google search trends, and the self-lagged target variable. In contrast, Adaptive Lasso selects only the lagged variable (`dengue`) and `date_num`.

Decision Trees In this case, the selection is based on variables that contribute the most to partitioning the decision tree. The feature `dengue` and `tmax` capture more than 96% of the total importance. Based on this, we manually selected the top 4 features from this model.

4.3.5 Sparse PCA + PFA

For these dimensionality-reduction-based methods, the selected features differ somewhat from those chosen by previous approaches. Air pollution variables seem more important in these methods, with `pm25` being consistently selected by both Sparse PCA and PFA.

4.3.6 VAR

The variables whose lags have an effect with a p-value < 0.05 are observed in Table 4.5. Here we see that a lot of covariates have effects through time. Mostly we see the strong autocorrelation of the series, but is also to be noted that the atmospheric variables are indicative of an effect on the target variable. This might be to the fact that changes in weather enable the proliferation of mosquitoes, that are the vehicle for the spread of this disease.

4.3.7 Selected Features - Epidemics

In the same way we collect the variables selected in Table 4.6. The variables with most votes are included in the final modelling. Note that there is a tie between two air pollutants: `pm25` and `no2`. The VAR methodology gives a vote to the `no2` based on distant lag, $\text{lag} = 5$. This is not very convincing, since the first lags are not included. Based on this information we decide to not include this variable. Additionally it would probably include similar information as the `pm25`, since both are trying to capture air pollution. To avoid increasing the complexity of the models we decide to keep it off the selected variables.

TABLE 4.4: Variable selection across different methods: COVID-19

Variable	Corr.	ANOVA	Back.	Forw.	A-D	Lasso	Adapt.	Tree	SPCA	PEA	VAR	Total
new_cases	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	11
google_symptom o3	✓	✓	✓	✓	✓	✓	✓	✓			✓	9
pop_vaccinated no2	✓	✓	✓	✓	✓		✓	✓				6
StringencyIndex_Average	✓	✓		✓	✓							4
date_num							✓		✓	✓	✓	3
google_physician		✓							✓	✓		2
google_hospital									✓	✓		2
pm25		✓										1
google_pharmacy									✓			1
co								✓				1
mob_index							✓					1
pm10									✓			1

TABLE 4.5: Significant lagged variables selected by the VAR model for Dengue (target: new_cases)

Variable	Coefficient	P-value
L1.dengue	-0.2202	0.027095
L1.humidity	-1.4178	0.003633
L2.dengue	0.2353	0.025915
L3.dengue	0.2453	0.025281
L3.pres	36.0925	0.001784
L3.google_hospital	-0.2872	0.012899
L5.dengue	0.2037	0.049074
L5.tavg	-1.7012	0.013546
L5.humidity	1.5581	0.000918
L5.no2	-0.2045	0.015240
L5.o3	0.1437	0.018303
L5.pm10	0.3762	0.033345

TABLE 4.6: Variable selection across different methods: Dengue

Variable	Corr.	ANOVA	Back.	Forw.	A-D	Lasso	Adapt.	Tree	SPCA	PFA	VAR	Total
dengue	✓	✓	✓	✓	✓	✓	✓	✓				9
pres	✓	✓	✓	✓	✓	✓				✓	✓	8
date_num	✓		✓			✓	✓		✓	✓		6
google_symptom		✓	✓	✓		✓						5
pn25		✓	✓						✓	✓		4
no2	✓	✓	✓							✓	✓	4
pn10		✓	✓	✓							✓	3
google_hospital		✓	✓	✓							✓	3
google_physician	✓	✓	✓									3
co		✓	✓									2
tmin								✓				2
cum_dengue	✓							✓				2
google_pharmacy		✓							✓			2
mob_index	✓	✓										2
so2		✓								✓		2
humidity		✓									✓	2
tavg									✓		✓	2
o3											✓	1
wdir		✓										1
wspd									✓			1
dew						✓						1

Chapter 5

Time Series Models

In this chapter we explore the time series models. Our goal is to find a model that can forecast to some extent the general behaviour of the infectious diseases in the long run. To achieve this goal we use the two packages of data stated in Chapter 3. The first application of the models is done on the COVID-19 data and the second one on the Epidemics data.

We will focus on the time series of **cumulative cases**. The motivation for this lies on the fact that some models like the diffusion models work internally using cumulative series. Additionally, we want to see the general trend and trajectory of the disease, so we can forecast in the future and use those forecast for business applications, like pricing and reserving of insurance policies. Using cumulative series allows us to know how many people will be sick in a given period and translate this directly to a business application of a policy recommendation. New cases time series are more volatile and the long run forecast of a volatile series is inherently more uncertain. So we stick to the goal at hand, forecast in the long run can have a more certain trajectory using cumulative time series.

For all time series we do a chronological train-test split using the 80%-20% partition. The evaluation of the model performances is done on the test set, that is the later 20% of the data.

We will do a summary of the models used starting from the Mechanistic Models, which are the traditional epidemiological models. Then we jump to diffusion models, which are also based on ordinary differential equations, that is, the evolution of some equations in time. Later we cover linear models such as the ARIMA and the Prophet. And finally we go into the Non-linear models: Recurrent Neural Networks.

5.1 Mechanistic Models

To be aligned with the most implemented family of models in the epidemiological theory, the first class of models we use to describe the spread of infectious diseases are the *Mechanistic* models. They are formulated as systems of ordinary differential equations with terms that encode the biological and contact-driven mechanisms of transmission.

5.1.1 SIR

The first mechanistic model of them all is the **Susceptible – Infected – Recovered (SIR)**.

The SIR model splits the population under study N into susceptible $S(t)$, infected $I(t)$, and recovered $R(t)$. Next we cover the step by step to get the values estimated by the model for cumulative series:

Model Equations

First we write down the model equations and identify the solution of the ordinary differential equations.

We assume

$$S(t) + I(t) + R(t) = N, \quad S(0) = S_0, \quad I(0) = I_0, \quad R(0) = R_0.$$

The dynamics are:

$$\frac{dS}{dt} = -\beta \frac{S I}{N}, \quad (5.1)$$

$$\frac{dI}{dt} = \beta \frac{S I}{N} - \gamma I, \quad (5.2)$$

$$\frac{dR}{dt} = \gamma I. \quad (5.3)$$

Here:

- β : transmission rate.
- γ : recovery rate ($1/\gamma$ is mean infectious period).

Cumulative Cases

The we make explicit the series for cumulative cases of infected by removing at each step in time the susceptibles (because at each step a fraction of the population is infected or recovered, the rest are susceptible) The cumulative number of individuals ever infected by

$$C(t) = N - S(t).$$

Numerical Simulation

For getting the numerical solution for a given set of parameters, we declare a function.

ODE Right-Hand Side In Python we implement the right-hand side of (5.1)–(5.3):

LISTING 5.1: SIR ODE system

```
def sir_model(y, t, beta, gamma, N):
    S, I, R = y
    dSdt = -beta * S * I / N
    dIdt = beta * S * I / N - gamma * I
    dRdt = gamma * I
    return dSdt, dIdt, dRdt
```

Simulation Function After the ODE system is implemented in the `sir_model` function, we can implement an integrator to get cumulative cases at each step in `simulate_sir`:

LISTING 5.2: Simulate cumulative SIR cases

```

from scipy.integrate import odeint

def simulate_sir(S0, I0, R0, t, beta, gamma, N):
    # Initial condition vector
    y0 = (S0, I0, R0)
    # Integrate ODE
    result = odeint(sir_model, y0, t, args=(beta, gamma, N))
    S, I, R = result.T
    # Cumulative infected = N - susceptibles
    cumulative = N - S
    return cumulative

```

Algorithm 1 Simulate cumulative SIR cases

Require: S_0, I_0, R_0 , time-grid t , parameters β, γ, N

- 1: $y_0 \leftarrow (S_0, I_0, R_0)$
 - 2: $[S(t), I(t), R(t)] \leftarrow \text{odeint}(\text{sir_model}, y_0, t; (\beta, \gamma, N))$
 - 3: **return** $C(t) = N - S(t)$
-

Pseudocode: SIR simulation**Parameter Fitting**

After having a code that simulates numerically the cumulative cases for the SIR model, based on some inputs that are related to the nature of the disease spread, we can try to use our data to find the parameters (inputs) that best describe how an infectious disease is simulated by a SIR model. For this we try to find the parameters that simulate a SIR model as close as possible to the real data, using optimization. We propose an optimization of parameters based on the minimization of a cost function. In this way we obtain the parameters that best fit the SIR model to the real data.

We have data points $\{(t_i, C_{\text{obs}}(t_i))\}_{i=1}^n$. Define the fit function:

$$\text{fit_function}(t; \beta, \gamma) = \text{simulate_sir}(S_0, I_0, R_0, t; \beta, \gamma, N).$$

We find

$$(\beta^*, \gamma^*) = \arg \min_{\beta, \gamma} \sum_{i=1}^n \left(C_{\text{obs}}(t_i) - \text{fit_function}(t_i; \beta, \gamma) \right)^2$$

via non-linear least squares:

LISTING 5.3: Curve fitting with SciPy

```

import numpy as np
from scipy.optimize import curve_fit

# t_data : 1D array of times t_i
# C_obs : 1D array of observed cumulative cases at t_i
# initial guesses for beta, gamma
p0 = [0.3, 0.1]

popt, pcov = curve_fit(

```


TABLE 5.1: Parameters SIR-COVID-19

	beta	gamma	R0
Brazil	3.841	3.603	1.066
Chile	1.706	1.560	1.094
Colombia	3.948	3.728	1.059
Dominican Republic	7.244	7.082	1.023
Germany	0.577	0.452	1.275
Italy	0.843	0.702	1.201
Mexico	7.266	7.119	1.021
Panama	1.474	1.353	1.089
US	1.699	1.513	1.123
Uruguay	0.812	0.697	1.165

```

fit_function ,      # maps (t, beta, gamma) -> C(t)
t_data ,           # observation times
C_obs ,            # observed cumulative cases
p0=p0
)
beta_opt, gamma_opt = popt
# Optional: compute confidence intervals from pcov
perr = np.sqrt(np.diag(pcov))

```

Results

When running the code in Listing 5.3 for each time series we obtain the optimal parameters for all use cases. The curve fitting is done on the training set of data and then a SIR model using the optimal parameters found is calculated for the whole range of data, in this way we obtain the estimated test values.

COVID-19 In Table 5.1 we see the optimal parameters fitted by the process to our data. To get some sense of our data remember that β is the average number of infectious contacts per week. γ is the fraction of the infectious people who either die or recover per week. R_0 is the reproduction number, meaning that a value above 1 is effectively creating more contagions than the population manages to recover, i.e, a pandemic/epidemic. In the table we see how the values are widely ranging in terms of β and γ but the reproduction number stays above and close to one. To make an example we take Brazil. We see a value of beta of 3.8, meaning that each infected individual has contact with 3.84 persons per week on average, creating the opportunity to infect them. For the value of gamma. For a more intuitive interpretation we get the $1/\gamma$, which is the average infectious period: $1/3.6 = 0.27$. Then the infection lasts 0.27 weeks (almost two days) in this country. Finally we see a reproduction number of 1.066, meaning that the transmission was only slightly self-sustaining: every infectious person generated, on average, ≈ 1.066 new cases.

Epidemics As for the COVID-19, we obtain the optimal parameters for the SIR model in each epidemic. The results are displayed in Table 5.2. We see very varying results. Two of the epidemics have a beta just below 1, while other two have a beta very large, above 46. For the lower part, we could interpret that the infected people

TABLE 5.2: Parameters for each Epidemic

	beta	gamma	R0
epidemic			
Dengue	0.898	0.894	1.005
Zika	46.904	46.884	1.000
Varicella	0.997	0.987	1.011
Chikungunya'	46.944	46.916	1.001

have contact with other individuals in average more than once a week, while for the upper part they have seldom contact with other individuals. Recall that Dengue, Zika and Chikungunya are transmittable by mosquito bites. So the beta parameters has not the same meaning. There is an approach on mosquito-borne diseases that stems from the SIR model: The Ross-MacDonald Model. This model originated to study the spread of Malaria and is similar to the SIR/SEIR compartment models. It has more parameters and interactions inside it since there must be an interaction between humans and mosquitoes to transmit the disease. The model includes further parameters to account for this interactions. If we use the SIR model for this type of diseases we must be aware that the beta parameter is including information about all the mosquito factors, collapsed into a single number. This number should not be constant, since some factors in it are variables from the ODE system of the Ross-MacDonald Model. A constant beta will include then a mixture of mosquito density, biting frequency, human and mosquito infectious periods, making it very hard to interpret. As for the gamma. In mosquito-borne diseases it can be interpreted the same, the fraction of people that recover, or $1/\gamma$, the recovery period. Again we see similar numbers, some below 1 and others close to 46. The diseases that are mostly eradicated are the ones with those extreme values of 46. Meaning that if we interpret the gamma, an individual would take $1/46$ weeks to recover, that is $1/46 = 0.21 \approx 3.6$ hours. This is the outcome of evaluating a disease that is mostly eradicated and was very contained. Overall all reproduction numbers are close to 1, which means that the disease maintains at a very low pace, but does not grow exponentially.

5.1.2 SEIR

Model Definition

The SIR is extended by adding an *Exposed* category $E(t)$ for individuals who have been infected but are not yet infectious. The population is

$$S(t) + E(t) + I(t) + R(t) = N, \quad S(0) = S_0, E(0) = E_0, I(0) = I_0, R(0) = R_0.$$

The parameters are:

- β : transmission rate
- σ : incubation rate ($\sigma = 1/\text{mean latent period}$)
- γ : recovery rate ($\gamma = 1/\text{mean infectious period}$)

Differential Equations

$$\dot{S} = -\beta \frac{S I}{N}, \quad (5.4)$$

$$\dot{E} = \beta \frac{S I}{N} - \sigma E, \quad (5.5)$$

$$\dot{I} = \sigma E - \gamma I, \quad (5.6)$$

$$\dot{R} = \gamma I. \quad (5.7)$$

Cumulative Cases

As before, we define the cumulative number infected up to time t by

$$C(t) = N - S(t).$$

Numerical Simulation

The numerical simulation is performed almost the same, as before, updating the code in ?? to include the new exposed portion of the population.

LISTING 5.4: SEIR ODE system

```
def seir_model(y, t, beta, gamma, sigma, N):
    S, E, I, R = y
    dSdt = -beta * S * I / N
    dEdt = beta * S * I / N - sigma * E
    dIdt = sigma * E - gamma * I
    dRdt = gamma * I
    return dSdt, dEdt, dIdt, dRdt
```

LISTING 5.5: Simulate SEIR and return cumulative cases

```
from scipy.integrate import odeint

def simulate_seir(S0, E0, I0, R0, t, beta, gamma, sigma, N):
    y0 = (S0, E0, I0, R0)
    result = odeint(seir_model, y0, t, args=(beta, gamma, sigma, N))
    S, E, I, R = result.T
    cumulative = N - S
    return cumulative, I, R
```

Algorithm 2 Simulate SEIR cumulative cases

Require: S_0, E_0, I_0, R_0 , time grid t , parameters β, γ, σ, N

- 1: $y_0 \leftarrow (S_0, E_0, I_0, R_0)$
 - 2: $[S, E, I, R] \leftarrow \text{odeint}(\text{seir_model}, y_0, t; (\beta, \gamma, \sigma, N))$
 - 3: **return** $(C(t) = N - S(t), I(t), R(t))$
-

Pseudocode

Curve Fitting

The curve fitting is performed in the same way, as the SIR model, just updating the cost function to include the relevant extra parameter.

We observe data $\{(t_i, C_{\text{obs}}(t_i))\}_{i=1}^n$. Define

$$\text{fit_function_seir}(t; \beta, \gamma, \sigma) = \text{simulate_seir}(S_0, E_0, I_0, R_0, t; \beta, \gamma, \sigma, N)_1$$

(the first output is the cumulative curve). Then

$$(\beta^*, \gamma^*, \sigma^*) = \arg \min \sum_{i=1}^n \left[C_{\text{obs}}(t_i) - \text{fit_function_seir}(t_i; \beta, \gamma, \sigma) \right]^2.$$

We write it down in Python in Listing 5.6 to obtain the optimal parameters based on the train data and then fit the SEIR model for the whole date range.

LISTING 5.6: Estimating β, γ, σ via nonlinear least squares

```
import numpy as np
from scipy.optimize import curve_fit

p0 = [0.3, 0.1, 0.2] # init. guess beta, gamma, sigma

f = lambda t, beta, gamma, sigma: \
    simulate_seir(S0, E0, I0, R0, t, beta, gamma, sigma, N)[0]

popt, pcov = curve_fit(f, t_data, C_obs, p0=p0)
beta_opt, gamma_opt, sigma_opt = pop
perr = np.sqrt(np.diag(pcov)) # param. uncertainties
```

Results

COVID-19 As done in the SIR we obtain the optimal parameters for the SEIR model and get the fitted values of the numerical simulation for the whole dates range. The optimal parameters for COVID-19 in all countries are observed in Table 5.3. The first thing we see is the beta: the transmission rate. This is interpreted like in the SIR: on average how many contacts does an infected person have with other people, per week. We see that all values for the beta are close to 1, and in some other cases more close to 2. That differs to some extent to the SIR results where the beta values were in a wider range. Then we see the gamma, which is the fraction of infectious that recovers or dies per week. To make it more interpretable we can take the inverse $1/\gamma$, in most cases is $1/0.9 \approx 1.1$ weeks of infectious period. Then we see the values of sigma. The sigma is interpreted as the incubation rate, and again it becomes more interpretable when we do the inverse $1/\sigma$: Their values range from less than 1 to over 30. This means that individuals incubate the disease from almost a week to even less than a day (when sigma is 30.9). This last value might be an overstretch and be the result of a poor fitting. But in all other cases we have very interpretable and actionable results that can be used in the health sciences for further research or policy recommendations. Lastly we see the reproduction number R_0 , that can be interpreted like in the SIR case: a value larger than 1 means that the epidemic grows, and lower than 1 means that it dies out. In our results we observe values larger than 1, pointing to an epidemic that would continue growing. Recall

TABLE 5.3: Parameters SEIR-COVID-19

	beta	gamma	sigma	R0
Brazil	1.190	0.979	1.646	1.216
Chile	1.626	1.451	3.522	1.121
Colombia	1.624	1.453	4.785	1.117
Dominican Republic	1.146	0.961	0.991	1.193
Germany	1.189	0.930	0.814	1.278
Italy	1.100	0.886	1.497	1.242
Mexico	1.167	0.985	1.181	1.185
Panama	1.338	1.214	3.643	1.103
US	1.585	1.407	30.999	1.127
Uruguay	0.946	0.809	2.630	1.169

TABLE 5.4: Parameters SEIR-Epidemics

	beta	gamma	sigma	R0
epidemic				
Dengue	45.949	45.649	0.394	1.007
Zika	8099.605	8101.606	21.454	1.000
Varicella	46.142	45.629	0.649	1.011
Chikungunya	46.674	46.661	48.247	1.000

that the optimal values are obtained in the training data, that can bias the results of the R_0 . But overall we saw that the pandemic had a behaviour similar to these results.

Epidemics The optimal parameters for the SEIR model fitted to the epidemics are displayed in Table 5.4. Again. The interpretation of the beta is not straightforward for the mosquito-borne diseases (Dengue, Zika, Chikungunya). So we do not bother to look for meaning in something that contains a mixture of factors inside it. Then for the gamma is equivalent to the SIR: the recovery or death rate of humans, while the sigma is the incubation rate of the disease in humans. We observe values of gamma close to 45 and in the case of Zika over 8.000. This would mean $1/8101.6 \approx 0.02$ hours. This is a very fast removal rate, but then the beta is very large as well. In turn we see sigma values from below 1 to above 48. This means that the incubation is in some cases over a week (in the cases where sigma is bellow 1) to under a day (3.5 hours), where sigma is 48. All epidemics result in reproduction numbers R_0 close to 1. Meaning a steady state for the diseases, where it does not die out completely but also does not grow in worrying rates.

5.2 Diffusion Models

As seen in the literature review, the diffusion models arise as an excellent tool to quickly model and understand the spread of infectious diseases. Particularly those arising from influenza viruses. In this section we review our two main tools in the diffusion models kit to understand both the COVID-19 and the different Colombian epidemics.

5.2.1 Bass diffusion Model

On his original work Bass, 1969 proposes a growth model for the adoption or purchase of new products. This was originally tested with consumer goods and has been widely since for many types of products and services. The main assumption is that the timing of a consumers initial purchase has to do with the number of previous buyers, making the adoption of the product dependent on the current state of adoption of the product.

The Bass diffusion model describes the adoption of a new product or innovation as a process driven by two forces:

- **Innovation (external influence):** driven by media or external sources.
- **Imitation (internal influence):** driven by word-of-mouth or peer interaction.

The model is driven by the time behaviour of the cumulative proportion of adopters $F(t)$ at time t , then the model is defined by the differential equation:

$$\frac{dF(t)}{dt} = (p + qF(t)) (1 - F(t))$$

where:

- p : coefficient of innovation (external influence),
- q : coefficient of imitation (internal influence),
- $F(t)$: cumulative adoption function, with $F(0) = 0$ and $F(\infty) = 1$.

Alternatively, in terms of total adopters $y(t) = mF(t)$, where m is the market potential (total number of possible adopters), the model becomes:

$$\frac{dy(t)}{dt} = \left(p + q \frac{y(t)}{m} \right) (m - y(t))$$

Adaptation to Infectious Diseases Modelling

The Bass model can be reinterpreted as a simple epidemiological model:

Bass Term	Epidemic Interpretation
$y(t)$	Cumulative number of infected individuals
m	Total susceptible population
p	Spontaneous infections: (e.g., imported cases from other geographies)
q	Transmission due to contacts with infected individuals

This form is analogous to the SI (Susceptible-Infected) model. Under this interpretation:

$$\frac{dy(t)}{dt} = \left(\beta_0 + \beta_1 \frac{y(t)}{m} \right) (m - y(t))$$

with $\beta_0 = p$ and $\beta_1 = q$ representing spontaneous and contact-based infection rates.

5.2.2 Remarks

- The Bass model captures the S-shaped adoption/infection curve observed in both marketing and epidemics.
- This same fact makes it a rigid model when the diseases spread does not form an S-curve.
- It is useful for early-stage modelling when recovery or immunity is not yet dominant.
- Unlike SIR models, it does not include a removal (recovery or death) term.

5.2.3 Implementation

The model is estimated using the `BM` function from the `DIMORA` package in R, using the train data to obtain the parameters. Then the fitted values for the whole data range are computed to obtain the data relevant for the test period.

5.2.4 Results

The idea in this part is to get some meaning out of the parameters given by the fitting of the Bass model to all our time series. For this we display some of the main results obtained from the fitting of the models.

COVID-19 The parameters found using the Bass Model for the COVID-19 train series are displayed in Table 5.5. Generally we see that $p < q$. In a Bass model for adoption this means that the product is innovative rather than imitative. But according to our re-interpretation of the parameters we can say that the disease is more transmittable than spontaneous. What we can do with the values of q , of transmissibility is compare them across countries. We see that Germany and Uruguay have large values of q , accompanied by low values of p respectively. It could mean that in these countries the disease was more often transmitted from human to human than new cases arising from people coming from other countries. The other thing we can do with the parameters is see the m market potential, or total susceptible population in the country. We see that some countries have larger values than their actual population, like Chile and Italy, while others like Mexico have a very small portion of their actual population estimated as susceptible. These results show the rigidity of the model and failure to obtain real actionable parameters for some countries.

Epidemics Table 5.6 displays the coefficients of the Bass Model fitted for the different epidemics. There we see a similar result to the models fitted for COVID-19. The $p < q$ across all series, indicating a more transmissible disease, rather than a disease that is composed of "imported" cases. The fitting of the Bass Model also results in the deterministic susceptible population m . Recall that Colombia is a country with over 50 million inhabitants. All four models result in series with either around 75 thousand susceptible individuals or 580 hundred thousand. This is only a fraction of the total real susceptible population, indicating that the diseases are contained or limited in growth in some extend.

TABLE 5.5: Parameters Bass-COVID-19

	country	m	p	q
1	Brazil	36,792,198	0.002	0.035
2	Chile	227,242,004	0.000	0.018
3	Colombia	6,270,131	0.001	0.061
4	Dominican Republic	886,405	0.002	0.023
5	Germany	54,780,651	0.000	0.065
6	Italy	677,363,835	0.000	0.031
7	Mexico	9,195,164	0.002	0.027
8	Panama	19,049,034	0.000	0.011
9	US	165,401,601	0.001	0.026
10	Uruguay	13,21,864	0.000	0.049

	Epidemic	m	p	q
1	Dengue	580,093	0.001	0.004
2	Zika	75,143	0.001	0.159
3	Chikungunya	76,884	0.005	0.052
4	Varicella	580,473	0.000	1.210

TABLE 5.6: BM coefficients by epidemic

5.2.5 Guseo Guidolin Model - GMM

This model is an extension of the classical Bass model, which generalizes it to have more flexibility. In their work Guseo and Guidolin, 2009 propose a co-evolutionary diffusion model where the adoption of new products is the result of the interaction between two forces:

- **Communication:** spreading awareness through a communication network.
- **Adoption:** actual uptake or use of the innovation.

In this model, the market potential $m(t)$ is dynamic and evolves over time rather than remaining constant like in the Bass model.

The main adoption equation is:

$$y'(t) = m(t) \left(-r_s \frac{y(t)}{m(t)} + \left(p_s + q_s \frac{y(t)}{m(t)} \right) \left(1 - \frac{y(t)}{m(t)} \right) \right) + y(t) \frac{m'(t)}{m(t)}$$

where:

- $y(t)$: cumulative adoptions,
- p_s, q_s : coefficients for innovation and imitation,
- r_s : decay parameter,
- $m(t)$: evolving market potential.

The market potential $m(t)$ is derived from communication dynamics modelled by a Riccati equation:

$$v'(t) = -(q_c + w_c)v^2(t) + (q_c - p_c - e_c)v(t) + p_c$$

and then:

$$m(t) = K\sqrt{v(t)}$$

where p_c, q_c model communication spread, w_c models negative word-of-mouth, and e_c models decay of information.

5.2.6 Adaptation to Model Infectious Diseases

Although the model was developed for innovation diffusion, it closely aligns with epidemic dynamics and can be reinterpreted as:

Innovation Model	Epidemic Interpretation
Adoptions $y(t)$	Cumulative infections or cases
Communication $v(t)$	Awareness, media attention, or mobility
Market potential $m(t)$	Susceptible population or contact potential
Innovation rate p_s	External seeding, imported cases
Imitation rate q_s	Person-to-person transmission

The availability of having more than one peak and having a dynamic "market potential" or peak of contagions $m(t)$ allows for an interesting take on the disease evolution.

5.2.7 Implementation

The same package as in the Bass model is used for the GGM. In this case the function GGM is used, finding the parameters of the model on the train data sets and forecasting for the whole date range to obtain the test estimations.

Note The market potential function argument `mt` has a default value of 'base'. In general the default value is used for the time series fitted in this work. In some particular cases, the the model fails to converge, then a χ^2 function with ten degrees of freedom is put in place, which results in appropriate convergence.

5.2.8 Results

COVID-19 In Table 5.7 we can see the results of the GGM model fitted to the train data. Overall we see a similar behaviour to the BM. Both p_c and p_s are smaller than their counterparts q_c and q_s . This indicates a larger rate of person-to-person transmission as per our previous interpretation of the GGM parameters for transmittable diseases. Other factor to take note is the initial rapid spread: The q_c parameter indicates that the speed of contagion through internal contact. Countries like Colombia, Germany and the US show larger values compared to other countries, indicating a faster spread of the disease. Then we put special attention in the negative values of q_c , which indicates probably a successful slowing down or containment of the disease with social distancing measures and vaccination rollouts. Brazil, Mexico, US and Uruguay show positive q_c values, indicating a probable slower containment of the disease. For the external influence denoted by p_s , we can conclude that countries like Colombia, which shows a slightly larger value than the other countries had a greater share of imported cases. Lastly we note the dynamic market potential, which in this case is interpreted as the susceptible population indicated by K in the same Table 5.7. The K is erratic in some countries like Chile, Germany, Panama, and Italy, where the fitting process ends up in a larger susceptible population than the

actual whole population of the country. Other countries like Colombia and Mexico result in an under estimation of the susceptible population, but this is more credible since the disease can limit its reach when the herd immunity is achieved.

	country	K	pc	qc	ps	qs
1	Brazil	257,633,612	0.000	0.017	0.004	0.067
2	Chile	698,813,046	0.000	0.043	0.004	-0.063
3	Colombia	7,747,291	0.000	0.088	0.015	-0.004
4	Dominican Republic	811,691	0.000	0.040	0.003	0.144
5	Germany	855,188,187	0.000	0.123	0.017	-0.409
6	Italy	944,023,875	0.000	0.070	0.004	-0.101
7	Mexico	8,443,210	0.000	0.044	0.002	0.137
8	Panama	104,858,050	0.000	0.027	0.002	-0.029
9	US	137,871,314	0.000	0.044	0.000	0.173
10	Uruguay	1,396,221	0.000	0.060	0.000	0.354

TABLE 5.7: GGM coefficients by country

Epidemics Table 5.8 show the results of the coefficients of the fitted GGM model on the epidemics data. As is observed in the results for COVID-19, we see how the p_c and p_s are smaller in magnitude to the q_c and q_s . From the q_c we can interpret that the fastest disease to spread is Chikungunya, while the slowest is Dengue or Zika. Then if we observe the values of q_c we can see if there was some containment in the disease, which was slow in all cases given their positive value. Then for p_s , we see the external influence, i.e., imported cases. Zika and Chikungunya appear to have a larger rate compared to the others of imported cases, but all four have low rates of imported cases and have a the same order of magnitude. To conclude this model we see that the Dynamic susceptible population was in line to the one estimated by the Bass Model for Zika and Chikungunya, but was way higher for Dengue and Varicella. Nevertheless they continue to probably underestimate the whole susceptible population which is 50 times higher.

	Epidemic	K	pc	qc	ps	qs
1	Dengue	2,363,614	0.000	0.002	0.002	0.003
2	Zika	75,301	0.000	0.404	0.003	0.121
3	Chikungunya	77,738	0.008	0.022	0.003	0.114
4	Varicella	1,373,578	0.007	0.013	0.001	0.005

TABLE 5.8: GGM coefficients by epidemic

5.3 Linear Models

5.3.1 ARIMA

The autoregressive integrated moving-average model $ARIMA(p, d, q)$ is used in this work since it is one of the pillars of time series modelling.

Comparison The ARIMA model is purely statistical, relying on past values of the data rather than any epidemiological mechanism. Unlike the SIR/SEIR models, it

does not encode transmission or recovery rates, and unlike diffusion models such as the Bass model or the GGM, it does not assume a specific sigmoidal or power-law growth shape—but it can flexibly fit observed autocorrelation patterns in the data.

Implementation The implemented version of the ARIMA models in this work is the `auto.arima` from the R package `forecast`.

Results The results of these models do not provide very useful insights other than the order AR and MA. So we do not delve deep into the particulars of each estimated model.

5.3.2 Prophet

Prophet is an additive time-series model with three components:

$$y(t) = g(t) + s(t) + h(t) + \varepsilon_t,$$

where

- $g(t)$ is a piecewise-linear (or logistic) trend with automatic change-point detection,
- $s(t)$ is periodic seasonality (e.g. weekly, yearly) modelled via Fourier series,
- $h(t)$ are user-provided holiday effects,
- ε_t is an error term.

Fitting proceeds by maximum-a-posteriori estimation with weakly informative priors on the trend and seasonality parameters.

Comparison Like ARIMA, Prophet is data-driven and does not have any assumptions or restrictions to satisfy in terms of shape. Its explicit change-points make it capable of capturing sudden shifts in epidemic trajectory, unlike a fixed-parameter SIR model. In contrast to diffusion-type models, Prophet can handle multiple overlapping seasonalities and known intervention dates but lacks a mechanistic interpretation in terms of transmission or incubation.

5.3.3 Implementation

The Prophet is implemented using Python, with the original library of Meta: Prophet. The model is trained on the train datasets and the forecast is done for the test datasets.

Results

Even if the Prophet can be similar to the ARIMA in the intuition, the results provided by the models are in turn more interpretable for our particular case. We can see if the series can be estimated using a combination of trends, seasonality and holiday effects.

COVID-19 For this example we show only the results for Colombia to see how the Prophet model can give some intuition of the diseases behaviour. Figure 5.1 shows how the COVID-19 has a clear upward trend. Besides we see a peak in cases on Tuesdays. Which could be explained in some sense by the social exposure people get during the weekends. Then we see that the first quarter of the years have a peak. Indicating, particularly in Colombia that this season (can be a rainy season), can lead to an increase in the contagions of COVID-19.

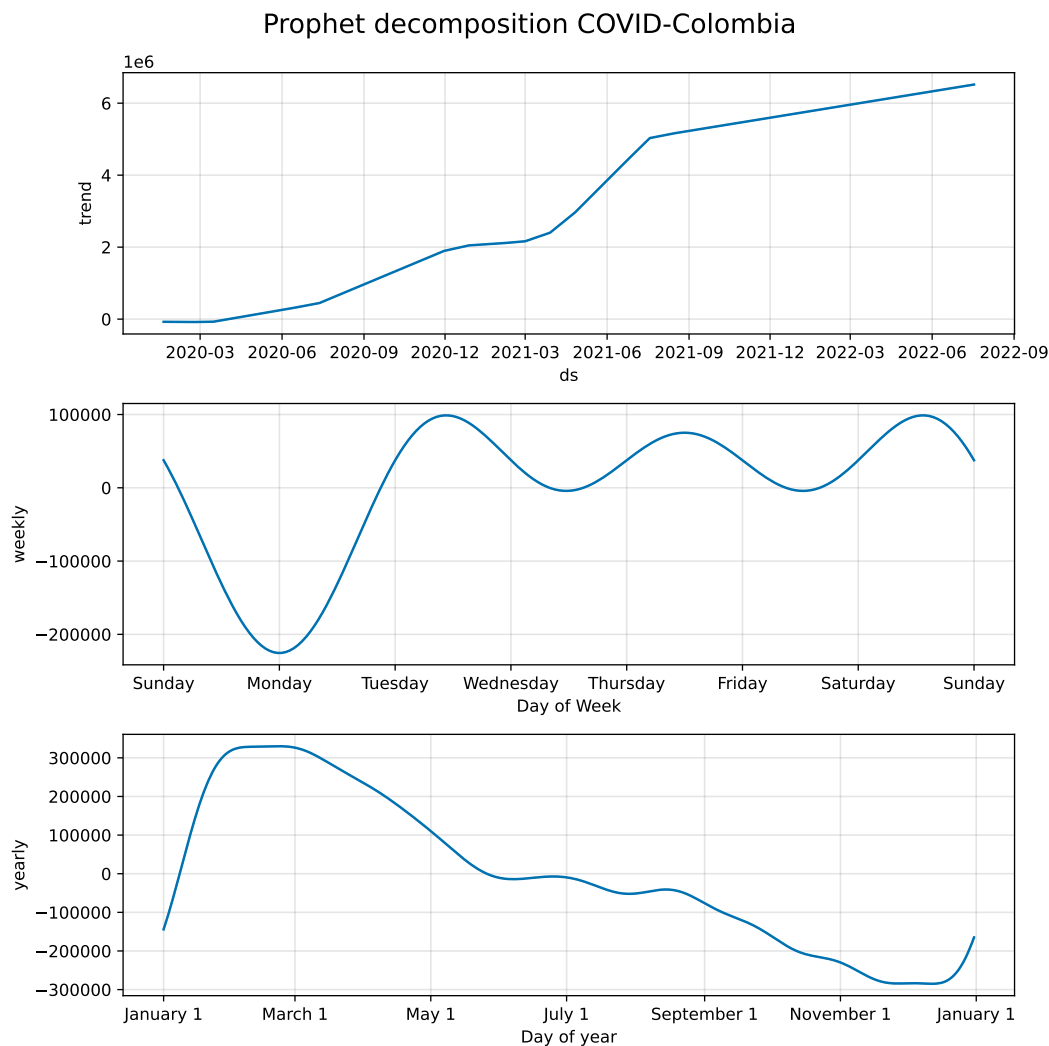


FIGURE 5.1: Prophet Components COVID cumulative cases (Colombia)

Epidemics In this case we show how the components of the fitted Prophet model on the Dengue time series. Figure 5.2 shows that the spread of Dengue has a clear trend. The trend slows down in 2017, to then pick up again later the same year. In the same plot we can also note a similar pattern to the COVID case in Colombia: the weekly peak seasonality is observed during Tuesday, but also Sunday. Finally the yearly seasonality shows that the disease has a peak in the middle of the year (July), showing almost a natural triangle where the lowest points are the beginning and the end of the year. This information is useful as a policy maker, since one can anticipate peaks of contagions during the month of July. With this information the

health centres and institutions can anticipate treatments and promote campaigns of prevention and generate awareness around the general population around this disease.

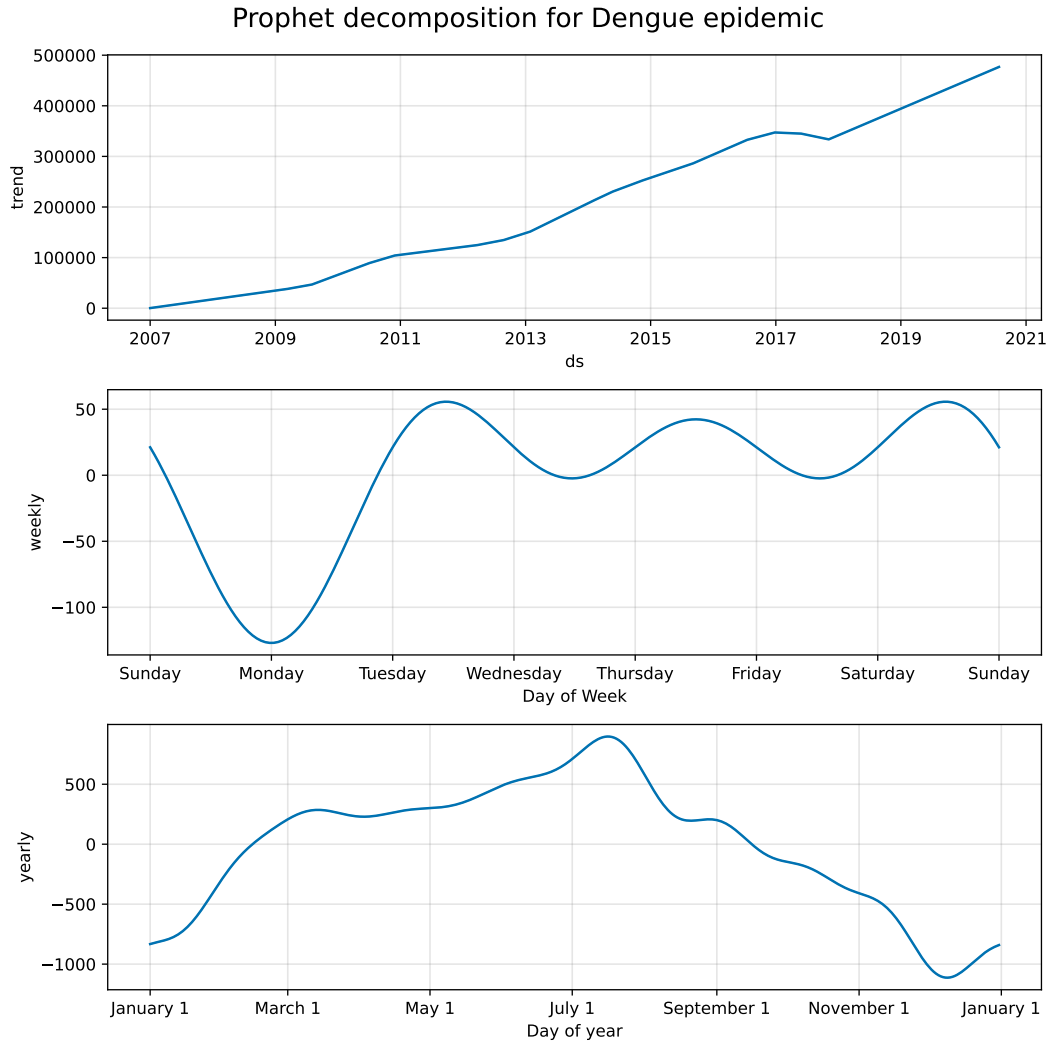


FIGURE 5.2: Prophet Components Dengue cumulative cases

5.4 Neural Networks

Recall the model review done by El Morr et al., 2024 where they note that Deep Learning models are extensively used due to their ability to recognize complex non-linear patterns and inter-temporal dependencies. Inspired by their extensive review, which covers mostly Multilayer Perceptrons and LSTM-RNNs, we chose to propose a slightly different architecture that has proven to be powerful in many other applications: the Gated Recurrent Unit (GRU).

5.4.1 GRU-based Recurrent Neural Network Architecture

The GRU architecture is an extension of standard Recurrent Neural Networks (RNN) that seeks to avoid the vanishing gradient issue. This is achieved by the having two

gates that control the memory of the unit. In this way the network selects which information about the past is useful to be filtered to the future and which can be forgotten.

- **Update Gate:** Determines how much of the previous hidden state to keep and how much new information to add.
- **Reset Gate:** Determines how much of the previous hidden state to forget.

Other authors have chosen a more complicated LSTM architecture that has three gates instead of two. This essentially makes it more complex, i.e., more parameters to learn. In the light of having very few data-points and a trending time series we think a GRU architecture would perform better in terms of having a lower variance, and a higher bias. These two factors are favourable to our learning goal of forecasting the long term spread of the diseases, which seen in the time series plots is not too volatile. To confirm the theoretical advantages of GRU over LSTM in our data, LSTM and GRU experiments were done using similar settings, to compare the pure inner architecture of the gates. The experiments confirmed that GRU was a better option in this particular case.

We denote by $\{x_t\}_{t=1}^T$ the time series of cumulative cases of a given disease. Before feeding into the network, Min-Max scaling is applied to the time series to address the vanishing-exploding gradient issues,

$$\tilde{x}_t = \frac{x_t - x_{\min}}{x_{\max} - x_{\min}}, \quad \tilde{x}_t \in [0, 1].$$

The next step is to form input-output pairs. For this we use a fixed look-back window of length $L = 5$:

$$\mathbf{x}^{(i)} = [\tilde{x}_i, \tilde{x}_{i+1}, \dots, \tilde{x}_{i+L-1}]^T \in \mathbb{R}^L, \quad y^{(i)} = \tilde{x}_{i+L},$$

for $i = 1, \dots, T - L$. We split at $i = T_{\text{train}} = \lfloor 0.8 T \rfloor$ into training and test sets.

GRU cell. At each time step t within a sequence, the GRU maintains a hidden state $h_t \in \mathbb{R}^d$ (here $d = 50$) updated by

$$\begin{aligned} z_t &= \sigma(W_z x_t + U_z h_{t-1} + b_z), \\ r_t &= \sigma(W_r x_t + U_r h_{t-1} + b_r), \\ \tilde{h}_t &= \tanh(W_h x_t + U_h (r_t \odot h_{t-1}) + b_h), \\ h_t &= (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t, \end{aligned}$$

where - $x_t \in \mathbb{R}$ is the input at step t , - $z_t, r_t \in \mathbb{R}^d$ are the update and reset gates, - W_*, U_* and b_* are learnable weight matrices and bias vectors, - σ is the logistic sigmoid, and $\odot$ denotes element-wise multiplication.

After processing the sequence of length L , the final hidden state h_L is passed through a dense output layer,

$$\hat{y} = V h_L + c,$$

with $V \in \mathbb{R}^{1 \times d}$ and $c \in \mathbb{R}$. The prediction $\hat{y}$ is our estimate of $\tilde{x}_{i+L}$.

Training objective

As in several machine learning problems, this one can be approached via minimizing the mean-squared error on the training set,

$$\mathcal{L}(\Theta) = \frac{1}{N_{\text{train}}} \sum_{i=1}^{N_{\text{train}}} (y^{(i)} - \hat{y}^{(i)})^2,$$

where Θ collects all GRU and dense layer parameters. Optimization on the learning parameters is performed via the Adam algorithm in an adaptive variation of traditional gradient descent.

Prediction After learning the weights and biases on the training data, we predict the full time series $\{\tilde{x}_t\}_{t=1}^T$ in sliding-window fashion to obtain predictions $\{\hat{y}_t\}$. This prediction is scaled, so we proceed to revert the Min-Max scaling and so the forecasts are done on the original scale.

Regularization For the Epidemics time series regularization was implemented to avoid over-fitting of the models. Particularly, L2 regularization was applied on the final dense layer. Achieving little improvement on the test set. Still, over-fitting was addressed to be able to conclude about the performance of the GRU architecture without having that issue at hand.

Advantages of the GRU Architecture

The GRU-based approach could outperform both classical mechanistic models (SIR/-SEIR) and purely statistical or diffusion models (ARIMA, Bass, GGM) for several reasons:

- **Universal Approximation theorem** Using multiple affine transformations and non-linear activations in the GRU cell and dense layer should give the neural network with the ability to approximate any given smooth function using the input data. Contrary to the SIR/SEIR, this approach does not assume a fixed parametric form derived from a particular ODE, or impose linear functions like in the ARIMA models.
- **Nonlinearity in Dynamics.** Each GRU update

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tanh(W_h x_t + U_h(r_t \odot h_{t-1}) + b_h)$$

is highly non-linear in the past state h_{t-1} and current input x_t . These non-linear interactions allow the network to capture complex feedback loops that are not easy to capture using more simple approaches like SIR/SEIR and ARIMA.

- **Adaptive Memory via Gates.** The update gate z_t and reset gate r_t dynamically control how much past information is retained or forgotten:

$$z_t = \sigma(W_z x_t + U_z h_{t-1} + b_z), \quad r_t = \sigma(W_r x_t + U_r h_{t-1} + b_r).$$

In theory, this mechanism allows the model to preserve long-term trends when needed, for example, a pandemic wave. But it also allows the model to reset and focus on recent shocks like sudden outbreaks. These gates result in a balance that fixed-lag or fixed-memory models cannot attain.

- **Data-Driven Adaptation.** During training, the Adam optimizer adjusts all weights $\{W_*, U_*, b_*\}$ to minimize observed prediction error. This allows the model to recognize patterns unique to each dataset and automatically discover input representations and time-scales without manual tuning of compartments or differencing orders.
- **Robustness to Noise and Non-stationarity.** The gating structure and the smoothing effect of recurrent hidden states help filter out high-frequency noise in daily case counts, while the network's capacity to re-weight past information makes it more resilient to regime changes (e.g. policy shifts) than models with fixed coefficients.

Results

COVID The training results of one of the series can be seen in Figure 5.3. Here we can note how the RNN achieves a steady state after approximately 30 epochs. After this the validation loss seems to stagnate and the training loss continues to decrease in a slow pace, which is almost not relevant. This small gap between training and validation loss lets us know that the model is not over-fitting on the training data. Based on these results we remain optimistic on the power of generalization of complex time series patterns of the GRU architecture.

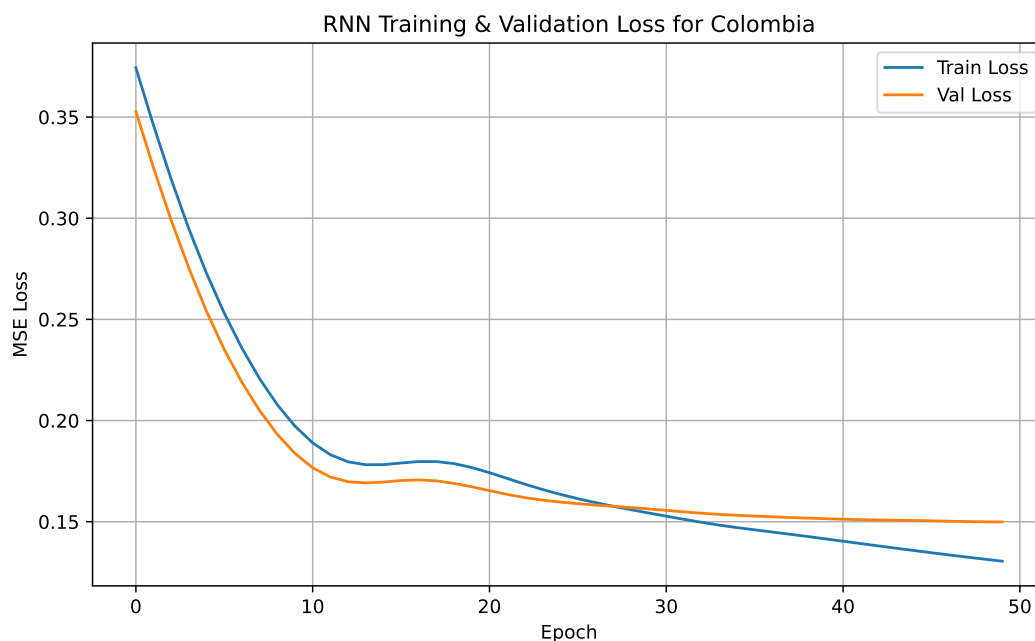


FIGURE 5.3: GRU Training and Validation Loss - COVID Colombia

Epidemics Figure 5.4 shows the training and validation loss of the Dengue time series. In this case we see how after around five epochs both curves are almost zero. This means that the model follows the data very closely. Even using regularization techniques, the model achieves a very low error, which is a good indicator of performance. But when it comes to forecasting ability we will see in the **Results** that the GRU architecture for the Epidemic time series is not the best model architecture.

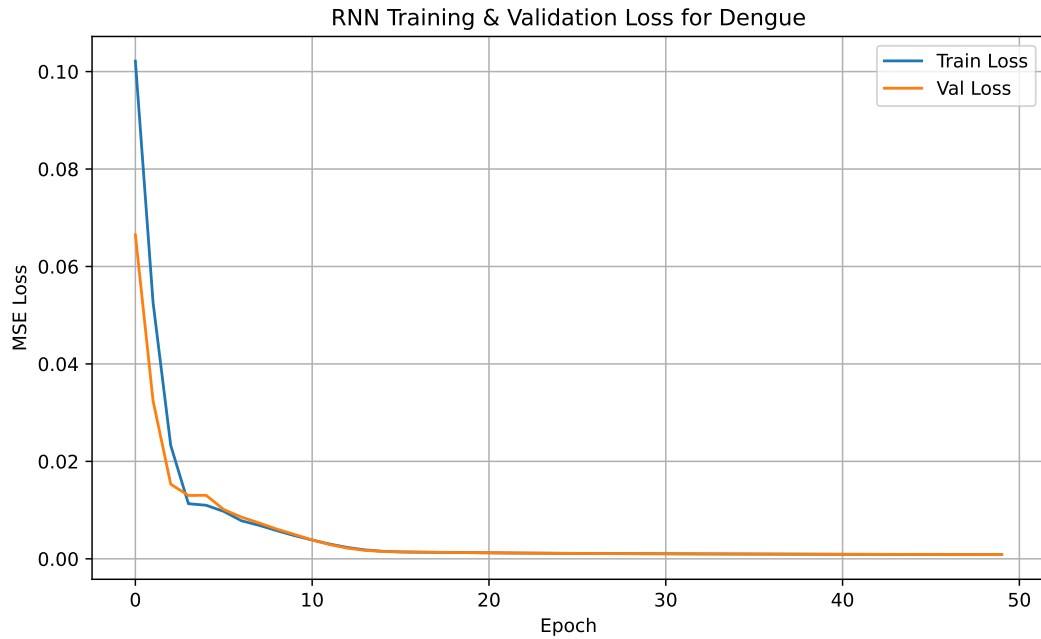


FIGURE 5.4: GRU Training and Validation Loss - Dengue

5.5 Results

Given the various settings of models and the fact that they are trained on 10 different countries, we summarize the results in this section. To grasp the performance of each separate model we present a figure displaying the fitted values of each model versus the actual series for both exemplary series: COVID-Colombia and Dengue. The goal is to be able to see the bias and variance introduced by each model fitted and how well do they forecast the data.

5.5.1 COVID

In Figure 5.5 the results of the COVID-Colombia time series. Not all country-model pairs behave in the same way, but this initial approach lets us know some general characteristics of the models. Something that we can conclude from this plot is the performance of the mechanistic and diffusion models against the other models. The mechanistic and diffusion models, as expected by construction, fit an S-shaped curve to describe the trajectory of the cumulative cases. In this particular case we can see how they cannot capture waves or increased spikes in the training phase. Whereas all the other models, both linear and non-linear "learn" the peaks and fit them in a more suitable way. This observation is key because for example we see how the Prophet model captures a seasonality factor which is estimated in the test set, but actually does not occur. Overall for the particular Colombian case, we can visibly see how diffusion models are the best tool to forecast the test set. Even if they do not capture the waves, they can capture the overall trend of the disease spread. In contrast the mechanistic models are the most rigid ones and the fitting of their parameters in the training phase either largely overestimates or underestimates the forecasted period.

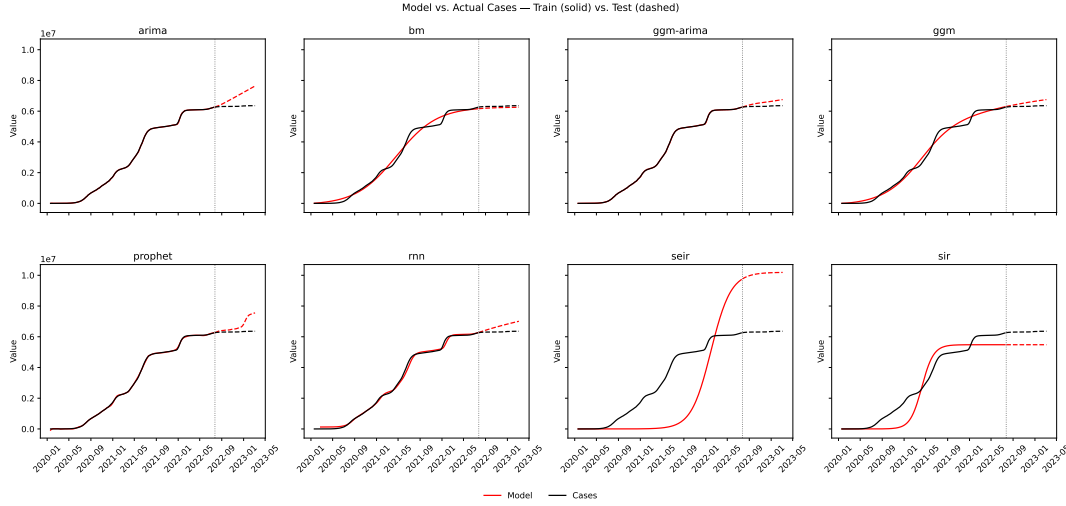


FIGURE 5.5: Model fitted values vs Actual COVID cumulative cases (Colombia)

Evaluation Metrics

To simplify the evaluation of all combinations of models and countries we propose a simple strategy to evaluate the model performance. For each model-country pair, we compute the Root Mean Square error (RMSE), and use it as input for a ranking methodology. The RMSE is a widely used metric to evaluate the accuracy of regression models. It measures the square root of the average squared differences between the predicted values and the actual observed values.

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i^{\text{true}} - y_i^{\text{pred}})^2}$$

- y_i^{true} : the actual (true) value for observation i
- y_i^{pred} : the predicted value for observation i
- n : the total number of observations

Both values y_i^{true} and y_i^{pred} are sliced from only the test part of the data to evaluate the performance on unseen data. A lower RMSE indicates a better fit of the model to the data.

We want to know which model is overall, the one that does a better job in the different countries. A strategy would be averaging the RMSE over all countries to see how each separate model behaves on average. The countries selected are very different in terms of population, hence the number of cumulative COVID cases is also different. This results in RMSE values between countries that are not directly comparable. Additionally, if we were to average the RMSE we would have to do a weighted average and the weighting factor is not obtainable in a straightforward way. For instance, the SEIR can largely overestimate the cases for one country with high number of cases giving a large relative RMSE but be very close in another, so a weight based on the population would be prone to over or underweight a selected country-model pair. To address this unbalance we propose a simpler methodology:

For each country rank the models based on their RMSE. Then based on those rankings deciding which models are the best performing ones overall.

Figure 5.6 displays how each model performs based on the rankings. A ranking of 1 means that this is the model with lowest RMSE for that country. Further simplification can be done by directly averaging the rankings in a "mean" column and rank the metrics matrix by this new mean column. Overall, we see that a simple ARIMA model is very powerful. It is the best model for four countries and the second best for two. In turn, it is also a bad model for the rest of the countries. Making it either very good or very inaccurate on the test set. The ARIMA performance does not seem to depend on regional characteristics, since it can perform well or poorly across different regions. For example, it is a powerful model in the US, Italy and the Dominican Republic; Countries with different weathers and different monetary resources so people could argue that testing frequency and climate patterns is altering the data. In contrast, the same model results underperforming in countries with the same diverse characteristics: Colombia, Germany, and Mexico. The RNN is tied with the ARIMA in first place showing similar behaviour. It can either be one of the best performing models or an average performing model (See Germany and Uruguay). Then we see that the diffusion models are on the same block: BM, GGM and GGM+refinement find themselves in the middle of the table. For some countries the simplicity of the BM achieves to capture the overall trend of the pandemic, while for others it is the GGM or the refined version of the GGM the one that does a great job. Finally, we see that the mechanistic models implemented in this work are not as powerful as they are believed to be. They either overestimate or underestimate the trajectory as we saw in Figure 5.5, which makes them a volatile tool to forecast the spread of any infectious diseases.

5.5.2 Epidemics

In this case we plot how the models fit the Dengue series, which has extensive observations and variability throughout the years. These characteristics make it ideal for us to check if the models learn any trends, seasonality or other patterns arising from a time series model. Individual model performances are displayed in Figure 5.7. In this figure we can see some divergence when compared to the COVID-Colombia models. First, we see that mechanistic models prove to be more powerful than before. This can be for a number of reasons. One could be that there is a larger amount of data to train the model. Other could be attributable to the fact that the disease spread is not yet shaped like an S-curve so simple models can fit a flat-ish curve which is more probable to be accurate than an S shape curve. Other interesting observation is that the RNN, which proved to be very capable in the COVID part, is more erratic in the test set of Dengue. It overestimates the increase in cases, which can result in a false alarm or wrong decision making by the model users. For the diffusion models, we see that they capture the overall trend but fail to predict on the test set the large spike. The BM is more rigid than the GGM, which is clear in the sense that the BM underestimates the cases in the test in a larger scale than the GGM. Finally, we see that the Prophet model captures the large spike by learning some strange pattern since about 2017.

Rank of RMSE for each model and country

Country	arima	4	1	6	2	6	1	7	2	1	1	3.1
	rnn	2	4	4	1	5	3	3	1	3	5	3.1
	bm	1	5	1	4	7	8	2	6	5	4	4.3
	ggm-arima	5	7	3	5	1	4	4	3	4	7	4.3
	ggm	6	8	2	3	2	7	1	7	2	8	4.6
	prophet	3	2	5	6	8	5	5	4	6	6	5
	seir	8	3	8	8	3	2	8	5	7	2	5.4
	sir	7	6	7	7	4	6	6	8	8	3	6.2
		Brazil	Chile	Colombia	Dominican Republic	Germany	Italy	Mexico	Panama	US	Uruguay	mean
		Model										

FIGURE 5.6: Ranking of RMSE by Country-Model - COVID

Evaluation Metrics

As before, we compute the RMSE of the predicted values against the actual values on the test set for all epidemics at hand. Figure 5.8 shows the ranking of the models in each epidemic. The ARIMA is again the best ranked model. In this case we can explain this by the behaviour of both Chikungunya and Zika. They have a very flat behaviour in the test set, which the ARIMA learns since the train set and just keeps propagating the forecast as an almost flat line. Then we see the GGM and its refined version as a very capable model to forecast the long term phase of the epidemic. Particularly, we see that in Dengue, this hybrid version is better than both stand-alone models. Here, we clearly identify the advantages including a diffusion model in our architecture. They efficiently capture the trend of the series. While the ARIMA part of the hybrid model, captures seasonal and instantaneous patterns. The combination of both proves to be very capable model. Overall, the mechanistic models are not very appealing. Even for Chikungunya and Zika, which are a clear S-shaped curve almost fully formed, the mechanistic models underperform other simple architectures. These models prove to be very rigid even for diseases that are in their last phase of spread.

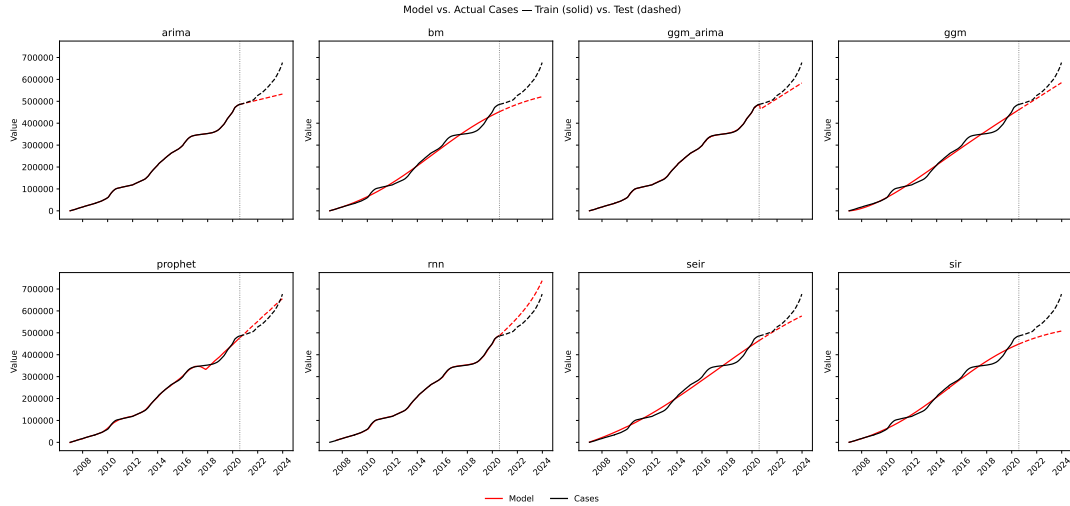


FIGURE 5.7: Model fitted values vs Actual Dengue cases

5.6 Conclusions

In both COVID and Epidemics sections we see how we can leverage the simple ARIMA model to predict the future behaviour of infectious diseases. We also see how diffusion models manage to capture the overall trend of the disease, which is very helpful because we can obtain the fitted coefficients and use them as input for forecasting scenarios in the outbreak of new diseases, while ARIMA models are data dependent so we have to have some data to be able to really capture the dynamic of the particular disease prior to forecasting with this model.

For long term forecasting of infectious disease spread, we recommend using diffusion models and calibrate their parameters based on prior findings. Then based on those parameters construct probable scenarios that can serve as a tool for policy makers and health experts to work on the long term control of the disease while understanding the real trajectory of the contagions. For a complementary strategy, we recommend using ARIMA models to capture the data dynamics as the disease spreads. This will help at least forecast some variability outside the general trend that can be of much use when preparing for a wave of cases or contagions.

Even if the models presented in this chapter show robust long-term forecast in the cumulative trends, they are short-sighted for the more near term behaviour of the diseases. On top of that, they do not provide any explanation of the external factors that could lead or anticipate a spike in contagions. To further understand the short term behaviour and drivers of infectious diseases we shift from univariate models to multivariate ones. External factors like government policy, public behaviour and meteorological drivers are included. We do so in the next chapter 6.

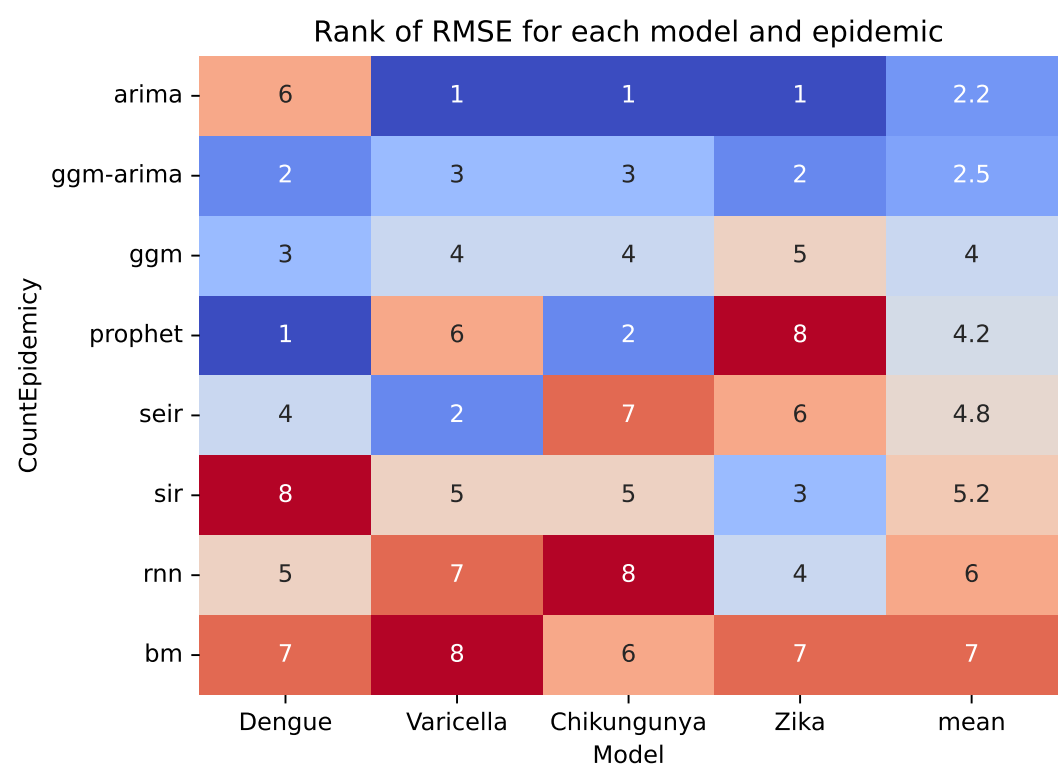


FIGURE 5.8: Ranking of RMSE by Epidemic-Model

Chapter 6

Multivariate Models

In this chapter we focus our attention on the models used for the multivariate regression framework. In this case we leverage external information for the regression problem of predicting new cases. Particularly, we are predicting the cases in $t + 1$ (next week), using a set of features at time t (the current week).

6.1 Data

Recall that in Chapter 4, the selection of features was performed to determine the set of variables included in all models.

The variables are used for each family of diseases:

- **COVID:**
 - `y_t`
 - `o3`
 - `no2`
 - `google_symptom`
 - `pop_vaccinated`
 - `StringencyIndex_Average`
- **Epidemics:**
 - `y_t`
 - `date_num`
 - `pres`
 - `google_symptom`
 - `pm25`

6.2 Linear Models

In this section, we present four variations of the linear regression model. Penalized and non-penalized versions of the basic Ordinary Least Squares model are used. They are all based on the regression task of predicting y_{t+1} (next-week cases) using the feature vector $X_t = (X_{1t}, X_{2t}, \dots, X_{nt})^\top$.

Ordinary Least Squares (OLS) We estimate coefficients $\beta \in \mathbb{R}^n$ by

$$\hat{\beta}^{\text{OLS}} = \arg \min_{\beta} \|y - X\beta\|_2^2,$$

where y stacks all observations of y_{t+1} and X stacks the corresponding rows X_t . Predictions are then

$$\hat{y}_{t+1} = X_t^\top \hat{\beta}^{\text{OLS}}.$$

Lasso Adds an ℓ_1 penalty to induce sparsity:

$$\hat{\beta}^{\text{Lasso}} = \arg \min_{\beta} \|y - X\beta\|_2^2 + \lambda \|\beta\|_1.$$

Ridge Penalizes the ℓ_2 -norm:

$$\hat{\beta}^{\text{Ridge}} = \arg \min_{\beta} \|y - X\beta\|_2^2 + \lambda \|\beta\|_2^2.$$

Elastic Net Combines both penalties:

$$\hat{\beta}^{\text{EN}} = \arg \min_{\beta} \|y - X\beta\|_2^2 + \lambda \left(\alpha \|\beta\|_1 + \frac{1-\alpha}{2} \|\beta\|_2^2 \right).$$

Lasso-Ridge-ElasticNet Implementation All three penalization models are developed following a similar procedure. For each country or epidemic dataset, the data is chronologically split into training and testing subsets, following an 80-20 ratio. Within the training subset, cross-validation is performed to identify the optimal regularization parameter λ . Specifically, the value of λ that minimizes the mean squared error (MSE) on the validation folds is selected. Once this optimal λ is determined, the final model is fitted using the entire training set, and predictions are subsequently generated using the test set features.

Coefficients

Here we present some examples of the coefficients estimated on the training time series COVID-Colombia and Dengue.

COVID-19 First let us check the statistical significance of the variables fitted using the `LinearRegression`. Table 6.1 shows the p-values of the beta coefficients. There we see that only `no2` shows a larger value than 0.05 so we refrain from interpreting this coefficient. Then in Table 6.2 we can see the coefficients of the same selected predictors and we can clearly see that all models arrive at very similar results. All models share the order of magnitude and direction of the coefficients, leaving only marginal differences between coefficients of one model and another. In order to interpret the results we can simply go with the `LinearRegression` coefficients, which are very similar to the rest.

- **intercept.** In this case we can say that when every predictor is zero, the new cases would account to a negative number. Given the nature of our predictors this scenario is not really plausible, so this is just an anchoring intercept in the plane of this equation.

TABLE 6.1: OLS parameter estimates COVID Colombia

	p_val
const	0.0000
new_cases	0.0000
google_symptom	0.0000
o3	0.0021
pop_vaccinated	0.0000
no2	0.3208
StringencyIndex_Average	0.0313
date_numeric	0.0000

- `new_cases` can be interpreted as follows: for every new case in last week, we can expect this week to have an additional 0.73 cases. This is practically the autocorrelation component of the series.
- `google_symptom` shows us how an increase in 1 unit in the index (recall it is in a 0-100 scale) will imply an increase in next week cases of COVID of over two thousand.
- `o3` has a negative coefficient in the order of magnitude of thousands, implying that an additional unit (micrograms per cubic meter) will be translated in a decrease of next week cases of over two thousand. This result is odd and can be a proxy for some other factors that are left behind in the variable elimination phase. More ozone should not directly impact COVID new cases, but it can be correlated to meteorological factors and seasonal patterns in the time series which leads us to have this negative relationship.
- `pop_vaccinated` shows that an increase in 1 unit in the index (0 is no one is vaccinated and 100 means everyone is vaccinated) translates in a decrease in next week cases of over 700. Or if we re-scale to percentage of population, an increase of 1% of the population vaccinated would lead to a decrease in over 70.000 cases.
- `StringencyIndex_Average` can be interpreted in the same way as before. The original scale is an index from 0 to 100. So an increase in 1 unit in the index (more stringency in the restrictions or social distancing) leads to an increase in about 300 cases in next week cases. This value is also peculiar, since social distancing should reduce the contagions. But we can conclude based on the temporality of the restrictions that: At the beginning there were more restrictive measures, which were slowly reduced, then in 2022 we saw a peak in new cases which coincides with lower restrictions. This positive relationship between the two variables hides a temporal alignment between them, we do not consider there is causality of this predictor on the observed direction.
- `date_numeric` shows a coefficient close to zero thanks to the unit of measure, but the p_value is positive, so we can conclude that there is a time component in the new cases for next week.

Epidemics Again, we check the statistical significance of the variables fitted using the `LinearRegression`. Table 6.3 shows the p-values of the beta coefficients. There

	intercept	new_cases	google_symptom	o3	pop_vaccinated	no2	StringencyIndex_Average	date_numeric
LinearRegression	-1964523.46	0.73	2681.98	-2174.14	-749.12	844.53	334.55	0.00
Lasso	-1640896.66	0.75	2558.79	-2155.94	-655.30	799.53	252.42	0.00
Ridge	-1747307.63	0.74	2618.46	-2254.69	-677.82	888.53	300.95	0.00
ElasticNet	-1574721.69	0.74	2566.79	-2324.90	-621.44	908.53	274.09	0.00

TABLE 6.2: Model coefficients (original scale) COVID-Colombia

TABLE 6.3: OLS parameter estimates Dengue

	p_val
const	0.0737
dengue	0.0000
pres	0.1403
google_symptom	0.0616
date_num	0.4757
pm25	0.5747

we see that only dengue and google_symptom shows p-values lower than 0.1, which is an α confidence also acceptable to be able to get some sense out of our data. We proceed to then only draw conclusions for statistical inference using these predictors, but this might fall short of a deep analysis as we would like to have for policy recommendation. The coefficients are analysed using the Table 6.4, where we can see the coefficients of the same selected predictors, and yet again, all models arrive at similar results. In order to interpret the results we can simply go with the LinearRegression coefficients, which are very similar to the rest.

- **intercept.** This case follows the same logic as in the COVID case, a negative coefficients that anchors the other variables.
- **dengue** Shows a coefficient of around 0.9. This means that for every dengue case this week we can expect almost a new case the next one, proving to be a very autocorrelated series. This can be more interpretable if we say that for every 10 cases of Dengue this week we will observe 9 the next one.
- **pres** Avoid interpreting this due to lack of statistical evidence of significance.
- **google_symptom** This coefficient tells us that for increment in 1 unit in the Google Trends index regarding symptoms we can expect a decrease in Dengue cases of up to 2 cases next week, which might be a bit strange and counter intuitive, when comparing to the results of COVID.
- **date_num** Avoid interpreting this due to lack of statistical evidence of significance.
- **pm25** Avoid interpreting this due to lack of statistical evidence of significance.

The aggregating of the series in a national level can lead to noise data, which does not directly imply that our estimates show a lack of causality between the predictors and the target variable. It would be smart to predict at a more granular level, to see the effect of our pre-selected covariates in the target, as will be performed in the next chapter.

	intercept	dengue	pres	google_symptom	date_num	pm25
LinearRegression	-6508.07	0.93	5.95	-2.31	0.00	0.37
Lasso	-5168.90	0.93	4.52	-1.68	0.00	0.03
Ridge	-6243.59	0.91	5.24	-2.29	0.00	0.50
ElasticNet	-6165.51	0.89	4.89	-2.28	0.00	0.59

TABLE 6.4: Model coefficients (original scale) Dengue

6.3 Non-Linear Models

Below, we give the core prediction rule for each method:

Random Forest An ensemble of M decision trees $\{T_m\}$ gives

$$\hat{y}_{t+1} = \frac{1}{M} \sum_{m=1}^M T_m(X_t).$$

Gradient Boosting Builds an additive model of trees $\{h_m\}$:

$$\hat{y}_{t+1}^{(0)} = \bar{y}, \quad \hat{y}_{t+1}^{(m)} = \hat{y}_{t+1}^{(m-1)} + \eta h_m(X_t),$$

where each h_m is fit to the negative gradient of the loss at step m .

Support Vector Regression (SVR) Finds w, b that minimize

$$\frac{1}{2} \|w\|_2^2 + C \sum_i (\xi_i + \xi_i^*)$$

subject to the ε -insensitive constraints

$$\begin{cases} y_i - (w^\top \phi(X_i) + b) \leq \varepsilon + \xi_i, \\ (w^\top \phi(X_i) + b) - y_i \leq \varepsilon + \xi_i^*, \\ \xi_i, \xi_i^* \geq 0, \end{cases}$$

and then $\hat{y}_{t+1} = w^\top \phi(X_t) + b$.

Feature Importance

Here we present the feature Importance of the Ensemble Methods (Random Forest and Gradient Boosting):

COVID-19 Figure 6.1 shows how the ensemble methods overweight the importance of `new_cases` in the past week to predict the ones in the current week. After that `google_symptom` and `StringencyIndex_Average` seem to have the largest importances in Gradient Boosting and Random Forest models respectively. These two variables proved to have enough statistical evidence to not reject the null hypothesis of their associated β coefficients to be equal to 0. So these two separate results, arising from different types of models align in the end.

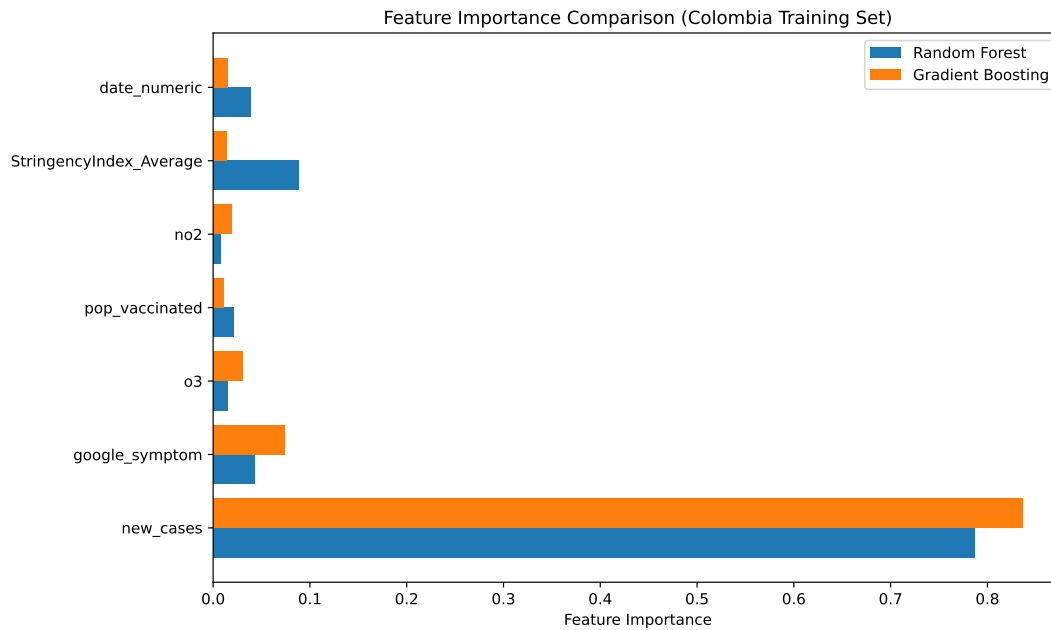


FIGURE 6.1: Feature Importance COVID-Colombia

Epidemics Figure 6.2 shows a similar story to the one encountered in the COVID exercise. We see how previous cases are the most importance variable for the ensemble models. In this case the date_num appears to have the second most important weight for the ensemble models. This results diverge from the COVID ones. Here we see how a feature that showed no evidence of statistical significance in the linear models, appears to be a strong predictor in the Non-linear models. One lesson is clear from both plots (Covid and Dengue): The previous cases are a strong predictor of future cases, showing how this time series has strong autocorrelation, confirming the performance of the models selected in Chapter 5.

6.3.1 Neural Networks

Recall again the work done by El Morr et al., 2024 were they point the extensive implementation of Neural Network architectures for the regression task at hand. For multivariate time series they cover extensively the Multilayer Perceptron, also called feedforward neural network. Neural networks have been used for several years for multiple tasks. Their popularity lies in their ability to capture non-linear relationships between the features and the target. This is due to a combination of architectural components that include:

- **Multiple layers:** Each composed of neurons with trainable weights and biases.
- **Non-linear activation functions** between hidden layers, which endow the network with the ability to approximate complex, non-linear mappings.
- **Gradient Descent-based optimization** of weights and biases, which minimizes a cost function by iteratively updating parameters in the direction of steepest descent.

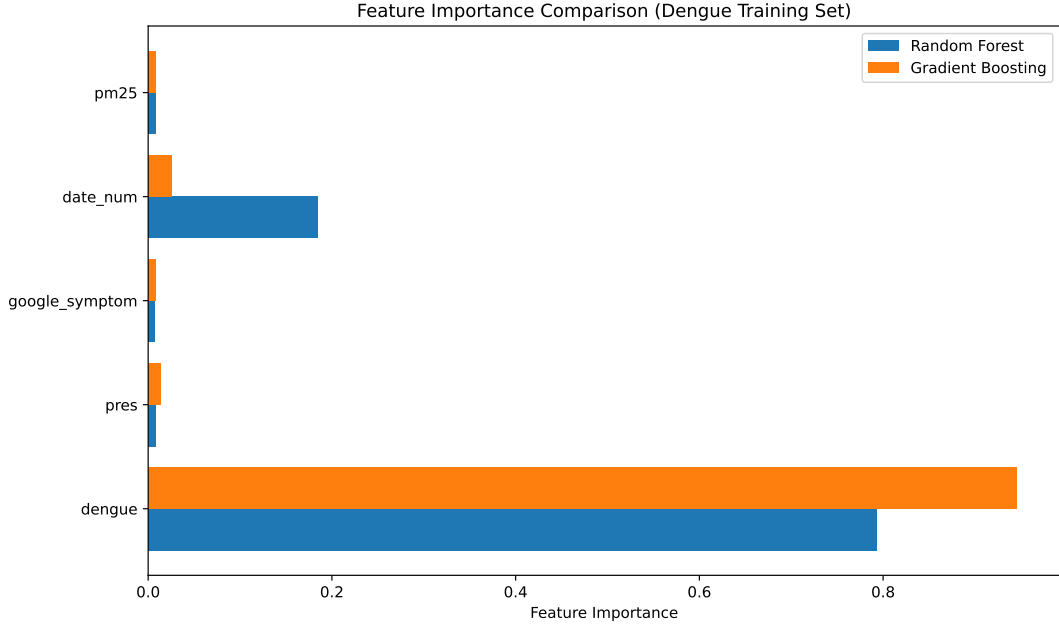


FIGURE 6.2: Feature Importance Dengue

A feedforward neural network computes predictions by passing the input $\mathbf{x} \in \mathbb{R}^d$ through L hidden layers. Each hidden layer performs a transformation of the form:

$$\mathbf{h}^{(l)} = \sigma^{(l)} \left(\mathbf{W}^{(l)} \mathbf{h}^{(l-1)} + \mathbf{b}^{(l)} \right), \quad l = 1, \dots, L,$$

where:

- $\mathbf{h}^{(0)} = \mathbf{x}$ is the input,
- $\mathbf{W}^{(l)}$ and $\mathbf{b}^{(l)}$ are the weight matrix and bias vector for layer l ,
- $\sigma^{(l)}(\cdot)$ is a non-linear activation function (e.g., ReLU or tanh).

The output layer typically computes a final transformation (possibly without an activation in regression settings) to obtain the prediction $\hat{y}$.

The model is trained by minimizing a loss function. In this case, the mean squared error (MSE) is used:

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2,$$

where N is the number of samples.

The optimization is carried out using the **Adam optimizer**, an adaptive learning rate variant of stochastic gradient descent, which combines momentum and adaptive scaling of gradients.

In our implementation, we use a simple feedforward Deep Neural Network (DNN) composed of 5 hidden layers, starting with 128 neurons and gradually reducing to 16 in the last layer. This architecture may be considered excessive given the relatively low dimensionality of the regression task, but remains computationally efficient and can be trained quickly on any modern personal computer.

The dataset is randomly split into 80% training and 20% validation, rather than chronologically ordered. As is common practice, the features are standardized using z-scaling to ensure stable gradient descent and avoid issues like exploding or vanishing gradients.

6.3.2 Results

To see how the FeedForward Network behaves, we can analyze, as we did in chapter 5, how the training curves evolve across the epochs. This will tell us if our model achieves a balanced generalization of the data in the learning phase.

COVID-19 In Figure 6.3 we can see how the FNN architecture implemented achieves to reduce the error in very few epochs. Then, both training and validation curves remain almost zero, with almost no gap between them, which indicates no over-fitting of the training data.

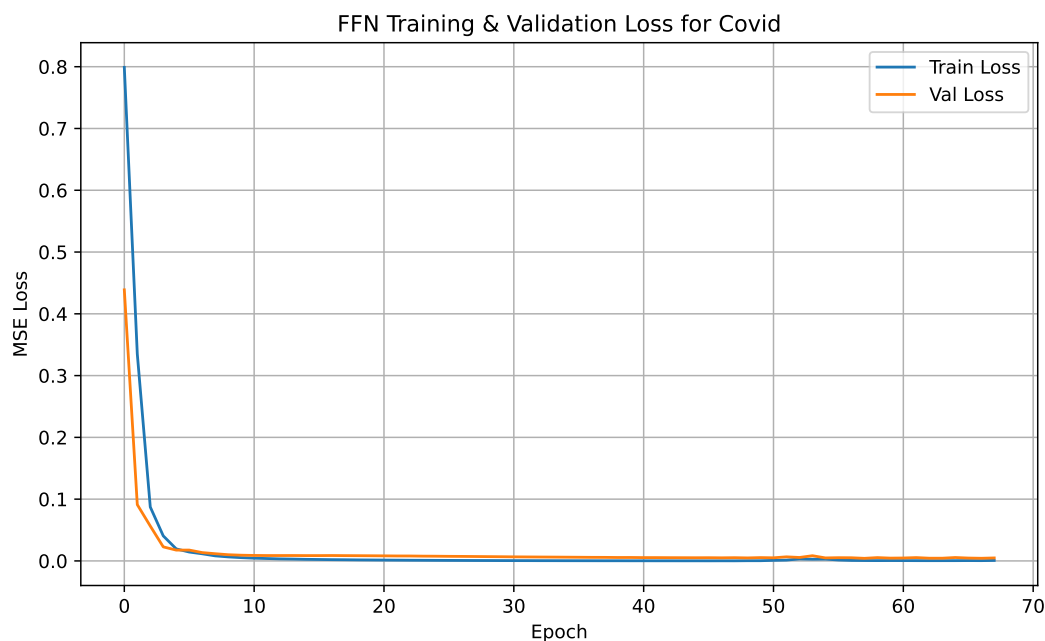


FIGURE 6.3: Training and Validation Loss FFN Covid- Colombia

Epidemics Figure 6.3 shows a very similar training behaviour. In this case, early stopping after a few epochs, given no significant improvement in the validation loss.

Remarks The multivariate neural network architecture proposed, shows great promise for being able to forecast next week cases. The error is minimized almost to zero. We will see how the model performs in the testing set in the next section.

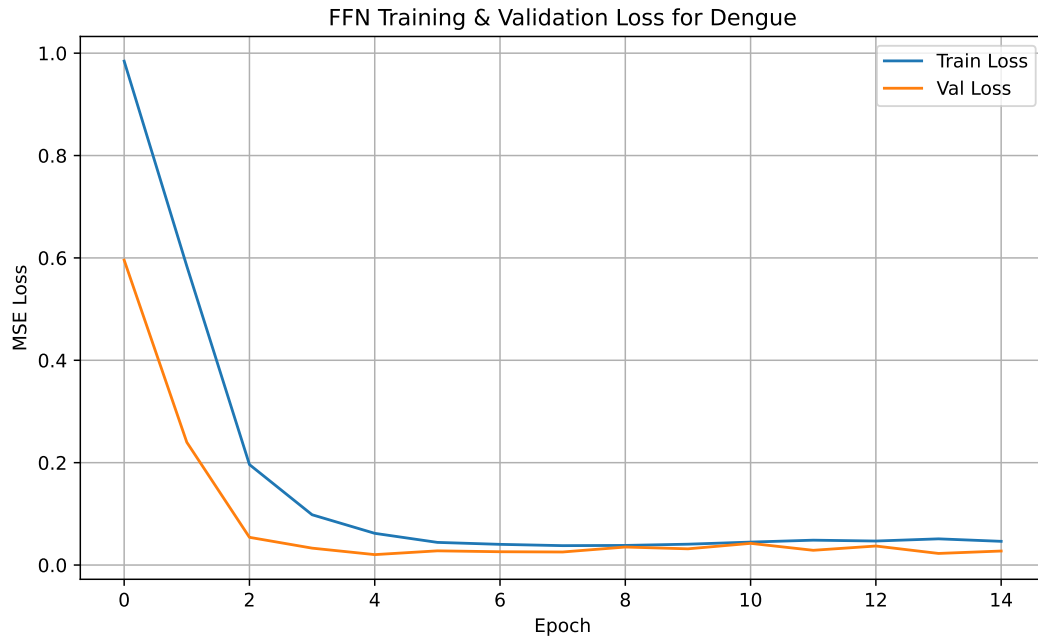


FIGURE 6.4: Training and Validation Loss FFN Dengue

6.4 Results

As done in the Chapter 5, we display the performance of the model for a selected case in each group of models. Again we select Colombia for the COVID series and Dengue for the epidemic series.

6.4.1 COVID

In Figure 6.5 we see the performance of the set of described models for the same series using the same train-test split and subset of selected features from Chapter 4. As we split the goal of modelling in two parts: forecasting the long term behaviour of the disease using time series models, and understanding the main drivers of the new cases for instantaneous forecast using multivariate regression models we see how different architectures that could have behaved well for the time series (cumulative) case, are now among the under-performers. First observation is the Linear Regression (OLS) and the penalized versions of it (Lasso, Ridge, ElasticNet) are comparatively underperforming the predictions if we see other models like the ensemble models and the neural networks, which capture non-linearities in the data. At first glance the Random Forest seems to be the most effective methodology for the task at hand, displaying the capabilities of the ensemble models and how they can out-perform deep learning architectures in smaller tubular datasets.

Evaluation Metrics Here again, the RMSE is used in the same fashion. The RMSE is computed for all Model-Country pair and the results are ranked based on the mean of those rankings. Figure 6.6 shows the computed ranking. We see how as expected, the DNN are the best performing models overall. But more importantly, we see how Lasso and ElasticNet are in the second and third place overall, showing a different behaviour than in the time series plots from the particular Colombian case. The fact that they are performing well is a good omen, because they are cheaper to

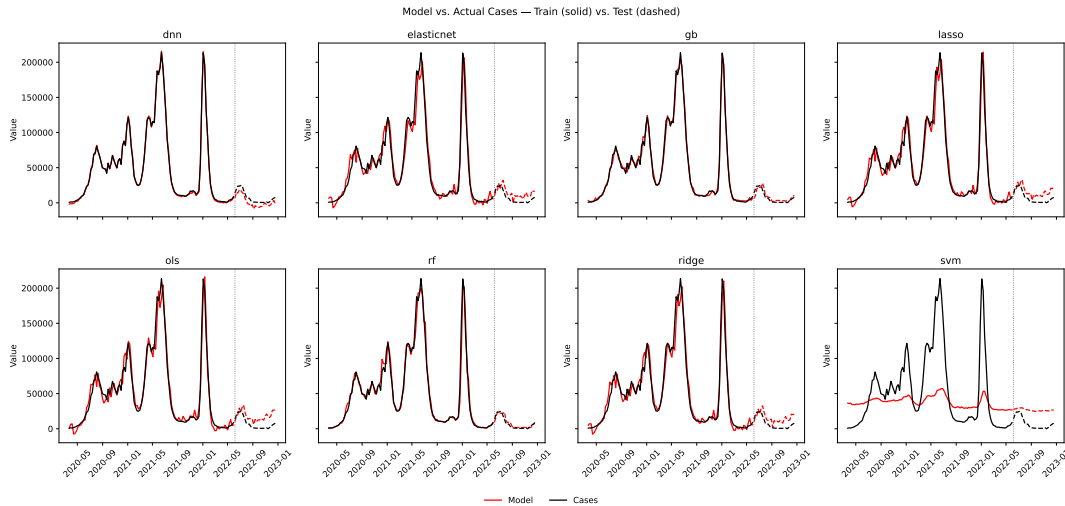


FIGURE 6.5: Model fitted values vs Actual COVID new cases (Colombia)

train and compute, and more importantly: they hold interpretable results. We can interpret the coefficients associated to each feature and do statistical inference, try to address the spread of the disease by focusing on policies that have a direct impact on the features. Say for example procure to have cleaner air to diminish new cases of COVID-19, or have social distancing to expect a certain outcome in new cases.

6.4.2 Epidemics

The behaviour of the models is similar in the Dengue example. But here we see a characteristic that is very important to take into consideration: The actual values in the test set are higher than the observed values in the training phase. This causes ensemble models like Random Forest and Gradient Boosting to struggle. In Figure 6.7 we see how ensemble models fail to capture the spike in new dengue cases in the test set, displaying on of the shortcomings of this type of models for the regression task at hand. Then we see how the DNNs seem to perform poorly in contrast with linear models like OLS, Lasso, Ridge and ElasticNet. Finally we see how the SVR performs very well in this case, while in the COVID section struggled to capture the new cases behaviour.

Evaluation Metrics As for the evaluation of the models, Figure 6.8 shows the ranking of the models by Epidemic-Model pairs. Overall, the SVR (or SVM) achieves good performance in the test set, followed by the block of linear models and the penalized linear models. Last we see the models that capture non-linearities better in theory, but also struggle with rigid trends or biased datasets. The fact that our data might have a large Bias component can play against these models and in favour of the linear ones.

6.5 emarks

Overall, we see how penalized linear models show good results. They prove to be a good tool for forecasting the new cases of infectious diseases, using the set of selected features from Chapter 4. The fact that linear models are among the most

Rank of RMSE for each model and country

	dnn	1	3	5	1	2	2.4
	lasso	3	6	1	3	4	3.4
	elasticnet	4	4	2	4	5	3.8
Country	rf	7	1	7	7	1	4.6
	gb	6	2	6	2	8	4.8
	ridge	5	5	3	5	6	4.8
	ols	2	7	4	6	7	5.2
	svm	8	8	8	8	3	7
		Chile	Colombia	Italy	Mexico	US	mean
		Model					

FIGURE 6.6: Ranking of RMSE by Country-Model (COVID)

capable ones is very good news, because they are easily implementable, provide interpretable results and are less computationally costly.

6.6 Conclusions

In this chapter we have evaluated a series of models for different scenarios, providing a rich set of heterogeneous data that forces us to avoid marrying to a single model because of some good of fitness metric like the RMSE. We have seen that Penalization in linear models are a really good choice for infectious disease modelling, even if we already selected a subset of features among all the available ones. They provide good performance in different scenarios: COVID across five different countries, and four different epidemics (Dengue, Chikungunya, Varicella and Zika) in Colombia. All these different time series have diverse dynamics which forces us to think of a model that generalizes well even if it sacrifices some variance minimization. SVM regression and Deep Neural Networks can work very well under specific scenarios, but they can also go to a complete opposite extreme and underperform compared to the most simple and widely implemented model in multivariate linear regression: Ordinary Least Squares.

This is why we propose that future infectious diseases, which are not mature enough to back-test, shall be modelled with some variation of the Penalized Least Squares model, depending on the available data Lasso, Ridge or the in-between (Elasti-Net) can surge as a great alternative to predict new cases in the short term, understand the impact of different covariates on the spread of the disease and provide governments, policy makers, health institutes with tools to mitigate the spread of the disease.

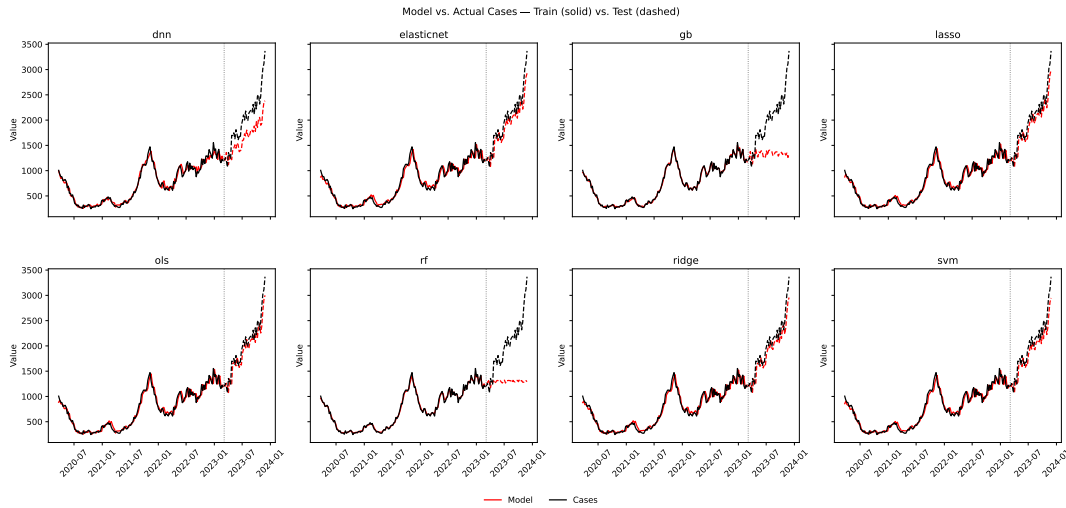


FIGURE 6.7: Model fitted values vs Actual Dengue new cases

The regression analysis at the country level could successfully identify variables like Google trends and the self lagged target as powerful covariates. Nevertheless, the lack of statistical significance for environmental factors in the dengue models, suggests that in the aggregation process, some impacts cease to be relevant. To address this, we jump to a more granular analysis on the city level, including other key drivers like coordinates to further try to explain the disease dynamics. We continue this analysis on the next chapter 7.

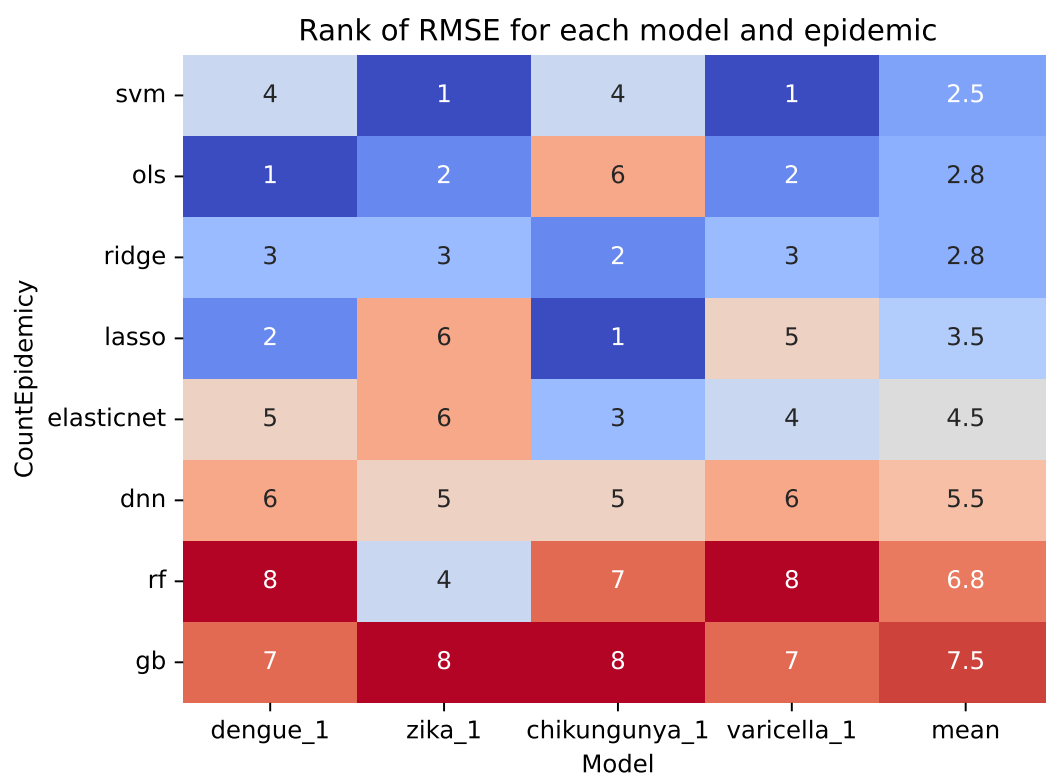


FIGURE 6.8: Ranking of RMSE by Epidemic-Model

Chapter 7

Spatial Statistics

7.1 Data

7.1.1 Target

The data utilized in this chapter is similar to the one used in chapter 6, particularly the one regarding Epidemics. In this case we go to a more specific setting of one of the epidemics. First, we select only one out the four epidemics: Dengue, which was covered extensively in the work so we have some intuition behind its behaviour. Then we only focus on a recent period: 2023. This focusing allows us to expand the dataset into a one more granular view. The original data from the web platform at Sistema Nacional de Vigilancia en Salud Pública (SIVIGILA), Instituto Nacional de Salud de Colombia, 2025 is reported at the testing clinic level, containing weekly new cases for each clinic. Among some of the available information we get the country, region and city of where the case was reported.

7.1.2 Geographic location

Thanks to the nature of the dataset we can expand a very general dataset into a more granular one that has different counts of new cases across the country. From these three pieces of information for each row we construct a table that groups the cases by city level. And using geopy package we use the Nominatim geolocator to extract the coordinates (Latitude, Longitude) from each city based on their names.

7.1.3 Weather

Given the geolocation of each city, we proceed to obtain weather data via the already used API Meteostat Development Team, 2025. The following variables were collected in a daily basis for all the cities in the dataset:

- tmax
- prcp
- wspd

The weather variables were averaged by week to match the periodicity of the target variable. Regretfully, not all cities in our new the dataset have a weather station. The consequence of this is that we got empty requests, so there was no weather information for several cities. Still, a large part of the cities, approximately 200, could be cross-referenced and weather information was obtained for them.

7.1.4 Summary

A summary of statistics of the data comprised in this section can be found in Table 7.1. We see that `pres` and `elevation` have less count than the rest, meaning they contain null values. We also see the overall magnitude and behaviour of the statistics. Beginning with the number of cases, which is the target for our problem. It seems to be very sparse, having more than 50% of the percentile with zero. But it also has some fat tail, with a max up to 456. All the covariates are less volatile in comparison to the target. It is important to keep in mind the sparsity of the target, because it will be a fact leveraged in the Modelling section of this chapter.

TABLE 7.1: Descriptive statistics of the dataset.

	count	mean	std	min	25%	50%	75%	max
N. Cases	13572	4.68	21.40	0.00	0.00	0.00	2.00	456.00
Latitude	13572	5.82	2.83	-4.21	4.18	5.02	7.83	12.58
Longitude	13572	-75.04	1.63	-81.72	-75.77	-75.24	-74.27	-67.48
tmax	13572	29.34	4.22	13.37	27.99	30.13	32.13	37.53
prcp	13572	6.84	7.09	0.00	1.63	5.04	9.86	61.87
wspd	13572	8.71	3.48	2.33	6.46	8.23	10.01	28.50

7.2 Motivation

The motivation for this approach comes from all the work done so far. We have found that in some cases for multivariate regression problems for the type of data we have, the linear models provide a powerful tool to assess the effect of a covariate on the target. Nevertheless, we also see that some non-linear models, or non-parametric ones can be insightful and find non linear relationships between covariates and the target. In this chapter we want to present an approach that comes in the middle ground between fully parametric models and fully non-parametric models. Azzalini and Scarpa, 2012 present in their book an excellent alternative to bridge the gap between the parametric and non parametric approach on regression tasks.

7.3 Mathematical Framework

In this chapter, we employ Generalized Additive Models, GAMs. They are an extension of the widely used linear regression model. In the Ordinary least squares framework we assume a strictly linear relationship between predictors and the target, GAMs in turn, offer a more flexible alternative that allows for non-linear effects while maintaining an interpretable, additive structure.

7.3.1 From Linear Model to GAM

Let us remember the multivariate linear regression model introduced in Chapter 6:

$$y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip} + \epsilon_i,$$

In this framework, the response variable y_i depends linearly on the predictors x_{ij} . Azzalini and Scarpa, 2012 present the Additive Models and the GAM as a middle ground between the linear (parametric) models and the non-linear (non-parametric)

ones. They build up from the definition of different basis functions that can explain the relationship between a predictor x with a target y via $g(x)$, a smooth function that manages to collect the non-linear relationships between x and y .

To introduce more flexibility in the parametric approach, we can replace the linear term $\beta_j x_{ij}$ with a smooth function $g_j(x_{ij})$ as is formulated in the book to arrive to the generalized additive model:

$$y_i = \beta_0 + \sum_{j=1}^p g_j(x_{ij}) + \epsilon_i,$$

where each g_j is a smooth function that is fitted to the data. A formulation like this one allows us to keep the additive structure of the linear model and at the same time allows each predictor to have a non-linear effect on the response. The functions g_j are constructed using basis functions. Below we outline some of the most popular and used basis functions, including the ones we will try in our modelling.

Basis Functions

Each smooth term is represented using a set of basis functions. The basis functions can be thought of as building blocks. We combine them to capture the relationship between a predictor and the response. The basis types relevant to our analysis and available in the open R packages are:

- **Thin Plate Regression Splines (tp):** A spline that optimally balances fit and smoothness. A nice analogy to think of this spline would be to think of it as one was bending a flexible sheet to pass through the data.
- **Cubic Regression Splines (cr):** Constructed by joining cubic polynomials at some fixed knots. This ensures a smooth curve across the predictor range.
- **Cubic Splines with Shrinkage (cs):** This one is an extension of the cubic spline. When we use this basis function the model shrinks the effect of the variable toward zero. By doing so, it effectively removes the term from the model if it is not influential.
- **P-Splines (Penalized B-splines, ps):** These use several smaller and local polynomial pieces, also called B-splines, and apply a penalty to their coefficients to ensure smoothness and prevent over-fitting.
- **Thin Plate Splines with Shrinkage (ts):** A version of thin plate splines but it incorporates an additional penalty to allow the entire smooth term to be shrunk out of the model.

Azzalini and Scarpa, 2012 make clear how the data is fitted in the GAM approach via a back-fitting algorithm. This is non-parametric estimation in an iterative way. In this algorithm each smoothing function is fitted inside a for loop, setting the others fixed, to converge to the best function of each of the smoothing functions.

7.4 Data Pre-Processing

Before using the GAM family of models we have to do some pre-processing. It is crucial to split the data into a training set for model fitting and a test set for evaluation, as we have done in previous chapters of this document. Spatial data has the

special consideration of **spatial autocorrelation**. Given that we already know the set of coordinates, we will overlook the spatial autocorrelation in the train-test split. This is because our model will not be used in different regions in the future, since Colombia is already a fixed country. We do not want to make a model that generalizes well for more countries, for that we would include more data in the training set. For these reasons, regular timeseries train test split is performed, using the first 80% of the data as training (at each city level) and the remaining 20% for testing.

7.5 Model Fitting

We employed the `mgcv` package in R to fit a Generalized Additive Model (GAM). Four candidate models were evaluated, including Poisson, Gaussian, Quasi-Poisson, and Log-Normal link functions. The data structure suggests to use a Poisson Link function for the estimation, since we have the count of new cases in each territory at each time.

7.5.1 Model Structure

We assume the number of observed cases at time $t + 1$, denoted as $Y_i = \text{count_t1}_i$, follows a given distribution to be determined by checking which distribution best models the data. The expected value μ_i is modelled directly by the additive predictor η_i via an identity link:

$$Y_i \mid \mu_i, \sigma^2 \sim \text{Normal}(\mu_i, \sigma^2) \quad \text{with} \quad \mu_i = \eta_i$$

The additive predictor η_i was defined as:

$$\begin{aligned} \eta_i = & \beta_0 + f_1(\text{tmax}_i) + f_2(\text{prcp}_i) \\ & + f_3(\text{wspd}_i) + f_4(\text{week_num}_i) + f_5(\text{Lon}_i, \text{Lat}_i) \end{aligned} \quad (7.1)$$

where:

- β_0 is the model intercept.
- $f_1, \dots, f_4$ are univariate smooth functions for the environmental predictors, and seasonality (`week_num`). The specific basis function for each term (e.g., P-splines, cubic regression splines) was determined by iteratively selecting the basis that minimized the Akaike Information Criterion (AIC).
- f_5 is a bivariate **tensor product smooth** of the geographical coordinates, capturing spatial dependencies.
- All models were fitted using Restricted Maximum Likelihood (REML) for smoothing parameter estimation, and the `select = TRUE` argument was enabled to allow for automatic term selection via extra penalization.

Remark

Up to now we have been including in several other models the lagged target as a covariate to explain the behaviour of infectious diseases. In this section we have opted to leave it out of the analysis on purpose because it has a large explanatory power, which shrinks the partial effects of the other covariates, so we cannot identify

the effect fully. The partial effects of almost all other covariates are shrunk to be negligible, and the only variable that has effect on the target appears to be its own lag.

In figures 7.1 and 7.2 we can see the difference in the effects of the terms in the case of a GAM using the Gaussian link function. We clearly see in the first plot how the count variable, which is the lagged value for the target, explains most of the dynamic of the spread when it is included, almost robbing the effect of other terms, like the precipitation, which in practice plays a crucial role to enable the proliferation of the mosquitoes that transmit Dengue. For this reason we think is best to limit the covariates in this section and continue the analysis using only external covariates, obviating the autoregressive nature of the phenomenon. Nevertheless, we include the count variable from time to time to evaluate the model predictive power and compare how the smooth terms vary if it is included or not.

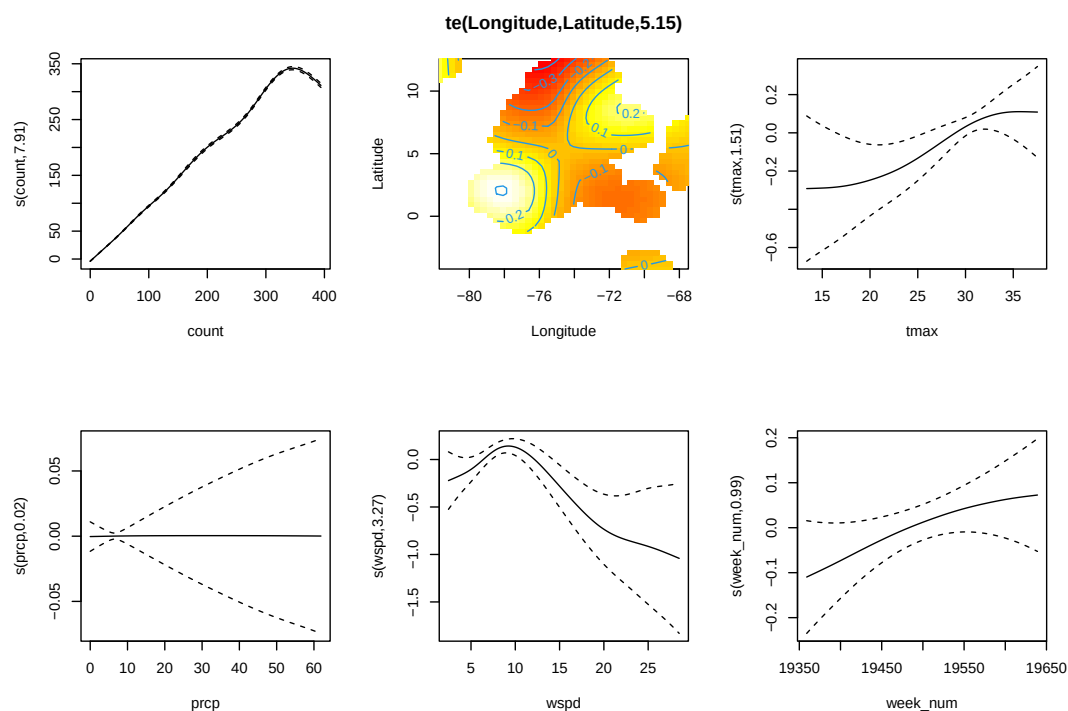


FIGURE 7.1: GAM Link Gaussian with count covariate

7.5.2 Link Function

To select the best link function for our problem we opted to fit four distinct models with the same base structure, only changing the link function between them. We also let the model estimate the smoothing parameter for each term automatically. Based on a combination of predictive performance, evaluated by Root Mean Squared Error, and the interpretability of the partial effects, a Gaussian model was selected for the final framework. In table 7.2 we can see the comparison of the different models implemented. We can see that the performance among models is pretty similar. The Log-normal was the best performing model based on RMSE, but the interpretability of the partial effects in the Gaussian version was much clearer and had a lot more sense with the natural drivers of Dengue spread.

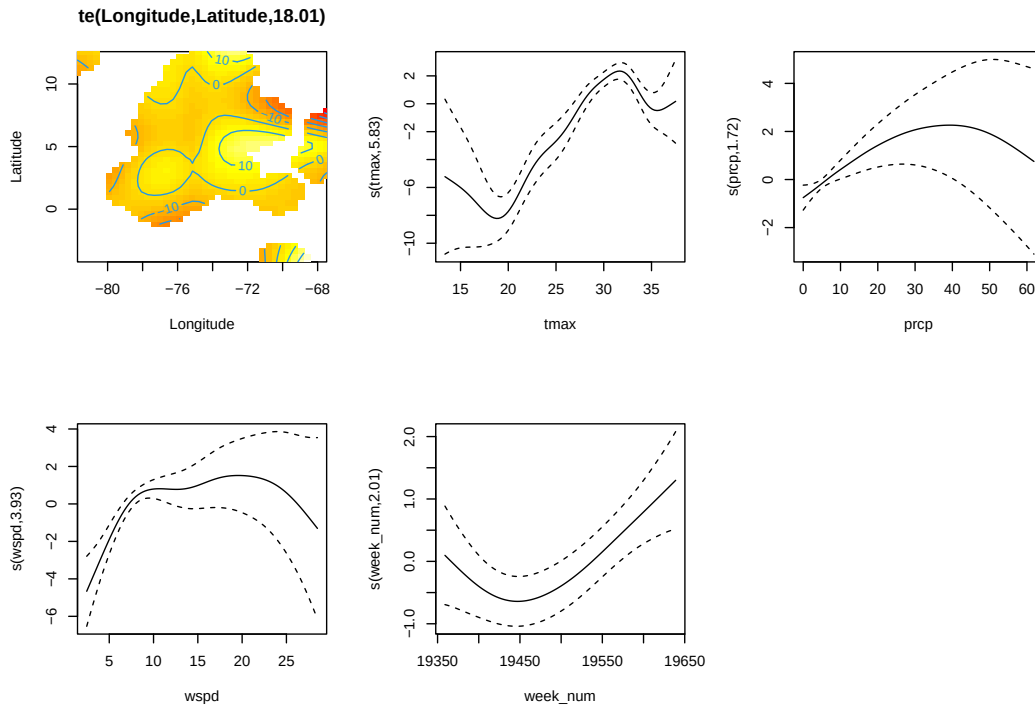


FIGURE 7.2: GAM Link Gaussian without count covariate

Model	RMSE
Log-Normal	29.9859
Quasipoisson	30.7570
Poisson	30.8909
Gaussian	31.8582

TABLE 7.2: Comparison of RMSE for GAM Models Link Functions

7.5.3 Final Predictor Specification

The structure of the additive predictor, η_i , was determined by identifying the optimal smoothing basis for each covariate through an iterative, AIC-based selection process. The final model specification is detailed below:

$s(tmax, bs='tp', k=6)$ A smooth for maximum temperature. The chosen smooth function was **thin plate regression spline** with a smoothing parameter of $k = 6$, for more reasonable interpretation of the interaction.

$s(prcp, bs='ts')$ Is the smooth function for precipitation. The best function found was a **thin plate regression spline with shrinkage**, with automatic smoothing parameter selection.

$s(wspd, bs='ps')$ Is the term for wind speed. The best smoother selected was a **P-spline basis**. The basis dimension was selected automatically.

$s(week_num, bs='ps')$ is the time feature included in the model. The smoothing basis found was **P-spline basis** to flexibly capture cyclical patterns. The smoothing degree k was also selected automatically.

`te(Longitude, Latitude)` A two-dimensional spatial effect. The smoothing function used is the **tensor product smooth**. This term captures the geographic interaction between longitude and latitude, with the basis dimension for each coordinate automatically found by the method.

7.6 Results

7.6.1 Smooth Terms

Table 7.3 shows the statistical significance of the smooth terms. We can see that all smooth terms have a low p-value, which suggests that a non-linear relationship exists between the covariates and the response variable. The **edf** column shows the estimated degrees of freedom, which indicates the complexity of the fitted function for each variable. The edf values are relatively low for `tmax`, `prcp`, `wspd`, and `week_num`, while the value for the spatial interaction term (`te(Longitude, Latitude)`) is considerably higher, reflecting a more complex spatial pattern. We also note that the basis dimension for `tmax` was set to $k = 6$ to ensure a smooth and interpretable partial effect plot.

Based on the deviance explained, the model's fit is modest at 6.02%. For comparison, a second model was fitted that also includes an autoregressive smooth term for the count variable. The inclusion of this term increases the deviance explained dramatically to 95.5%. The results of this second model are presented in Table 7.4. However, we see that other terms, such as `week_num`, are no longer statistically significant based on their p-value. This is expected, as the temporal component of the data is now largely captured by the autoregressive count term rather than by the explicit time variable.

	edf	Ref.df	F	p-value
<code>s(tmax)</code>	3.7402	5.00	37.82	0.0000
<code>s(prcp)</code>	0.9425	9.00	1.48	0.0001
<code>s(wspd)</code>	3.3928	9.00	3.98	0.0000
<code>s(week_num)</code>	2.3510	9.00	1.69	0.0003
<code>te(Longitude,Latitude)</code>	18.0437	24.00	19.46	0.0000
Deviance Explained: 6.02%				

TABLE 7.3: Significance of Smooth Terms and Model Fit - Base Model

	edf	Ref.df	F	p-value
<code>s(tmax)</code>	1.5081	5.00	1.70	0.0019
<code>s(prcp)</code>	2.9523	9.00	0.80	0.0396
<code>s(wspd)</code>	3.0118	9.00	2.67	0.0000
<code>s(week_num)</code>	1.0536	9.00	0.28	0.0836
<code>te(Longitude,Latitude)</code>	5.0167	24.00	0.77	0.0003
<code>s(count)</code>	7.9139	9.00	24850.97	0.0000
Deviance Explained: 95.54%				

TABLE 7.4: Significance of Smooth Terms and Model Fit - Model with count variable

7.6.2 Parametric Effects

In table 7.5 we can see the parametric effects of the intercept, which also displays a low p-value. The interpretation here is not straightforward so we avoid delving into it.

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.0578	0.1666	24.36	0.0000

TABLE 7.5: Parametric Coefficients of the GAM

7.6.3 Smooth Interactions

In Figures 7.3 and 7.4 we see the smooth interactions between the target variable and the covariates.

1. The temperature term `tmax` shows a similar behaviour in both plots. The low temperatures appear to reduce the risk of spreading Dengue. Then as the temperature rises the risk increases. The difference between the two models is that the confidence intervals appear to be wider in the autoregressive version, and the magnitude of the effect also seems lower. The count could be absorbing most of the effect.
2. The precipitation `prcp` interaction changes from one version of the model to the other. When the count variable is included, the effect seems to be low at low levels of precipitation, then increases slightly but the confidence intervals are too wide for this term to be properly interpreted. On the other hand, in the base model the precipitation seems to have an increasing effect on the Dengue cases. The more precipitation, the more risk of Dengue outbreaks and spread. In this version the confidence intervals also widen as the value of precipitation increases but the effect is much clearer than in the autoregressive version.
3. The wind speed `wspd` also shows different behaviours from one model version to the other. In the base case, the wind speed shows a reducing effect at low values, and then the effect of the wind speed increases up to approximately a speed of ten km/h. Then it makes a slight undulation, to later decrease at higher values. This can be interpreted as follows: whenever the wind speed is low, the mosquitoes cannot travel as far, hence the dengue spread is mildly contained. At higher speeds, the mosquitoes can go further and infect more individuals, up to a certain speed when the mosquitoes might probably have trouble flying due to intense wind speeds. For the autoregressive version, the plot makes a sine-like wave, which would imply that at some ideal wind speed of 10 the mosquitoes can infect more actively. Then we see a downturn and recover, but the error margin is too large to make this curve interpretable.
4. The time variable `week_num` shows a decline in the base model. Going from around 0 effect to negative in the first part of the plot, implying a period in which the cases were contained due to some factors. Then we see an increasing effect as the time variable value increases. In contrast, in the autoregressive version, we see an increasing trend effect. As the time value is higher, the more the effect of dengue contagion, but in the extremes the error margins are too large to have interpretable effects.

5. The lagged target count shows an increasing effect of the variable on the target. The effect is always positive indicating that the more cases of Dengue there are, the more probable it is to have higher Dengue cases in the following week. The error margin is low and the effect saturates at the end, around 350, which might be given some particular situation in the data, rather than the nature of the disease spread.

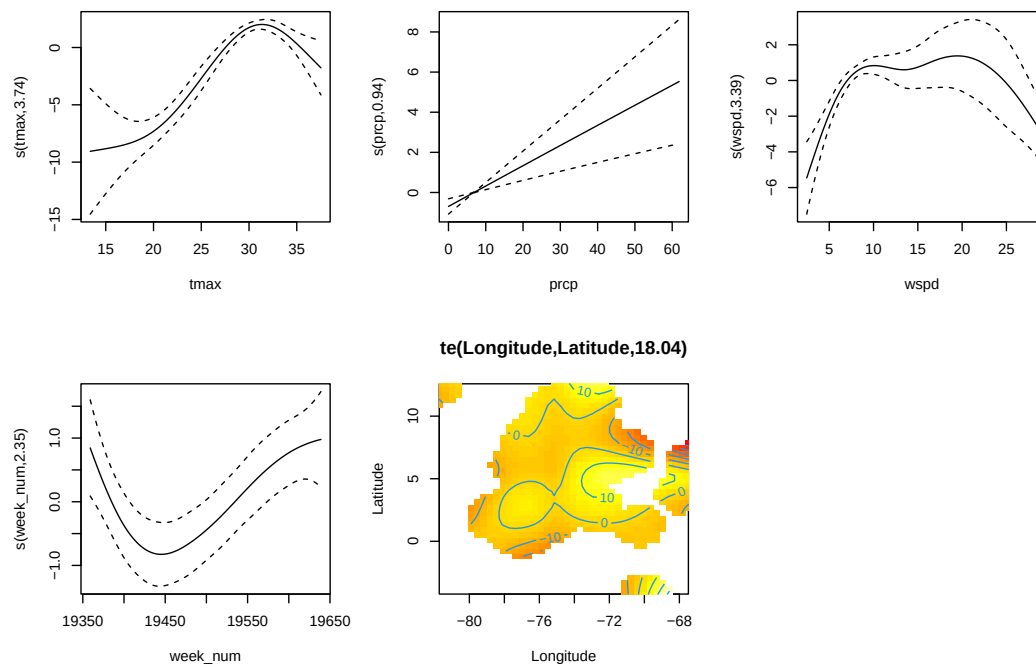


FIGURE 7.3: GAM Effects Base

Longitude and Latitude: Longitude and Latitude: The effect of this variable can be seen better in Figures 7.5 and 7.6. In both maps we see some similarities. The first one is that the lower left side and upper right side (south west and north east) seem to have a larger effect than the rest of the country. The south western side makes perfect sense, since there is a large metropolitan area, with heat and precipitation which would result in a region with higher propensity to Dengue spread. The north eastern part of the map is somehow different across both maps. The map with the base effects, shows how there is a large portion with higher effect on dengue spread. This region is called the eastern flatlands "llanos orientales" in spanish. There are seasonal rain patterns that cause flooding. This regions is also less populated and no large metropolitan areas are found there. The climate is hot and humid, and there is a lot of nature. It makes perfect sense to have this region lighted in bright red in this map, because of the climatological and demographical characteristics of the region. The map with the autoregressive term, appears to avoid giving too much importance to said regions. It might be because of the low Dengue cases found there, comparative to other regions. Which is only natural since this is a less populated section of the country. Having the autoregressive term can imply absorbing some spatial effect. Other comparison can be made for the north of the country. The map with the lagged count variable shows negative spatial effect in that region, while the

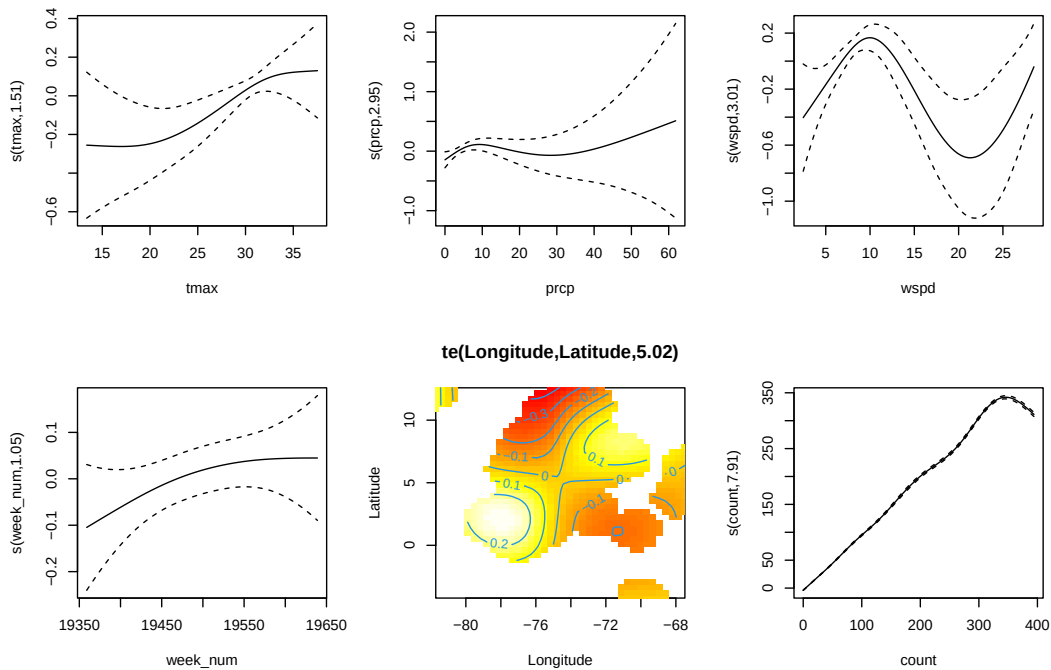


FIGURE 7.4: GAM Effects autoregressive

map of the model without the count variable shows a positive effect, specially in the most northern region called La Guajira. This is also a region with low population, but has a lot of heat and poor sanitary conditions. This regions has low precipitation, which could cause to low dengue proliferation. But there is the sea, which can cause for salted water pools to form where mosquitoes lay their eggs.

Both versions capture slightly the fact that the Andean mountains have lower effect on the dengue spread. This might be because of the cold weather and the scraped terrain that hinders the formation of rain pools and mosquito proliferation.

The map for the base model also shows how in the deep south there is a larger spatial effect of Dengue incidence. This region is the Amazon basin, with a city in it which limits with neighbouring cities in other countries (Brazil and Peru).

The map for the spatial effects without the inclusion of the autoregressive term seems more appropriate overall, based on the fact that it better displays the nature of the climatological, topographical and demographical characteristics of the regions, while the map with the autoregressive interaction included leaves aside some of the regions where it makes sense for the disease to be more likely to spread.

7.6.4 Predictive power

Here we analyse the predictive power of the model we have been adjusting. 7.6 shows how both versions of the model, including and leaving aside the count variable. We see how the Base model (which does not include the variable) performs poorly when compared to its more robust version in the test set. Both RMSE and R-squared are lower for the base model, while the Autoregressive one achieves great performance.

We also wanted to check directly how does the GAM approach compare to the previous multivariate approaches seen in chapter 6. In this case, we fitted the same

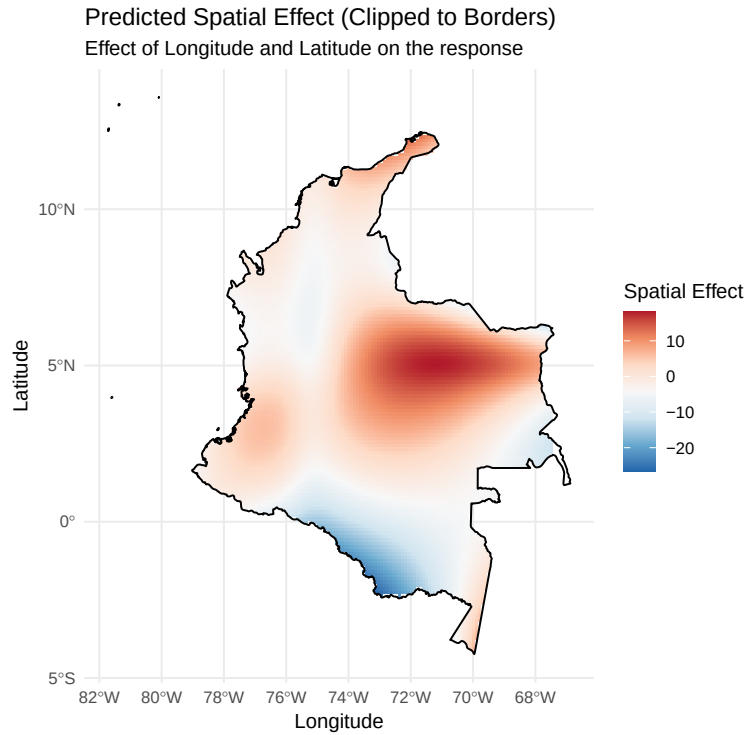


FIGURE 7.5: Coordinates Effect Base

Model	RMSE	R_squared
Base Model	31.8881	0.0358
Autoregressive Model	7.6691	0.9442

TABLE 7.6: Comparison of Model Performance on the Test Set

model architectures as before, using the same hyper-parameter optimization, the thing that changes is that now we fit the models to the same data we are using in the current chapter. The train and test split is done in the same way so that we can compare the performance metric of all models on the test set. In table 7.7 we see that the GAM outperforms almost all models in the case when the count variable is not included, except the Random Forest, which is highly non linear. Then for the version where the variable count is included as regressor, the model seems to be the worst performing one. For this analysis, where we are focusing on the term interaction between latitude and longitude (on the model with no autoregressive component) we think that is this very interaction that provides the GAM with the additional performance and predictive power.

7.7 Conclusions

In this chapter we successfully showed the powerful attributes of the Generalized Additive Models in the context the infectious disease modelling. Particularly, we used them to find the non-linear relationships between environmental and spatial variables with the count cases of Dengue in many regions of Colombia. The main goal of leaving the interpretable world of linear models. This sacrifice allows us to capture the spatial and climatic drivers of the disease.

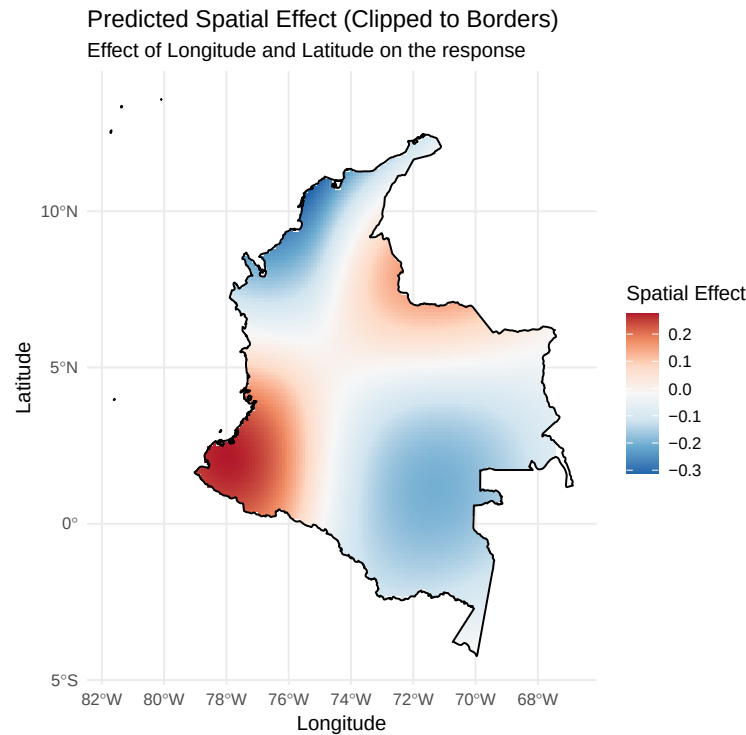


FIGURE 7.6: Coordinates Effect autoregressive

The key epidemiological insight from the modelling approach is the confirmation of how Dengue risk is concentrated in specific regions within Colombia, which have hot temperatures and their precipitation higher than average. We also found that regions that are known for their poor social conditions have a higher effect on the spread of Dengue. The results clearly associate higher risk with the low-altitude, high-temperature, and high-pressure conditions. These specific characteristics are found in coastal regions and regions with flooding. This synthesis, which acknowledges the confounding nature of these variables, aligns with established knowledge and provides a data-driven confirmation of risk zones.

Nevertheless, this section also showed the difficulties of trying to capture the time series nature of the spread diseases and interpreting the other features, where the time series nature can absorb the effects of the other variables. We saw that not including the time series nature results in much lower performance and ability to

(A) No count			(B) Count		
Model	Test R2	Test RMSE	Model	Test R2	Test RMSE
OLS	0.2%	32.44	OLS	97.3%	5.356
Lasso	-0.7%	32.59	Lasso	97.3%	5.362
Ridge	0.2%	32.44	Ridge	97.3%	5.362
Elastic Net	-0.8%	32.6	Elastic Net	97.1%	5.493
Random Forest	49.9%	22.97	Random Forest	95.5%	6.873
Gradient Boosting	14.6%	30.08	Gradient Boosting	96.6%	6.005
SVM	-1.5%	32.7	SVM	97.0%	5.62
Best GAM	3.6%	31.88	Best GAM	94.4%	7.66

TABLE 7.7: Comparison of models with and without count

explain the variations in the data.

We also found that GAM models might outperform linear models when the autoregressive feature is not included, but it does not achieve as high performance as other architectures like ensemble models. Still, the analysis of the interactions between the target and the covariates via non linearities are extremely useful to understand the particular infection drivers, which can be monitored for disease surveillance and policy recommendation by the national health institute of Colombia and the insurance companies that want to look after unexpected policy claims. In particular, those agents should look for:

- **Maximum temperature ranges:** whenever the maximum temperature in a city oscillates in the observed ranges in this chapter, that boost Dengue spread, prevention and control campaigns should take place in said cities to contain the spread of Dengue.
- **Wind Speed:** Whenever the wind exceeds speeds of around seven km/h, the relevant players should look after the spread in the riskier areas
- **Coordinates:** Insurance companies could use the location information as a pricing input, to recharge the price of riskier areas like the "Llanos orientales", La Guajira and Cali. In this way they can have a more accurate pricing and less unexpected losses in the health policies, particularly for Dengue claims. As for the national health institute, it could use the maps provided in this thesis to look for focused health campaigns in said areas. In this way scarce resources are effectively allocated to areas where Dengue or other similar diseases are more prone to happen, and the spread can be contained from the very source.

Looking forward, the following paths for future research can build on top of the current research:

- **Alternative Distributions:** Given the sparse nature of the case counts (many zeros) and the fat tail observed in the data, moving to a **Negative Binomial** or **Tweedie** distribution could provide a better fit for the overdispersed data and potentially improve model performance.
- **Enriching Predictors:** The model could be enhanced by incorporating additional data sources that might explain regional variations, such as population density, socioeconomic indicators, disease control efforts, and human mobility data.

Chapter 8

Conclusions

Closing Remarks

This thesis set out to explore and identify robust statistical and machine learning models capable of explaining and forecasting the spread of infectious diseases. Through an in-depth analysis of diverse datasets for COVID-19 and several Colombian epidemics, particularly Dengue, this work confirms that a multi-faceted modelling approach is essential for capturing the complex dynamics of disease transmission. The basis for this document was both academic intrigue and industry related.

The exploration of **time series models** in Chapter 5 revealed that no single model is universally superior. While traditional mechanistic models like SIR and SEIR provide epidemiological interpretability, their rigid structure often fails to capture the multi-wave nature of modern pandemics. In contrast, data-driven models demonstrated greater flexibility. The simple ARIMA model proved surprisingly effective for long-term forecasting in some scenarios, and diffusion models like the GGM showed promise in capturing overall trends. This highlights a key takeaway: for long-term forecasting, simpler, trend-focused models often provide more reliable trajectories than highly flexible ones that may over-fit short-term noise. In literature review and industry application we see a lot of mechanistic models, but some of the other more flexible and sometimes better performing models like the diffusion ones are not so used. This document proves that said simple yet powerful models can serve as a very good tool for long term forecasting of epidemics and pandemics.

The investigation of **multivariate models** in Chapter 6 shifted the focus to short-term prediction and identifying key drivers of new cases. This analysis consistently displayed the importance of lagged case counts, confirming the strong temporal dependency inherent in disease spread. Additionally, external factors such as Google Trends search data, air quality metrics, and policy indicators were identified as significant predictors. Notably, penalized linear models like Lasso and ElasticNet emerged as highly effective, providing a valuable balance of strong predictive performance and interpretability.

Finally, the **spatial analysis** in Chapter 7 provided the most granular insights, confirming that the risk of Dengue in Colombia is not uniform but is concentrated in specific climatic and geographic niches and regions where socioeconomic development is poor. By employing a Generalized Additive Model, we showed that factors such as **high temperature**, medium to high **wind speed** and high **precipitation** are significantly associated with increased Dengue incidence. This confirms that low-altitude regions are high-risk zones. However, this analysis also revealed a critical challenge: despite its excellent explanatory power (94% test R²), it is only achievable by incorporating the lagged target. When trying to explain the spread of the disease only with external regressors we found low explanatory power in the GAM framework. When compared to other regression approaches it proved to be a balanced

way to achieve the goal. When no autoregressive component was included, the performance was superior to all linear models but inferior to ensemble models. In this way we see that the non-linearities and the term interaction between the coordinates showed to be of great use in this particular task.

In conclusion, this research validates the thesis that a combination of statistical and machine learning models is instrumental in dissecting the behaviour of infectious diseases. We found that lagged information, coupled with meteorological, environmental, behavioural (Google Trends), and geospatial data, provides a powerful toolkit for both explanation and prediction. The optimal modelling strategy depends on the specific goal: diffusion and linear time series models for long-term trends, and interpretable multivariate or spatial models for identifying actionable, short-term risk factors. The findings here not only contribute to the field of epidemiological modelling but also provide a practical framework for public health agencies and private entities, such as insurance carriers, to better anticipate and manage the risks posed by infectious diseases.

Using diffusion and ARIMA models for long-term resource planning, while employing interpretable Lasso models for weekly monitoring of risk factors to guide immediate policy interventions. This is based on the findings of chapters 5 and 6.

Remark: Trade-Off Between Prediction and Explanation In chapter 6 we saw that Lasso models effectively served for the goal of being a good explanatory model and also being capable of predicting the near term cases for COVID, Dengue, Zika, Chikungunya and Varicella. When we made the jump to a higher spatial resolution in Colombia in chapter 7 we encountered that The GAM modelling approach with the inclusion of the autoregressive component was the best performing model in terms of Test RMSE and R2. Nevertheless, the explanatory power of the environmental and spatial features was poor. When making dropping the count variable we saw a decrease from over 94% to about 3.6% in the R2 metric. Here we see the classic trade-off between explanation and predictive power. For this document we opted to reduce the predictive power in order to comprehend further the effect of the covariates on the infectious diseases. Other case uses, might worry about being able to predict more accurately, as could be an insurance company trying to estimate the claims in a health business.

8.0.1 Applications of this work

The diverse strengths of each modelling approach can be integrated into a multilayered framework for public health and insurance applications. We propose using the findings of this Thesis as follows:

Risk Mapping: Using the GAM finding of chapter 7, interested parties could create a baseline map of endemic risk zones. This would be a visual insight for insurance actuaries who are pricing and targeting their sales efforts. For public health use, this map could serve as a guide for allocating resources into health preventive campaigns, like vector control in the high risk zone of "Llanos orientales."

Long term analysis: Diffusion models and classic ARIMA models from chapter 5 should be used for forecasts of infectious diseases in a 6 to 12 month period ahead.

This forecasting gives a high level understanding of how could an epidemic spread which can be useful for the insurance reserving (saving money to pay for claims). In terms of public health, the state could plan the hospital capacity, order key medical equipment and impose mobility restrictions to control the spread if they see fit based on the forecasts done.

Weekly Monitoring: From chapter 6 gained knowledge on how to adapt a weekly early-warning system. This system would be made by a Lasso model that has interpretable parameters. The interested parties like policy makers and actuaries at insurance companies could focus on monitoring the features of the model to anticipate sudden outbreaks. For example a spike in Google trends indices could indicate a coming spike in contagions, giving the public health authorities and insurance companies some time to address the infections in advance.

With the joint integration of these models we are leveraging the tools we have for the right task, rather than choosing a single model to be our Swiss-knife. In this way we have a system that can do medium to long term planning and also be aware of the short term volatilities provide more appropriate responses.

Limitations

One limitation in this thesis document could be the very data used for the analysis. The data was obtained via different sources. In the case of disease data, we are inferring that the testing for COVID is comparable across countries, which might not be true and could cause the results to be biased. At the national Colombian level, the Dengue data is obtained in the SIVIGILA portal which provides cases at testing level. This disjoint of testing clinic and actual dengue contagion could cause some bias. For instance, if a large city receives peripheral areas' residents for the testing, the results in said city could be inflated. Some cities might be neighbouring in distance, but the meteorological and socio-demographic characteristics could vary tremendously. Even at the same city level large cities like Medellin have some areas located above 2.000 meters above the sea level, with colder climate, while some other areas are around 1.300 meters. This difference in micro climates could be biasing the results in some way or another. Regarding the climatological data, it was recovered via APIs. These data could be prone to measurement error since some data in the repository where it is originally collected could be measured via satellite and not via weather stations directly. The difference in measurement could also be a cause of bias and volatility in the collected data.

Future Work

Looking forward, several avenues for future research emerge directly from the limitations identified in this work:

- **Alternative Distributions:** Given the sparse nature of the case counts (many zeros) and the fat tail observed in the data, moving from a Poisson to a **Negative Binomial** or **Tweedie** distribution could provide a better fit for the overdispersed data and potentially improve model performance.
- **Enriching Predictors:** The model could be enhanced by incorporating additional data sources that might explain regional variations, such as population density, socioeconomic indicators, vector control efforts, and human mobility data.

- **Network Science Approach:** The network science has proven to be a powerful tool to explain the spread of information, people and diseases in our society. The limitations of this field of knowledge lied on computing power. For future works, with more budget and computing power, modelling networks and simulating mechanistic models around them could prove to be another useful tool to explain the dynamics of infectious diseases as the ones covered in this Thesis.
- **Graph Neural Networks:** As we saw in chapter 7 the spatial information proves to be useful, so we might go further and model the single cities as nodes in a graph and include the different features as information on the nodes. In this way we could apply Graph Neural Networks to use both the spatial information and the network based approach on epidemiology.

Appendix A

EDA Details

This Appendix contains details on the methods used for the EDA on Chapter 4.

A.1 Analysis of Variance - ANOVA

The ANOVA (Analysis of Variance) is used to find the statistical relationship between the target variable and each individual feature. Specifically, the **F-test** is used to evaluate whether there is a significant linear relationship between each feature and the target.

For each feature x_i , a linear regression model is fit:

$$y = \beta_0 + \beta_1 x_i + \epsilon$$

Then we compute the **F-test statistic** as:

$$F = \frac{\text{explained variance by } x_i}{\text{unexplained variance (residuals)}} = \frac{MSR}{MSE}$$

where:

- MSR is the mean square regression, or variance explained by the regression,
- MSE is the mean square error, or variance not explained by the regression.

To check if the covariates have an effect on the target we perform an hypothesis test. For this test the F-statistic follows an F-distribution under the null hypothesis. This means that under the null hypothesis the feature has no effect on the target. So if we reject the null hypothesis, we keep the covariates as candidates for the modelling.

Then the corresponding **p-values** for each F-statistic are computed. Features with a p-value below the threshold set at $\alpha = 0.05$ are selected as relevant predictors. Because we reject the hypothesis of them having no effect on the target, i.e. we reject $h_0 : \beta_1 = 0$.

A.2 Variance Inflation Factor - VIF

The VIF gives a measure of how much the variance of a regression coefficient β_i is inflated due to its linear dependence with other features. In practice, this means checking if a given feature can be linearly predicted from the other ones, hence it is redundant.

A.3 Wrapper methods

AIC vs. Deviance: The deviance measures the goodness-of-fit of a model. This metric is often inflated by the addition of variables that add little information. The inclusion of a variable that marginally increases the R^2 , will result in a lower deviance. We need to account for this marginal effect so we avoid including unnecessary covariates that bring in noise. The AIC is built on top of deviance and introduces a penalty for model complexity. Specifically, AIC is defined as:

$$\text{AIC} = 2p - 2\log(L)$$

where p is the number of parameters and L is the maximum value of the likelihood function for the model evaluated. This penalty avoids overfitting by balancing model complexity and goodness of fit.

Backward Elimination: In this method one starts with a full regression, that includes all the predictors. Then an iterative method of predictor elimination begins. If we have m covariates: at the first iteration fit the full model removing each of the m covariates, so each model has $m - 1$ covariates. For each compute the AIC, compare them all and select the one that has a minimum AIC. When the AIC is no longer minimized, then the loop breaks and the model with the minimum AIC is returned.

Forward Selection: This approach starts from an empty model and adds features one at a time. The empty model is just the target regressed on the intercept. At each step add one of the m covariates. Compare them all and choose the one with the minimum AIC. The loop stops when adding more variables no longer improves the AIC.

Add-Delete (Stepwise Selection): This method combines forward selection and backward elimination. It allows both the addition and removal of variables at each step, offering more flexibility and potentially better models than either of the strategies alone.

A.4 Embedded Methods

A.4.1 Lasso

The Lasso (Least Absolute Shrinkage and Selection Operator) is an extension of the ordinary least squares (OLS) regression. The idea is to penalize the regression coefficients β_i via $L1$ norm. The method minimizes the residual sum of squares while adding a penalty proportional to the sum of the absolute values of the coefficients. The problem can be expressed as:

$$\min_{\beta} \left\{ \frac{1}{2n} \sum_{i=1}^n (y_i - \mathbf{x}_i^T \beta)^2 + \lambda \sum_{j=1}^p |\beta_j| \right\}$$

where $\lambda \geq 0$ is a regularization parameter that controls the strength of the penalty. The $L1$ penalty ($\sum |\beta_j|$) tends to shrink some coefficients exactly to zero, thus performing variable selection.

The steps used for variable selection with the Lasso model are as follows:

1. **Standardization:** All features are scaled to have zero mean and unit variance to ensure the regularization is applied fairly.

2. **Cross-validation:** We perform cross-validation on the training set to determine the optimal value of the regularization parameter λ .
3. **Model fitting:** Using the best λ , we fit the Lasso model on the training set.
4. **Variable selection:** Only the features with non-zero coefficients in the final model are retained.

A.4.2 Adaptive Lasso

This method is a variation of the Lasso method above. In this case different penalty weights are assigned to different coefficients, not like the standard lasso that applies the same penalty to all coefficients. The weights are determined by the coefficients of a Ridge or OLS model fitted on the data. The idea is to allow more flexibility by penalizing less the coefficients that are likely to be important, which would take a large absolute value in the prior model. The optimization is formulated as:

$$\min_{\beta} \left\{ \frac{1}{2n} \sum_{i=1}^n (y_i - \mathbf{x}_i^\top \beta)^2 + \lambda \sum_{j=1}^p w_j |\beta_j| \right\}$$

where w_j are data-dependent weights, often set as $w_j = 1/|\hat{\beta}_j^{\text{init}}|^\gamma$, with $\hat{\beta}^{\text{init}}$ being a preliminary estimate, such as from OLS or Ridge regression. This method improves the consistency of variable selection, especially when some true coefficients are small.

A.4.3 Decision Trees

Tree-based models such as regression trees and random forests naturally perform feature selection during the training process. They are a type of ensemble models which uses little weak learners as building blocks to create a bigger model that results in a strong regression predictor. They are based on decision trees, at each split, the algorithm chooses the feature that results in the greatest reduction in impurity. Features that are never chosen for splitting are excluded from the final model. In this way, trees themselves are a way to select the features that do the most reduction of impurity (minimization of variance)

A.5 Principal Component Analysis - PCA

Principal Component Analysis (PCA) is a dimensionality reduction technique that transforms a dataset into a set of orthogonal vectors (variables) that explain the variance of the data in the best way possible. It is usually used for dimensionality reduction, which can be used in our favour to select some columns (features) of matrix of data. Take note that for the PCA approach we are interested in the features that generate most of the variation, for this we exclude the target from the data matrix to be analysed.

Let $A \in \mathbb{R}^{n \times p}$ be the data matrix, where n is the number of observations and p the number of features. The data is assumed to be centered (i.e., the mean of each column is zero).

Step 1: Compute the covariance matrix. The sample covariance matrix $\Sigma \in \mathbb{R}^{p \times p}$ is given by:

$$\Sigma = \frac{1}{n-1} A^\top A$$

This matrix captures how the features in A co-vary with one another.

Step 2: Perform Singular Value Decomposition (SVD). We apply SVD to the covariance matrix:

$$\Sigma = U \Lambda U^\top$$

where:

- $U \in \mathbb{R}^{p \times p}$ is an orthogonal matrix whose columns are the eigenvectors of Σ ,
- $\Lambda \in \mathbb{R}^{p \times p}$ is a diagonal matrix containing the eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p \geq 0$.

This decomposition can be interpreted geometrically as:

$$\text{Rotation (via } U^\top) \rightarrow \text{Scaling (via } \Lambda) \rightarrow \text{Rotation (via } U)$$

Step 3: Principal components. The principal components of A are obtained by projecting the original data onto the eigenvectors:

$$Z = AU$$

The matrix $Z \in \mathbb{R}^{n \times p}$ contains the principal components, where:

- Each column of Z is a linear combination of the original features,
- The variance explained by each component is proportional to its corresponding eigenvalue λ_i ,
- The first k columns of Z capture the directions of maximum variance in the data, allowing for dimensionality reduction if $k < p$.
- So these first k columns of Z are in essence the k principal components of our original matrix A (without the target)

By choosing k columns that explain, for example a determined percentage of the total variance (95% in our case) we can reduce the dimensionality of our original matrix.

The problem here is that these k columns are not directly interpretable, because they are a linear combination of all features in matrix A . We do not have the individual variance attributed by each feature. That is why we resort to an extension of PCA to retain interpretability on the original features, but still reduce the number of features, via explained variance.

A.5.1 Sparse PCA

The limitations of traditional PCA for feature selection leads us to use the **Sparse PCA**. Here we place a sparsity constraint on the loading vectors. In contrast to standard PCA, where each principal component can contain information from all

features, Sparse PCA encourages many of the coefficients in each loading vector to be zero. As a result, the resulting principal components are linear combinations of a subset of features: feature selection.

Sparse PCA modifies the original optimization problem of PCA to include an $L1$ -penalty on the loadings matrix U . This constraint enforces sparsity in the principal directions. The method still looks to maximize the variance explained:

$$\max_U \text{Var}(AU) \quad \text{s.t. sparsity on } U$$

As a result, the method gives a low-dimensional representation of the data like in standard PCA, but also allows for feature-level interpretability. One can identify which original variables contribute to each sparse component. With a value of $\lambda = 1$ on the $L1$ -penalty we obtain a set of interpretable features using Sparse PCA.

A.5.2 Principal Feature Analysis (PFA)

An alternative to Sparse PCA that also results in feature selection is **Principal Feature Analysis (PFA)**. Instead of constructing sparse principal components, PFA uses the structure of PCA to directly select a subset of original features. These features capture the variance in the data in the best way possible.

First we need to standardize the input matrix A , and then apply standard PCA to obtain the top k components that explain a predetermined amount of variance, in our case 95%. In this way we obtain a set of principal axes—linear combinations of the original features that span a lower-dimensional subspace containing most of the information in the data.

These components are not used directly. PFA projects each feature (column of the standardized matrix) into the principal component space. Then one clusters the projected features using a clustering algorithm, in our case K-means setting 5 clusters. The idea is that features with similar projection profiles (loadings) on the principal components are redundant, ergo can be grouped together.

From each group, a single representative feature is selected, we select the one with the largest absolute loading on the first principal component. This feature is then retained as a proxy for the others in the cluster.

Mathematically, if $V_k \in \mathbb{R}^{p \times k}$ contains the top k principal directions (loadings), then the projected features are given by:

$$\text{PC-loadings} = V_k^\top \in \mathbb{R}^{k \times p}$$

These are clustered into m groups, where m is the desired number of features to retain.

PFA provides a computationally efficient and interpretable approach to dimensionality reduction. We can select actual columns from the original dataset, instead of forming new combinations of all features. Still, we make use of PCA to identify where most of the variance comes from. Thanks to this interpretability we can get the features that explain most of the variance of our data matrix A .

A.6 Vector Autoregression (VAR)

In mathematical terms we can stack p variables into a vector $y_t \in \mathbb{R}^p$. The simplest VAR model is the VAR(1), which means that the vector of variables only depends on

its first lag. The VAR(1) model is written as:

$$y_{t+1} = Ay_t + \varepsilon_t,$$

where $A \in \mathbb{R}^{p \times p}$ is the transition matrix estimated using least squares. That is, estimate the A using multivariate least squares, which is equivalent to the ordinary least squares estimator for each separate equation. ε_t is a vector of error terms.

Order of integration One of the main assumptions in the VAR is that the variables have the same order of integration. For addressing this, we log-differentiate all variables before fitting a VAR model on them, to obtain stationary variables, i.e., variables with integration order zero $I(0)$: stationary.

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